

Feature catalogue printout of S201

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Table of Contents

FeatureTypes

1. [Aids to Navigation](#)
2. [Structure Object](#)
3. [Equipment](#)
4. [Electronic Aton](#)
5. [Generic Beacon](#)
6. [Generic Buoy](#)
7. [Generic Light](#)
8. [Landmark](#)
9. [Lateral Beacon](#)
10. [Lateral Buoy](#)
11. [Navigation Line](#)
12. [Recommended Track](#)
13. [Light Sector](#)
14. [Light All Around](#)
15. [Light Air Obstruction](#)
16. [Light Fog Detector](#)
17. [Radar Reflector](#)
18. [Fog Signal](#)
19. [Environment Observation Equipment](#)
20. [Radio Station](#)
21. [Daymark](#)
22. [Retroreflector](#)
23. [Radar Transponder Beacon](#)
24. [Virtual AIS Aid to Navigation](#)
25. [Physical AIS Aid to Navigation](#)
26. [Synthetic AIS Aid to Navigation](#)
27. [Power Source](#)
28. [Isolated Danger Beacon](#)
29. [Cardinal Beacon](#)
30. [Isolated Danger Buoy](#)
31. [Cardinal Buoy](#)
32. [Installation Buoy](#)
33. [Mooring Buoy](#)
34. [Emergency Wreck Marking Buoy](#)
35. [Lighthouse](#)
36. [Light Float](#)
37. [Light Vessel](#)
38. [Offshore Platform](#)
39. [Silo/Tank](#)
40. [Pile](#)
41. [Building](#)
42. [Bridge](#)
43. [Sinker Anchor](#)
44. [Mooring Shackle](#)
45. [Cable Submarine](#)
46. [Swivel](#)
47. [Bridle](#)
48. [Counter Weight](#)
49. [Topmark](#)
50. [Safe Water Beacon](#)
51. [Special Purpose/General Beacon](#)
52. [Safe Water Buoy](#)
53. [Special Purpose/General Buoy](#)
54. [Dangerous Feature](#)
55. [Aton Aggregation](#)
56. [Aton Association](#)
57. [Quality of Non-Bathymetric Data](#)

58. Data Coverage
59. Local Direction of Buoyage
60. Navigational System of Marks
61. Sounding Datum
62. Vertical Datum of Data

InformationTypes

1. AtoN Fixing Method
2. Aton Status Information
3. Positioning Information
4. Spatial Quality

Simple attributes

1. Category of Association
2. Seasonal Action Required
3. Counter Weight Type
4. Swivel Type
5. Legs Details
6. BridleLink Type
7. Sinker Type
8. Weight
9. Cable Length
10. Diameter
11. Horizontal Accuracy
12. Manufacturer
13. IsSlatted
14. Candela
15. Effective Intensity
16. Peak Intensity
17. Vertical Accuracy
18. Quality of Horizontal Measurement
19. Positioning Device
20. Change Types
21. Height/Length Units
22. Positioning Procedure
23. Horizontal Datum
24. Reference Point
25. Signal Status
26. Signal Duration
27. Sector Line Length
28. Sector Bearing
29. Wave Length Value
30. Radar Band
31. Number of Features
32. Multiplicity Known
33. Sector Extension
34. Moire Effect
35. Category of Cable
36. Category of Installation Buoy
37. Shackle Type
38. Category of Pile
39. Category of Silo/Tank
40. Building Shape
41. Product
42. Category of Offshore Platform
43. Horizontal Width
44. Horizontal Length
45. Category of Cardinal Mark
46. Type of Light
47. NMEAString
48. Light Characteristic
49. Category Of Power Source
50. Category Of Synthetic AIS Aid To Navigation
51. Category Of Physical AIS Aid To Navigation
52. Virtual AIS Aid to Navigation Type
53. MMSI Code
54. Category of Radar Transponder Beacon
55. Topmark/Daymark Shape
56. Orientation Value
57. Category of Special Purpose Mark
58. Estimated Range of Transmission
59. Category of Radio Station
60. Type of Environmental Observation Equipment
61. Value of Maximum Range

62. Signal Period
63. Signal Output
64. Signal Group
65. Signal Frequency
66. Flare Bearing
67. Category of Fog Signal
68. Value of Nominal Range
69. Value of Luminous Range
70. Value of Geographic Range
71. Major Light
72. Light Visibility
73. Signal Generation
74. Exhibition Condition of Light
75. Category of Light
76. Traffic Flow
77. Technique of Vertical Measurement
78. Quality of Vertical Measurement
79. Maximal Permitted Draught
80. Depth Range Minimum Value
81. Based On Fixed Marks
82. Category of Navigation Line
83. Category of Lateral Mark
84. Manned Structure
85. Vertical Datum
86. Function
87. Category of Landmark
88. Type of Buoy
89. Buoy Shape
90. Visual Prominence
91. Vertical Length
92. Status
93. Radar Conspicuous
94. Nature of Construction
95. Marks Navigational - System Of
96. Height
97. Elevation
98. Colour Pattern
99. Colour
100. Beacon Shape
101. Date Start
102. Date End
103. Installation Date
104. AtoN Number
105. ID Code
106. Scale Minimum
107. Source Date
108. Source
109. Pictorial Representation
110. Inspection Frequency
111. Inspection Requirements
112. AtoN Maintenance Record
113. Display Name
114. Name
115. Aid Availability Category
116. Condition
117. Remote Monitoring System
118. Delivery Point
119. City Name
120. Administrative Division
121. Country Name
122. Postal Code
123. Category of Aggregation
124. Text
125. Language
126. Headline
127. File Reference
128. File Locator
129. Uncertainty Variable Factor
130. Uncertainty Fixed
131. Information in National Language
132. Horizontal Distance Uncertainty
133. Orientation Uncertainty
134. Category of Temporal Variation
135. Maximum Display Scale
136. Minimum Display Scale
137. Aton Commissioning
138. Aton Removal

139. Aton Replacement
140. Fixed Aton Change
141. Floating Aton Change
142. Audible Signal Aton Change
143. Lighted Aton Change
144. Electronic Aton Change
145. Positioning Equipment

Complex attributes

1. Contact Address
2. Directional Character
3. Feature Name
4. Fixed Date Range
5. Light Sector
6. Multiplicity of Features
7. Orientation
8. Periodic Date Range
9. Radar Wave Length
10. Rhythm of Light
11. Sector Characteristics
12. Sector Information
13. Sector Limit
14. Sector Limit One
15. Sector Limit Two
16. Shape Information
17. Signal Sequence
18. Spatial Accuracy
19. Cable Dimensions
20. Change Details
21. Obscured Sector
22. Sinker Dimensions
23. Positioning Method
24. Horizontal Position Uncertainty
25. Information
26. Textual Description
27. Vertical Uncertainty

Roles

1. Aton Part
2. Status Part
3. Buoy Part
4. Topmark part
5. Parent
6. Child
7. Physical AIS Broadcast By
8. Physical AIS Broadcasts
9. Synthetic AIS Broadcast By
10. Synthetic AIS Broadcasts
11. Virtual AIS Broadcast By
12. Virtual AIS Broadcasts
13. Buoy Attached
14. Counter Weightholds
15. Buoy Hangs
16. Bridle Holds
17. Shackle To Cableconnected To
18. Shackle To Cableconnected
19. Swivel Attached
20. Bridle Attached
21. Cable Holds
22. Shackle To Bridle Connecteda
23. Shackle To Bridle Connected To
24. Shackle To Buoy Connected
25. Shackle To Buoy Connected To
26. Shackle To Swivel Connected
27. Shackle To Swivel Connected To
28. Shackle To Anchor Connected To
29. Shackle To Anchor Connected
30. Bridle Hangs
31. Swivel Holds
32. Peer Aton Aggregation
33. Aton Aggregation By
34. Peer Aton Association
35. Aton Association By
36. Navigable Track

37. [Navigation Line](#)
38. [Fixing Method](#)
39. [Fixed By](#)
40. [Positioning Method](#)
41. [Positioned By](#)
42. [Danger](#)
43. [Marking Aton](#)

Information associations

1. [Aton Status](#)
2. [Aton Fixing Method Association](#)
3. [Aton Positioning Information Association](#)

Feature associations

1. [Buoy Topmark](#)
2. [Structure/Equipment](#)
3. [Physical AIS](#)
4. [Synthetic AIS](#)
5. [Virtual AIS](#)
6. [Buoy Counter Weight](#)
7. [Bridle Connection](#)
8. [Shackle Connection](#)
9. [Shackle Connection From Cable](#)
10. [Swivel Cable Connection](#)
11. [Bridle Cable Connection](#)
12. [Shackle To Bridle Connection](#)
13. [Shackle To Swivel Connection](#)
14. [ShackleToAnchorConnection](#)
15. [Swivel Connection](#)
16. [Aton Aggregations](#)
17. [Aton Associations](#)
18. [Range System](#)
19. [Dangerous Feature Association](#)

1 Feature Types

1.1 Aids to Navigation

Type: Abstract FeatureType

Definition: A visual, acoustical, or radio device, external to a ship, designed to assist in determining a safe course or a vessel's position, or to warn of dangers and/or obstructions. Aids to navigation usually include buoys, beacons, fog signals, lights, radio beacons, leading marks, radio position fixing systems and GNSS which are chart-related and assist safe navigation.

CamelCase: AidsToNavigation

Super type: AbstractFeatureType

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
ID Code			TE	0, 1
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1

Text			(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name			(S) TE	1, 1
Scale Minimum			IN	0, 1
Source Date			DA	0, 1
Source			TE	0, 1
Pictorial Representation			TE	0, 1
Inspection Frequency			TE	0, 1
Inspection Requirements			TE	0, 1
AtoN Maintenance Record			TE	0, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Periodic Date Range			C	0, 1
Date End			(S) TD	1, 1
Date Start			(S) TD	1, 1
Seasonal Action Required			TE	0, *

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	AtonAggregations	Aids to Navigation	atonAggregationBy	0..*	AtonAggregation	peerAtonAggregation	0..*
	--FeatureType-- Aids to Navigation 0..* AtonAggregations 0..* atonAggregationBy peerAtonAggregation --FeatureType-- AtonAggregation						
Association	AtonAssociations	Aids to Navigation	atonAssociationBy	0..*	AtonAssociation	peerAtonAssociation	0..*
	--FeatureType-- Aids to Navigation 0..* AtonAssociations 0..* atonAssociationBy peerAtonAssociation --FeatureType-- AtonAssociation						

Information bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	Atonstatus	Aids to Navigation			AtonStatusInformation	Statuspart	0..1
	--FeatureType-- Aids to Navigation <i>Atonstatus</i> 0..1 --InformationType-- AtonStatusInformation						

1.2 Structure Object

Type: Abstract FeatureType

Definition: Something (such as a house, tower, bridge, etc.) that is built by putting parts together and that usually stands on its own.

CamelCase: StructureObject

Super type: AidsToNavigation

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			UI	1, 1
Aid Availability Category		1. Category 1 2. Category 2 3. Category 3	EN	0, 1
Condition		1. Under Construction 2. Ruined 3. Under Reclamation 4. Wingless 5. Planned Construction	EN	0, 1
Contact Address			C	0, 1
Delivery Point			(S) TE	0, 1
City Name			(S) TE	0, 1
Administrative Division			(S) TE	0, 1
Country Name			(S) TE	0, 1
Postal Code			(S) TE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	StructureEquipment	Structure Object	parentFihoXyz	1..1	Equipment	child	0..*
	--FeatureType-- Structure Object 1..1 StructureEquipment 0..* parentFihoXyz child --FeatureType-- Equipment						

Information bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	AtonPositioningInformationAssociation	Structure Object			PositioningInformation	positioningMethodFihoXyz	0..*
	--FeatureType-- Structure Object AtonPositioningInformationAssociation 0..* positioningMethodFihoXyz --InformationType-- PositioningInformation						
Association	AtonFixingMethodAssociation	Structure Object			AtoNFixingMethod	fixingMethod	0..*
	--FeatureType-- Structure Object AtonFixingMethodAssociation 0..* fixingMethod --InformationType-- AtoNFixingMethod						

1.3 Equipment

Type: Abstract FeatureType

Definition: The implements used in an operation or activity.

CamelCase: Equipment

Super type: AidsToNavigation

Primitive: -

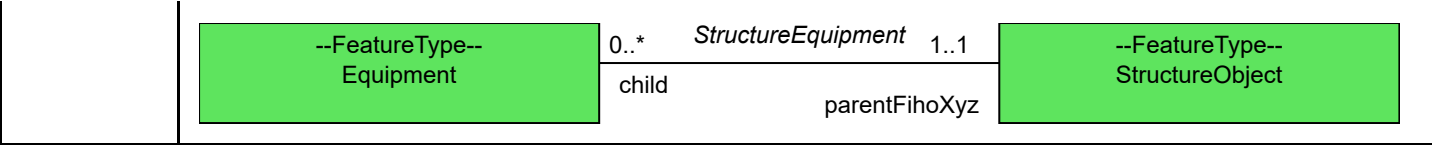
Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Remote Monitoring System			TE	0, *

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	StructureEquipment	Equipment	child	0..*	StructureObject	parentFihoXyz	1..1



1.4 Electronic Aton

Type: Abstract FeatureType

Definition: TBD

CamelCase: ElectronicAton

Super type: AidsToNavigation

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			UI	0, 1
MMSI Code			TE	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away	EN	0, *

		21. Rising		
		22. Increasing		
		23. Decreasing		
		24. Strong		
		25. Good		
		26. Moderately		
		27. Poor		
		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.5 Generic Beacon

Type: Abstract FeatureType

Definition: A fixed artificial navigation mark that can be recognized by its shape, colour, pattern, topmark or light character, or a combination of these. It may carry various additional aids to navigation.

CamelCase: GenericBeacon

Super type: StructureObject

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Beacon Shape		1. Stake, Pole, Perch, Post 2. Withy 3. Beacon Tower 4. Lattice Beacon 5. Pile Beacon 6. Cairn 7. Buoyant Beacon	EN	1, 1
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	1, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes	EN	0, *

		4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle		
Elevation			RE	0, 1
Height			RE	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed	EN	0, *

		12. Glass 13. Fiberglass 14. Plastic		
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed	EN	0, *

		29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

1.6 Generic Buoy

Type: Abstract FeatureType

Definition: A floating object moored to the bottom in a particular (charted) place, as an aid to navigation or for other specific purposes.

CamelCase: GenericBuoy

Super type: StructureObject

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Buoy Shape		1. Conical 2. Can 3. Spherical	EN	1, 1

		4. Pillar 5. Spar 6. Barrel 7. Superbuoy 8. Ice Buoy		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	1, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown)	EN	0, *

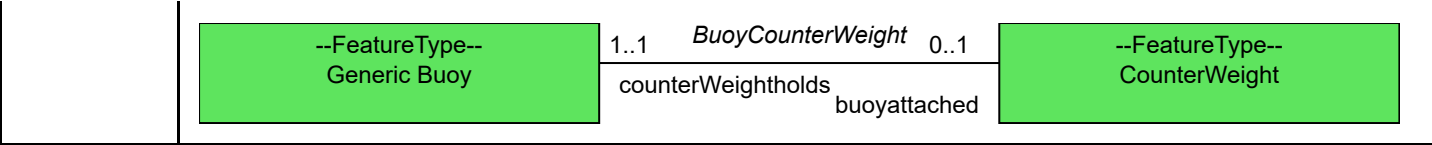
		6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle		
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, *

Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting	EN	0, *

		32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Type of Buoy			TE	0, 1
Vertical Length			RE	0, 1
Vertical Accuracy			RE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BuoyTopmark	Generic Buoy	buoyPart	1..1	Topmark	topmarkPart	0..*
	--FeatureType-- Generic Buoy 1..1 BuoyTopmark 0..* --FeatureType-- Topmark buoyPart topmarkPart						
Association	ShackleConnection	Generic Buoy	shackleToBuoyconnectedTo	0..1	MooringShackle	shackleToBuoyconnected	1..1
	--FeatureType-- Generic Buoy 0..1 ShackleConnection 1..1 --FeatureType-- MooringShackle shackleToBuoyconnectedTo shackleToBuoyconnected						
Association	BridgeConnection	Generic Buoy	bridgeholds	1..1	Bridge	buoyhangs	0..1
	--FeatureType-- Generic Buoy 1..1 BridgeConnection 0..1 --FeatureType-- Bridge bridgeholds buoyhangs						
Association	BuoyCounterWeight	Generic Buoy	counterWeights	1..1	CounterWeight	buoyattached	0..1



1.7 Generic Light

Type: Abstract FeatureType

Definition: -

CamelCase: GenericLight

Super type: Equipment

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	1, *

Height			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting	EN	0, *

		32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water 17. Mean High Water Springs 18. High Water	EN	0, 1

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes
Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference
Level

32. Local High Water Reference
Level

33. Local Mean Water Reference
Level

34. Equivalent Height of Water
(German GIW)

35. Highest Shipping Height of
Water (German HSW)

36. Reference Low Water Level
According to Danube
Commission

37. Highest Shipping Height of
Water According to Danube
Commission

38. Dutch River Low Water
Reference Level (OLR)

39. Russian Project Water Level

40. Russian Normal Backwater
Level

41. Ohio River Datum

43. Dutch High Water Reference
Level

44. Baltic Sea Chart Datum 2000

45. Dutch Estuary Low Water
Reference Level (OLW)

Vertical Length			RE	0, 1
Effective Intensity			RE	0, 1
Peak Intensity			RE	0, 1

1.8 Landmark

Type: FeatureType

Definition: A prominent object at a fixed location on land which can be used in determining a location or a direction.

CamelCase: Landmark

Super type: StructureObject

Primitive: point,curve,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Landmark		1. Cairn 2. Cemetery 3. Chimney 4. Dish Aerial 5. Flagstaff 6. Flare Stack 7. Mast 8. Windsock 9. Monument 10. Column/Pillar 11. Memorial Plaque 12. Obelisk 13. Statue 14. Cross 15. Dome 16. Radar Scanner 17. Tower 18. Windmill 19. Windmotor	EN	1, *

		20. Spire/Minaret 21. Large Rock or Boulder on Land 22. Triangulation Mark 23. Boundary Mark 24. Observation Wheel 25. Torii 26. Bridge 27. Dam		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes	EN	0, *

		3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle		
Function		2. Harbour-Masters Office 3. Customs Office 4. Health Office 5. Hospital 6. Post Office 7. Hotel 8. Railway Station 9. Police Station 10. Water-Police Station 11. Pilot Office 12. Pilot Lookout 13. Bank Office 14. Headquarters for District Control 15. Transit Shed/Warehouse 16. Factory 17. Power Station 18. Administrative 19. Educational Facility 20. Church 21. Chapel 22. Temple 23. Pagoda 24. Shinto Shrine	EN	0, *

		25. Buddhist Temple 26. Mosque 27. Marabout 28. Lookout 29. Communication 30. Television 31. Radio 32. Radar 33. Light Support 34. Microwave 35. Cooling 36. Observation 37. Timeball 38. Clock 39. Control 40. Airship Mooring 41. Stadium 42. Bus Station 43. Passenger Terminal Building 44. Sea Rescue Control 45. Observatory 46. Ore Crusher 47. Boathouse 48. Pumping Station		
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal	EN	0, *

		8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong	EN	0, *

		25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water	EN	0, 1

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| 13. Low Water |
| 14. Approximate Mean Low Water |
| 15. Approximate Mean Lower Low Water |
| 16. Mean High Water |
| 17. Mean High Water Springs |
| 18. High Water |
| 19. Approximate Mean Sea Level |
| 20. High Water Springs |
| 21. Mean Higher High Water |
| 22. Equinoctial Spring Low Water |
| 23. Lowest Astronomical Tide |
| 24. Local Datum |
| 25. International Great Lakes Datum 1985 |
| 26. Mean Water Level |
| 27. Lower Low Water Large Tide |
| 28. Higher High Water Large Tide |
| 29. Nearly Highest High Water |
| 30. Highest Astronomical Tide |
| 31. Local Low Water Reference Level |
| 32. Local High Water Reference Level |
| 33. Local Mean Water Reference Level |
| 34. Equivalent Height of Water (German GIW) |
| 35. Highest Shipping Height of Water (German HSW) |
| 36. Reference Low Water Level According to Danube Commission |
| 37. Highest Shipping Height of Water According to Danube Commission |
| 38. Dutch River Low Water Reference Level (OLR) |

		39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	1, 1
Elevation			RE	0, 1
Height			RE	0, 1
Manned Structure			BO	0, 1
Vertical Length			RE	0, 1
Vertical Accuracy			RE	0, 1

1.9 Lateral Beacon

Type: FeatureType

Definition: A lateral beacon is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark		1. Port-Hand Lateral Mark 2. Starboard-Hand Lateral Mark 3. Preferred Channel to Starboard Lateral Mark 4. Preferred Channel to Port Lateral Mark 5. Right-Hand Side of the Waterway	EN	1, 1

		6. Left-Hand Side of the Waterway		
		7. Right-Hand Side of the Channel		
		8. Left-Hand Side of the Channel		
		9. Bifurcation of the Waterway		
		10. Bifurcation of the Channel		
		11. Channel Near the Right Bank		
		12. Channel Near the Left Bank		
		13. Channel Cross-Over to the Right Bank		
		14. Channel Cross-Over to the Left Bank		
		15. Danger Point or Obstacles at the Right-Hand Side		
		16. Danger Point or Obstacles at the Left-Hand Side		
		17. Turn Off at the Right-Hand Side		
		18. Turn Off at the Left-Hand Side		
		19. Junction at the Right-Hand Side		
		20. Junction at the Left-Hand Side		
		21. Harbour Entry at the Right-Hand Side		
		22. Harbour Entry at the Left-Hand Side		
		23. Bridge Pier Mark		
		24. Entry From a Lake to a Narrower Waterway, Right Bank		
		25. Entry From a Lake to a Narrower Waterway, Left Bank		
		26. Change Bank		
		27. Continue Along Bank		

1.10 Lateral Buoy

Type: FeatureType

Definition: A lateral buoy is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well-defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark		1. Port-Hand Lateral Mark 2. Starboard-Hand Lateral Mark 3. Preferred Channel to Starboard Lateral Mark 4. Preferred Channel to Port Lateral Mark 5. Right-Hand Side of the Waterway 6. Left-Hand Side of the Waterway 7. Right-Hand Side of the Channel 8. Left-Hand Side of the Channel 9. Bifurcation of the Waterway 10. Bifurcation of the Channel 11. Channel Near the Right Bank 12. Channel Near the Left Bank 13. Channel Cross-Over to the Right Bank 14. Channel Cross-Over to the Left Bank 15. Danger Point or Obstacles at the Right-Hand Side 16. Danger Point or Obstacles at the Left-Hand Side 17. Turn Off at the Right-Hand Side 18. Turn Off at the Left-Hand Side 19. Junction at the Right-Hand Side 20. Junction at the Left-Hand Side 21. Harbour Entry at the Right-Hand Side	EN	1, 1

		22. Harbour Entry at the Left-Hand Side 23. Bridge Pier Mark 24. Entry From a Lake to a Narrower Waterway, Right Bank 25. Entry From a Lake to a Narrower Waterway, Left Bank 26. Change Bank 27. Continue Along Bank		
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1.11 Navigation Line

Type: FeatureType

Definition: A straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing.

CamelCase: NavigationLine

Super type: AidsToNavigation

Primitive: curve,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Navigation Line		1. Clearing Line 2. Transit Line 3. Leading Line Bearing a Recommended Track	EN	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated	EN	0, *

		13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Orientation			C	1, 1

Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	RangeSystem	Navigation Line	navigationLineFihoXyz	1..*	RecommendedTrack	navigableTrack	0..*
	--FeatureType-- Navigation Line 1..* RangeSystem 0..* navigationLineFihoXyz navigableTrack --FeatureType-- RecommendedTrack						

1.12 Recommended Track

Type: FeatureType

Definition: A route which has been specially examined to ensure so far as possible that it is free of dangers and along which ships are advised to navigate.

CamelCase: RecommendedTrack

Super type: AidsToNavigation

Primitive: curve,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Based On Fixed Marks			BO	1, 1
Depth Range Minimum Value			RE	0, 1
Maximal Permitted Draught			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished	EN	0, *

		12. Illuminated		
		13. Historic		
		14. Public		
		15. Synchronized		
		16. Watched		
		17. Unwatched		
		18. Existence Doubtful		
		19. On Request		
		20. Drop Away		
		21. Rising		
		22. Increasing		
		23. Decreasing		
		24. Strong		
		25. Good		
		26. Moderately		
		27. Poor		
		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water 17. Mean High Water Springs 18. High Water 19. Approximate Mean Sea Level 20. High Water Springs 21. Mean Higher High Water 22. Equinoctial Spring Low Water 23. Lowest Astronomical Tide 24. Local Datum 25. International Great Lakes Datum 1985 26. Mean Water Level 27. Lower Low Water Large Tide 28. Higher High Water Large Tide 29. Nearly Highest High Water	EN	0, 1
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		30. Highest Astronomical Tide 31. Local Low Water Reference Level 32. Local High Water Reference Level 33. Local Mean Water Reference Level 34. Equivalent Height of Water (German GIW) 35. Highest Shipping Height of Water (German HSW) 36. Reference Low Water Level According to Danube Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Vertical Uncertainty			C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Quality of Vertical Measurement		1. Depth Known 2. Depth or Least Depth Unknown 3. Doubtful Sounding 4. Unreliable Sounding	EN	0, *

		5. No Bottom Found at Value Shown 6. Least Depth Known 7. Least Depth Unknown, Safe Clearance at Value Shown 8. Value Reported (Not Surveyed) 9. Value Reported (Not Confirmed) 10. Maintained Depth 11. Not Regularly Maintained		
Technique of Vertical Measurement		1. Found by Echo Sounder 2. Found by Side Scan Sonar 3. Found by Multi Beam 4. Found by Diver 5. Found by Lead Line 6. Swept by Wire-Drag 7. Found by Laser 8. Swept by Vertical Acoustic System 9. Found by Electromagnetic Sensor 10. Photogrammetry 11. Satellite Imagery 12. Found by Levelling 13. Swept by Side Scan Sonar 14. Computer Generated 15. Found by LIDAR 16. Synthetic Aperture Radar 17. Hyperspectral Imagery	EN	0, *
Traffic Flow		1. Inbound 2. Outbound 3. One-Way 4. Two-Way	EN	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	RangeSystem	Recommended Track	navigableTrack	0..*	NavigationLine	navigationLineFihoXyz	1..*
	--FeatureType-- Recommended Track 0..*RangeSystem1..*NavigationLine navigableTracknavigationLineFihoXyz --FeatureType-- NavigationLine						

1.13 Light Sector

Type: FeatureType

Definition: A light presenting different appearances (in particular, different colours) over various parts of the horizon of interest to maritime navigation.

CamelCase: LightSector

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Light		1. Directional Function 4. Leading Light 5. Aero Light 6. Air Obstruction Light 8. Flood Light 9. Strip Light 10. Subsidiary Light 11. Spotlight 12. Front 13. Rear 14. Lower 15. Upper 17. Emergency 18. Bearing Light 19. Horizontally Disposed 20. Vertically Disposed	EN	0, *

Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, *
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Obscured Sector			C	0, *
Sector Limit			(S) C	1, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1

Sector Line Length			(S) IN	0, 1
Sector Information			(S) C	0, 1
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Characteristics			C	1, *
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light 31. Composite Group-Occulting Light	(S) EN	1, 1

		32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light		
Light Sector			(S) C	1, 10
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	(S) EN	1, *
Directional Character			(S) C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	(S) EN	0, *
Sector Limit			(S) C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Value of Nominal Range			(S) RE	0, 1
Sector Information			(S) C	0, *
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Extension			(S) BO	0, 1
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Candela			(S) RE	0, 1

1.14 Light All Around

Type: FeatureType

Definition: An all around light is a light that is visible over the whole horizon of interest to marine navigation and having no change in the characteristics of the light.

CamelCase: LightAllAround

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Light		1. Directional Function 4. Leading Light 5. Aero Light 6. Air Obstruction Light 8. Flood Light 9. Strip Light 10. Subsidiary Light 11. Spotlight 12. Front 13. Rear 14. Lower 15. Upper 17. Emergency 18. Bearing Light 19. Horizontally Disposed 20. Vertically Disposed	EN	0, *
Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, *
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint	EN	0, 1

		4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range		
Major Light			BO	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Value of Nominal Range			RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1
Rhythm of Light			C	1, 1

Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light 31. Composite Group-Occulting Light 32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light	(S) EN	1, 1
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Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Flare Bearing			IN	0, 1

1.15 Light Air Obstruction

Type: FeatureType

Definition: An air obstruction light is a light marking an obstacle which constitutes a danger to air navigation.

CamelCase: LightAirObstruction

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	EN	0, *
Value of Geographic Range			RE	0, 1

Value of Luminous Range			RE	0, 1
Value of Nominal Range			RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1
Rhythm of Light			C	1, 1
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light	(S) EN	1, 1

		31. Composite Group-Occulting Light 32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light		
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Flare Bearing			IN	0, 1

1.16 Light Fog Detector

Type: FeatureType

Definition: A fog detector light is a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal.

CamelCase: LightFogDetector

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Rhythm of Light			C	1, 1
Light Characteristic		1. Fixed	(S) EN	1, 1

		2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light 31. Composite Group-Occulting Light 32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light		
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1

Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.17 Radar Reflector

Type: FeatureType

Definition: A device capable of, or intended for, reflecting radar signals.

CamelCase: RadarReflector

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful	EN	0, *

		19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs	EN	0, 1

7. Approximate Mean Low Water Springs

8. Indian Spring Low Water

9. Low Water Springs

10. Approximate Lowest Astronomical Tide

11. Nearly Lowest Low Water

12. Mean Lower Low Water

13. Low Water

14. Approximate Mean Low Water

15. Approximate Mean Lower Low Water

16. Mean High Water

17. Mean High Water Springs

18. High Water

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference Level

32. Local High Water Reference Level

33. Local Mean Water Reference Level

34. Equivalent Height of Water (German GIW)

		35. Highest Shipping Height of Water (German HSW) 36. Reference Low Water Level According to Danube Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Vertical Accuracy			RE	0, 1

1.18 Fog Signal

Type: FeatureType

Definition: A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal.

CamelCase: FogSignal

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Fog Signal		1. Explosive 2. Diaphone 3. Siren 4. Nautophone 5. Reed 6. Tyfon 7. Bell	EN	1, 1

		8. Whistle 9. Gong 10. Horn		
Signal Frequency			IN	0, 1
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Signal Group			TE	0, 1
Signal Output			RE	0, 1
Signal Period			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful	EN	0, *

		19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Value of Maximum Range			RE	0, 1
Signal Sequence			C	0, 1
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.19 Environment Observation Equipment

Type: FeatureType

Definition: A sensor used to observe the environment.

CamelCase: EnvironmentObservationEquipment

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong	EN	0, *

		25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Type of Environmental Observation Equipment			TE	1, *

1.20 Radio Station

Type: FeatureType

Definition: A place equipped to transmit radio waves. Such a station may be either stationary or mobile, and may also be provided with a radio receiver.

CamelCase: RadioStation

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Radio Station		1. Circular (Non-Directional) Marine or Aero-Marine Radiobeacon 2. Directional Radiobeacon	EN	1, 1

		3. Rotating Pattern Radiobeacon 4. Consol Beacon 5. Radio Direction-Finding Station 6. Coast Radio Station Providing QTG Service 7. Aeronautical Radiobeacon 8. Decca 9. Loran C 10. Differential GNSS 11. Toran 12. Omega 13. Syledis 14. Chaika 19. Radio Telephone Station 20. AIS Base Station		
Estimated Range of Transmission			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched	EN	0, 1

		17. Unwatched		
		18. Existence Doubtful		
		19. On Request		
		20. Drop Away		
		21. Rising		
		22. Increasing		
		23. Decreasing		
		24. Strong		
		25. Good		
		26. Moderately		
		27. Poor		
		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	PhysicalAIS	Radio Station	physicalAIS broadcasts	1..*	PhysicalAIS AidToNavigation	physicalAIS broadcastBy	0..1

	--FeatureType-- Radio Station 1..* PhysicalAIS 0..1 physicalAISbroadcasts physicalAISbroadcastBy --FeatureType-- PhysicalAISaidToNavigation						
Association	SyntheticAIS	Radio Station	syntheticAISbroadcasts	1..*	SyntheticAISaidToNavigation	syntheticAISbroadcastBy	0..1
	--FeatureType-- Radio Station 1..* SyntheticAIS 0..1 syntheticAISbroadcasts syntheticAISbroadcastBy --FeatureType-- SyntheticAISaidToNavigation						
Association	VirtualAIS	Radio Station	virtualAISbroadcasts	0..1	VirtualAISaidToNavigation	virtualAISbroadcastBy	0..*
	--FeatureType-- Radio Station 0..1 VirtualAIS 0..* virtualAISbroadcasts virtualAISbroadcastBy --FeatureType-- VirtualAISaidToNavigation						

1.21 Daymark

Type: FeatureType

Definition: The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid.

CamelCase: Daymark

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark 7. Spoil Ground Mark 8. Outfall Mark 9. ODAS 10. Recording Mark 11. Seaplane Anchorage Mark 12. Recreation Zone Mark	EN	0, 1

- | | | |
|--|--|--|
| | 13. Private Mark | |
| | 14. Mooring Mark | |
| | 15. LANBY | |
| | 16. Leading Mark | |
| | 17. Measured Distance Mark | |
| | 18. Notice Mark | |
| | 19. TSS Mark | |
| | 20. Anchoring Prohibited Mark | |
| | 21. Berthing Prohibited Mark | |
| | 22. Overtaking Prohibited Mark | |
| | 23. Two-Way Traffic Prohibited Mark | |
| | 24. Reduced Wake Mark | |
| | 25. Speed Limit Mark | |
| | 26. Stop Mark | |
| | 27. General Warning Mark | |
| | 28. Sound Ship's Siren Mark | |
| | 29. Restricted Vertical Clearance Mark | |
| | 30. Maximum Vessel's Draught Mark | |
| | 31. Restricted Horizontal Clearance Mark | |
| | 32. Strong Current Warning Mark | |
| | 33. Berthing Permitted Mark | |
| | 34. Overhead Power Cable Mark | |
| | 35. Channel Edge Gradient Mark | |
| | 36. Telephone Mark | |
| | 37. Ferry Crossing Mark | |
| | 39. Pipeline Mark | |
| | 40. Anchorage Mark | |
| | 41. Clearing Mark | |
| | 42. Control Mark | |
| | 43. Diving Mark | |

		44. Refuge Beacon 45. Foul Ground Mark 46. Yachting Mark 47. Heliport Mark 48. GNSS Mark 49. Seaplane Landing Mark 50. Entry Prohibited Mark 51. Work in Progress Mark 52. Mark With Unknown Purpose 53. Wellhead Mark 54. Channel Separation Mark 55. Marine Farm Mark 56. Artificial Reef Mark 57. Ice Mark 58. Nature Reserve Mark 59. Fish Aggregating Device 60. Wreck Mark 61. Customs Mark 62. Causeway Mark 63. Wave Recorder 64. Jetski Prohibited		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet	EN	1, *

		11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Elevation			RE	0, 1
Height			RE	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted	EN	0, *

		10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Orientation Value			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately	EN	0, *

		27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Topmark/Daymark Shape		1. Cone (Point Up) 2. Cone (Point Down) 3. Sphere 4. 2 Spheres 5. Cylinder 6. Board 7. X-Shaped 8. Upright Cross 9. Cube (Point Up) 10. 2 Cones (Point to Point) 11. 2 Cones (Base to Base) 12. Rhombus 13. 2 Cones (Points Upward) 14. 2 Cones (Points Downward) 15. Besom (Point Up)	EN	1, 1

		16. Besom (Point Down) 17. Flag 18. Sphere Over a Rhombus 19. Square 20. Rectangle (Horizontal) 21. Rectangle (Vertical) 22. Trapezium (Up) 23. Trapezium (Down) 24. Triangle (Point Up) 25. Triangle (Point Down) 26. Circle 27. Two Upright Crosses (One Over the Other) 28. T-Shape 29. Triangle Pointing Up Over a Circle 30. Upright Cross Over a Circle 31. Rhombus Over a Circle 32. Circle Over a Triangle Pointing Up 33. Other Shape (See Shape Information) 34. Tubular		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs	EN	0, 1

10. Approximate Lowest
Astronomical Tide

11. Nearly Lowest Low Water

12. Mean Lower Low Water

13. Low Water

14. Approximate Mean Low Water

15. Approximate Mean Lower
Low Water

16. Mean High Water

17. Mean High Water Springs

18. High Water

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes
Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference
Level

32. Local High Water Reference
Level

33. Local Mean Water Reference
Level

34. Equivalent Height of Water
(German GIW)

35. Highest Shipping Height of
Water (German HSW)

36. Reference Low Water Level
According to Danube
Commission

		37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Vertical Length			RE	0, 1
Shape Information			C	0, 1
Language			(S) TE	0, 1
Text			(S) TE	1, 1
IsSlatted			BO	1, 1

1.22 Retroreflector

Type: FeatureType

Definition: A means of distinguishing unlighted marks at night. Retro-reflective material is secured to the mark in a particular pattern to reflect back light.

CamelCase: Retroreflector

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow	EN	0, *

		7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two	EN	0, 1

		Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed	EN	0, *

		29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water	EN	0, 1

- | |
|---|
| 16. Mean High Water |
| 17. Mean High Water Springs |
| 18. High Water |
| 19. Approximate Mean Sea Level |
| 20. High Water Springs |
| 21. Mean Higher High Water |
| 22. Equinoctial Spring Low Water |
| 23. Lowest Astronomical Tide |
| 24. Local Datum |
| 25. International Great Lakes
Datum 1985 |
| 26. Mean Water Level |
| 27. Lower Low Water Large Tide |
| 28. Higher High Water Large Tide |
| 29. Nearly Highest High Water |
| 30. Highest Astronomical Tide |
| 31. Local Low Water Reference
Level |
| 32. Local High Water Reference
Level |
| 33. Local Mean Water Reference
Level |
| 34. Equivalent Height of Water
(German GIW) |
| 35. Highest Shipping Height of
Water (German HSW) |
| 36. Reference Low Water Level
According to Danube
Commission |
| 37. Highest Shipping Height of
Water According to Danube
Commission |
| 38. Dutch River Low Water
Reference Level (OLR) |
| 39. Russian Project Water Level |
| 40. Russian Normal Backwater
Level |
| 41. Ohio River Datum |

		43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Height			RE	0, 1
Vertical Accuracy			RE	0, 1

1.23 Radar Transponder Beacon

Type: FeatureType

Definition: A transponder beacon transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder.

CamelCase: RadarTransponderBeacon

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Radar Transponder Beacon		1. Ramark, Radar Beacon Transmitting Continuously 2. Racon, Radar Transponder Beacon 3. Leading Racon/Radar Transponder Beacon	EN	1, 1
Radar Wave Length			C	0, 1
Radar Band			(S) TE	1, 1
Wave Length Value			(S) RE	1, 1
Signal Group			TE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary	EN	0, *

	8. Private	
	9. Mandatory	
	11. Extinguished	
	12. Illuminated	
	13. Historic	
	14. Public	
	15. Synchronized	
	16. Watched	
	17. Unwatched	
	18. Existence Doubtful	
	19. On Request	
	20. Drop Away	
	21. Rising	
	22. Increasing	
	23. Decreasing	
	24. Strong	
	25. Good	
	26. Moderately	
	27. Poor	
	28. Buoyed	
	29. Fully Operational	
	30. Partially Operational	
	31. Drifting	
	32. Broken	
	33. Offline	
	34. Discontinued	
	35. Manual Observation	
	36. Unknown Status	
	37. Confirmed	
	38. Candidate	
	39. Under Modification	

		41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Value of Nominal Range			RE	0, 1
Manufacturer			TE	0, 1
Sector Limit One			C	0, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			C	0, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Signal Sequence			C	0, 1
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.24 Virtual AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station to simulate on navigation systems an Aid to Navigation which does not physically exist.

CamelCase: VirtualAIS AidToNavigation

Super type: ElectronicAton

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Virtual AIS Aid to Navigation Type		1. North Cardinal 2. East Cardinal 3. South Cardinal 4. West Cardinal 5. Port Lateral 6. Starboard Lateral	EN	1, 1

		7. Preferred Channel to Port 8. Preferred Channel to Starboard 9. Isolated Danger 10. Safe Water 11. Special Purpose 12. New Danger Marking	
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	VirtualAIS	Virtual AIS Aid to Navigation	virtualAISbroadcastBy	0..*	RadioStation	virtualAISbroadcasts	0..1
	--FeatureType-- Virtual AIS Aid to Navigation 0..* <i>VirtualAIS</i> 0..1 --FeatureType-- RadioStation virtualAISbroadcastByvirtualAISbroadcasts						

1.25 Physical AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from a physical Aid to Navigation, or transmitted from an AIS station for an Aid to Navigation which physically exists.

CamelCase: PhysicalAIS AidToNavigation

Super type: ElectronicAton

Primitive: point,

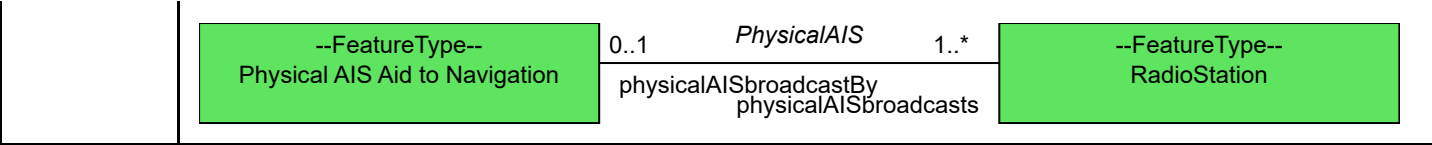
Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Physical AIS Aid To Navigation		1. Physical AIS Type 1 2. Physical AIS Type 2 3. Physical AIS Type 3	EN	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	PhysicalAIS	Physical AIS Aid to Navigation	physicalAISbroadcastBy	0..1	RadioStation	physicalAISbroadcasts	1..*



1.26 Synthetic AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station located remotely from the intended physical Aid to Navigation.

CamelCase: SyntheticAISaidToNavigation

Super type: ElectronicAton

Primitive: point,

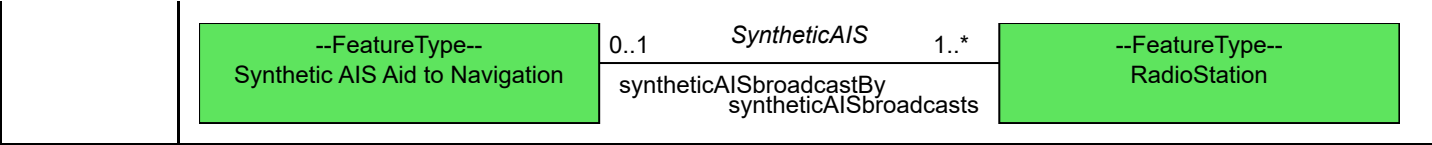
Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Synthetic AIS Aid To Navigation		1. predicted 2. monitored	EN	1, 1
Virtual AIS Aid to Navigation Type		1. North Cardinal 2. East Cardinal 3. South Cardinal 4. West Cardinal 5. Port Lateral 6. Starboard Lateral 7. Preferred Channel to Port 8. Preferred Channel to Starboard 9. Isolated Danger 10. Safe Water 11. Special Purpose 12. New Danger Marking	EN	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	SyntheticAIS	Synthetic AIS Aid to Navigation	syntheticAIS broadcastBy	0..1	RadioStation	syntheticAIS broadcasts	1..*



1.27 Power Source

Type: FeatureType

Definition: -

CamelCase: PowerSource

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Power Source		1. Battery 2. generator 3. solar panel 4. electrical service	EN	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched	EN	0, *

		18. Existence Doubtful	
		19. On Request	
		20. Drop Away	
		21. Rising	
		22. Increasing	
		23. Decreasing	
		24. Strong	
		25. Good	
		26. Moderately	
		27. Poor	
		28. Buoyed	
		29. Fully Operational	
		30. Partially Operational	
		31. Drifting	
		32. Broken	
		33. Offline	
		34. Discontinued	
		35. Manual Observation	
		36. Unknown Status	
		37. Confirmed	
		38. Candidate	
		39. Under Modification	
		41. Under Removal / Deletion	
		42. Removed / Deleted	
		43. Candidate for Modification	

1.28 Isolated Danger Beacon

Type: FeatureType

Definition: An isolated danger beacon is a beacon erected on an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.29 Cardinal Beacon

Type: FeatureType

Definition: A cardinal beacon is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark		1. North Cardinal Mark 2. East Cardinal Mark 3. South Cardinal Mark 4. West Cardinal Mark	EN	1, 1

1.30 Isolated Danger Buoy

Type: FeatureType

Definition: An isolated danger buoy is a buoy moored on or above an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.31 Cardinal Buoy

Type: FeatureType

Definition: A cardinal buoy is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark		1. North Cardinal Mark 2. East Cardinal Mark 3. South Cardinal Mark 4. West Cardinal Mark	EN	1, 1

1.32 Installation Buoy

Type: FeatureType

Definition: An installation buoy is a buoy used for loading tankers with gas or oil.

CamelCase: InstallationBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Installation Buoy		1. Catenary Anchor Leg Mooring 2. Single Buoy Mooring	EN	1, 1

1.33 Mooring Buoy

Type: FeatureType

Definition: A buoy secured to the bottom by permanent moorings with means for mooring a vessel by use of its anchor chain or mooring lines.

CamelCase: MooringBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.34 Emergency Wreck Marking Buoy

Type: FeatureType

Definition: An emergency wreck marking buoy is a buoy moored on or above a new wreck, designed to provide a prominent (both visual and radio) and easily identifiable temporary first response.

CamelCase: EmergencyWreckMarkingBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.35 Lighthouse

Type: FeatureType

Definition: A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.

CamelCase: Lighthouse

Super type: Landmark

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.36 Light Float

Type: FeatureType

Definition: A boat-like structure used instead of a light buoy in waters where strong streams or currents are experienced, or when a greater elevation than that of a light buoy is necessary.

CamelCase: LightFloat

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta	EN	1, *

		13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Horizontal Length			RE	0, 1
Horizontal Width			RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework	EN	0, *

		11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor	EN	0, *

		28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1
Horizontal Accuracy			RE	0, 1

1.37 Light Vessel

Type: FeatureType

Definition: A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation. By night, it displays a characteristic light(s) and is usually equipped with other devices, such as fog signal, submarine sound signal, and radio-beacon, to assist navigation.

CamelCase: LightVessel

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	1, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Horizontal Length			RE	0, 1

Horizontal Width			RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, *
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public	EN	0, *

		15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous	EN	0, 1

		3. Prominent		
Vertical Accuracy			RE	0, 1
Horizontal Accuracy			RE	0, 1

1.38 Offshore Platform

Type: FeatureType

Definition: A permanent offshore structure, either fixed or floating.

CamelCase: OffshorePlatform

Super type: StructureObject

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Offshore Platform		1. Oil Rig 2. Production Platform 3. Observation/Research Platform 4. Articulated Loading Platform 5. Single Anchor Leg Mooring 6. Mooring Tower 7. Artificial Island 8. Floating Production, Storage and Off-Loading Vessel 9. Accommodation Platform 10. Navigation, Communication and Control Buoy 11. Floating Oil Tank	EN	0, *
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey	EN	0, *

		8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Height			RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden	EN	0, *

		7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Product		1. Oil 2. Gas 3. Water 4. Stone 5. Coal 6. Ore 7. Chemicals 8. Drinking Water 9. Milk 10. Bauxite 11. Coke 12. Iron Ingots 13. Salt 14. Sand 15. Timber 16. Sawdust/Wood Chips 17. Scrap Metal 18. Liquefied Natural Gas 19. Liquefied Petroleum Gas 20. Wine 21. Cement 22. Grain 23. Electricity	EN	0, *

		24. Ice		
		25. Clay		
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational	EN	0, *

		30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water	EN	0, 1

17. Mean High Water Springs
18. High Water
19. Approximate Mean Sea Level
20. High Water Springs
21. Mean Higher High Water
22. Equinoctial Spring Low Water
23. Lowest Astronomical Tide
24. Local Datum
25. International Great Lakes Datum 1985
26. Mean Water Level
27. Lower Low Water Large Tide
28. Higher High Water Large Tide
29. Nearly Highest High Water
30. Highest Astronomical Tide
31. Local Low Water Reference Level
32. Local High Water Reference Level
33. Local Mean Water Reference Level
34. Equivalent Height of Water (German GIW)
35. Highest Shipping Height of Water (German HSW)
36. Reference Low Water Level According to Danube Commission
37. Highest Shipping Height of Water According to Danube Commission
38. Dutch River Low Water Reference Level (OLR)
39. Russian Project Water Level
40. Russian Normal Backwater Level
41. Ohio River Datum
43. Dutch High Water Reference Level

		44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

1.39 Silo/Tank

Type: FeatureType

Definition: A large storage structure used for storing loose materials, liquids and/or gases.

CamelCase: SiloTank

Super type: StructureObject

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Building Shape		5. High-Rise Building 6. Pyramid 7. Cylindrical 8. Spherical 9. Cubic	EN	0, 1
Category of Silo/Tank		1. Silo in General 2. Tank in General 3. Grain Elevator 4. Water Tower	EN	0, 1
Colour		1. White 2. Black 3. Red 4. Green	EN	0, *

		5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Elevation			RE	0, 1
Height			RE	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders	EN	0, *

		4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Radar Conspicuous			BO	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away	EN	0, *

		21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water	EN	0, 1

9. Low Water Springs

10. Approximate Lowest
Astronomical Tide

11. Nearly Lowest Low Water

12. Mean Lower Low Water

13. Low Water

14. Approximate Mean Low Water

15. Approximate Mean Lower
Low Water

16. Mean High Water

17. Mean High Water Springs

18. High Water

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes
Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference
Level

32. Local High Water Reference
Level

33. Local Mean Water Reference
Level

34. Equivalent Height of Water
(German GIW)

35. Highest Shipping Height of
Water (German HSW)

36. Reference Low Water Level
According to Danube

		Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

1.40 Pile

Type: FeatureType

Definition: A long heavy timber or section of steel, wood, concrete, etc., forced into the earth or sea floor to serve as a support, as for a pier, or to resist lateral pressure; or as a free standing pole within a marine environment.

CamelCase: Pile

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Pile		1. Stake 3. Post 4. Tripodal 5. Piling 6. Area of Piles	EN	0, 1

		7. Pipe		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *

Height			RE	0, 1
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water 17. Mean High Water Springs 18. High Water 19. Approximate Mean Sea Level 20. High Water Springs 21. Mean Higher High Water 22. Equinoctial Spring Low Water 23. Lowest Astronomical Tide 24. Local Datum 25. International Great Lakes Datum 1985 26. Mean Water Level 27. Lower Low Water Large Tide 28. Higher High Water Large Tide	EN	0, 1

		29. Nearly Highest High Water 30. Highest Astronomical Tide 31. Local Low Water Reference Level 32. Local High Water Reference Level 33. Local Mean Water Reference Level 34. Equivalent Height of Water (German GIW) 35. Highest Shipping Height of Water (German HSW) 36. Reference Low Water Level According to Danube Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW)		
Vertical Length			RE	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

1.41 Building

Type: FeatureType

Definition: A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation.

CamelCase: Building

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.42 Bridge

Type: FeatureType

Definition: A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.

CamelCase: Bridge

Super type: StructureObject

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.43 Sinker Anchor

Type: FeatureType

Definition: A heavy weight (of concrete, cast-iron, etc..) that rests on the sea bed and to which a mooring line can be attached. (IALA Dictionary, 8-5-025)

CamelCase: SinkerAnchor

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted	EN	0, 1

		10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Sinker Dimensions			C	0, 1
Height/Length Units		1. Metres 2. Feet 3. Kilometres 4. Hectometres 5. Statute Miles 6. Nautical Miles	(S) EN	1, 1
Horizontal Length			(S) RE	0, 1
Horizontal Width			(S) RE	0, 1
Vertical Length			(S) RE	0, 1
Weight			RE	1, 1
Sinker Type			TE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	ShackleToAnchorConnection	Sinker Anchor	shackleToAnchorconnectedTo	0..1	MooringShackle	shackleToAnchorconnected	1..1
	--FeatureType-- Sinker Anchor 0..1ShackleToAnchorConnection1..1 shackleToAnchorconnectedTo shackleToAnchorconnected --FeatureType-- MooringShackle						

1.44 Mooring Shackle

Type: FeatureType

Definition: A shackle at the lower end of a mooring chain, for attachment to an anchor or sinker. (IALA Dictionary, 8-5-150)

CamelCase: MooringShackle

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

Association	ShackleToBridleConnection	Mooring Shackle	shackleToBridleconnected	1..1	Bridle	shackleToBridleconnectedTo	0..1
	--FeatureType-- Mooring Shackle 1..1 ShackleToBridleConnection 0..1 shackleToBridleconnected shackleToBridleconnectedTo --FeatureType-- Bridle						
Association	BridleCableConnection	Mooring Shackle	cableholds	1..1	CableSubmarine	bridleattached	0..*
	--FeatureType-- Mooring Shackle 1..1 BridleCableConnection 0..* cableholds bridleattached --FeatureType-- CableSubmarine						
Association	ShackleToSwivelConnection	Mooring Shackle	shackleToSwivelconnected	1..1	Swivel	shackleToSwivelconnectedTo	0..*
	--FeatureType-- Mooring Shackle 1..1 ShackleToSwivelConnection 0..* shackleToSwivelconnected shackleToSwivelconnectedTo --FeatureType-- Swivel						
Association	ShackleToAnchorConnection	Mooring Shackle	shackleToAnchorconnected	1..1	SinkerAnchor	shackleToAnchorconnectedTo	0..1
	--FeatureType-- Mooring Shackle 1..1 ShackleToAnchorConnection 0..1 shackleToAnchorconnected shackleToAnchorconnectedTo --FeatureType-- SinkerAnchor						

1.45 Cable Submarine

Type: FeatureType

Definition: An assembly of wires or fibres, or a wire rope or chain, which has been laid underwater or buried beneath the sea floor.

CamelCase: CableSubmarine

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Cable Dimensions			C	0, 1
Cable Length			(S) RE	1, 1
Height/Length Units		1. Metres 2. Feet 3. Kilometres 4. Hectometres 5. Statute Miles	(S) EN	1, 1

		6. Nautical Miles		
Diameter			(S) RE	1, 1
Category of Cable		1. Power Line 3. Transmission Line 4. Telephone 5. Telegraph 6. Mooring Cable 7. Ferry 8. Fibre Optic Cable	EN	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing	EN	0, *

Association	ShackleCon nectionFrom Cable	Cable Submarine			MooringSha ckle	shackleToC ableconnect ed	1..1
	--FeatureType-- Cable Submarine		ShackleConnectionFromCable1		shackleToCableconnected		

1.46 Swivel

Type: FeatureType

Definition: A chain link that provides for rotary motion between the lengths of chain that it connects. (IALA Dictionary, 8-5-165)

CamelCase: Swivel

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, 1
Weight			RE	0, 1
Swivel Type			TE	0, 1

Feature bindings:

Type	Association Name	Association Ends

		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	SwivelConnection	Swivel	bridlehangs	0..1	Bridle	swivelholds	1..1
	--FeatureType--Swivel 0..1SwivelConnection1..1 bridlehangs swivelholds --FeatureType--Bridle						
Association	SwivelCableConnection	Swivel	cableholds	1..1	CableSubmarine	swivelattached	0..*
	--FeatureType--Swivel 1..1SwivelCableConnection0..* cableholds swivelattached --FeatureType--CableSubmarine						
Association	ShackleToSwivelConnection	Swivel	shackleToSwivelconnectedTo	0..*	MooringShackle	shackleToSwivelconnected	1..1
	--FeatureType--Swivel 0..*ShackleToSwivelConnection1..1 shackleToSwivelconnectedTo shackleToSwivelconnected --FeatureType--MooringShackle						

1.47 Bridle

Type: FeatureType

Definition: Two lengths of chain connected by a central ring and used for lifting wide loads. (IALA Dictionary,8-3-195)

CamelCase: Bridle

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
BridleLink Type			TE	0, 1
Legs Details			TE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BridleConnection	Bridle	buoyhangs	0..1	GenericBuoy	bridleholds	1..1
	--FeatureType--Bridle 0..1BridleConnection1..1 buoyhangs bridleholds --FeatureType--GenericBuoy						
Association	SwivelConnection	Bridle	swivelholds	1..1	Swivel	bridlehangs	0..1

	--FeatureType-- Bridle 1..1 SwivelConnection 0..1 --FeatureType-- Swivel swivelholds bridlehangs						
Association	ShackleToBridleConnection	Bridle	shackleToBridleconnectedTo	0..1	MooringShackle	shackleToBridleconnected	1..1
	--FeatureType-- Bridle 0..1 ShackleToBridleConnection 1..1 --FeatureType-- MooringShackle shackleToBridleconnectedTo shackleToBridleconnected						
Association	BridleCableConnection	Bridle	cableholds	1..1	CableSubmarine	bridleattached	0..*
	--FeatureType-- Bridle 1..1 BridleCableConnection 0..* --FeatureType-- CableSubmarine cableholds bridleattached						

1.48 Counter Weight

Type: FeatureType

Definition: -

CamelCase: CounterWeight

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass	EN	0, 1

		14. Plastic		
Weight			RE	1, 1
Counter Weight Type			TE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BuoyCounterWeight	Counter Weight	buoyattached	0..1	GenericBuoy	counterWeightholds	1..1
	--FeatureType-- Counter Weight 0..1 BuoyCounterWeight 1..1 --FeatureType-- GenericBuoy buoyattached counterWeightholds						

1.49 Topmark

Type: FeatureType

Definition: A characteristic shape secured at the top of a buoy or beacon to aid in its identification.

CamelCase: Topmark

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *

		14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized	EN	0, *

		16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Topmark/Daymark Shape		1. Cone (Point Up) 2. Cone (Point Down) 3. Sphere 4. 2 Spheres	EN	1, 1

- | | | |
|--|--|--|
| | 5. Cylinder | |
| | 6. Board | |
| | 7. X-Shaped | |
| | 8. Upright Cross | |
| | 9. Cube (Point Up) | |
| | 10. 2 Cones (Point to Point) | |
| | 11. 2 Cones (Base to Base) | |
| | 12. Rhombus | |
| | 13. 2 Cones (Points Upward) | |
| | 14. 2 Cones (Points Downward) | |
| | 15. Besom (Point Up) | |
| | 16. Besom (Point Down) | |
| | 17. Flag | |
| | 18. Sphere Over a Rhombus | |
| | 19. Square | |
| | 20. Rectangle (Horizontal) | |
| | 21. Rectangle (Vertical) | |
| | 22. Trapezium (Up) | |
| | 23. Trapezium (Down) | |
| | 24. Triangle (Point Up) | |
| | 25. Triangle (Point Down) | |
| | 26. Circle | |
| | 27. Two Upright Crosses (One Over the Other) | |
| | 28. T-Shape | |
| | 29. Triangle Pointing Up Over a Circle | |
| | 30. Upright Cross Over a Circle | |
| | 31. Rhombus Over a Circle | |
| | 32. Circle Over a Triangle Pointing Up | |
| | 33. Other Shape (See Shape Information) | |
| | 34. Tubular | |

Vertical Length			RE	0, 1
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BuoyTopmark	Topmark	topmarkPart	0..*	GenericBuoy	buoyPart	1..1
	--FeatureType-- Topmark		0..*	<i>BuoyTopmark</i>	1..1	--FeatureType-- GenericBuoy	
			topmarkPart		buoyPart		

1.50 Safe Water Beacon

Type: FeatureType

Definition: A safe water beacon is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.51 Special Purpose/General Beacon

Type: FeatureType

Definition: A special purpose beacon is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark	EN	1, *

- | | | | | |
|--|--|--|--|--|
| | | 7. Spoil Ground Mark | | |
| | | 8. Outfall Mark | | |
| | | 9. ODAS | | |
| | | 10. Recording Mark | | |
| | | 11. Seaplane Anchorage Mark | | |
| | | 12. Recreation Zone Mark | | |
| | | 13. Private Mark | | |
| | | 14. Mooring Mark | | |
| | | 15. LANBY | | |
| | | 16. Leading Mark | | |
| | | 17. Measured Distance Mark | | |
| | | 18. Notice Mark | | |
| | | 19. TSS Mark | | |
| | | 20. Anchoring Prohibited Mark | | |
| | | 21. Berthing Prohibited Mark | | |
| | | 22. Overtaking Prohibited Mark | | |
| | | 23. Two-Way Traffic Prohibited Mark | | |
| | | 24. Reduced Wake Mark | | |
| | | 25. Speed Limit Mark | | |
| | | 26. Stop Mark | | |
| | | 27. General Warning Mark | | |
| | | 28. Sound Ship's Siren Mark | | |
| | | 29. Restricted Vertical Clearance Mark | | |
| | | 30. Maximum Vessel's Draught Mark | | |
| | | 31. Restricted Horizontal Clearance Mark | | |
| | | 32. Strong Current Warning Mark | | |
| | | 33. Berthing Permitted Mark | | |
| | | 34. Overhead Power Cable Mark | | |
| | | 35. Channel Edge Gradient Mark | | |
| | | 36. Telephone Mark | | |

		37. Ferry Crossing Mark		
		39. Pipeline Mark		
		40. Anchorage Mark		
		41. Clearing Mark		
		42. Control Mark		
		43. Diving Mark		
		44. Refuge Beacon		
		45. Foul Ground Mark		
		46. Yachting Mark		
		47. Heliport Mark		
		48. GNSS Mark		
		49. Seaplane Landing Mark		
		50. Entry Prohibited Mark		
		51. Work in Progress Mark		
		52. Mark With Unknown Purpose		
		53. Wellhead Mark		
		54. Channel Separation Mark		
		55. Marine Farm Mark		
		56. Artificial Reef Mark		
		57. Ice Mark		
		58. Nature Reserve Mark		
		59. Fish Aggregating Device		
		60. Wreck Mark		
		61. Customs Mark		
		62. Causeway Mark		
		63. Wave Recorder		
		64. Jetski Prohibited		

1.52 Safe Water Buoy

Type: FeatureType

Definition: A safe water buoy is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.53 Special Purpose/General Buoy

Type: FeatureType

Definition: A special purpose buoy is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark 7. Spoil Ground Mark 8. Outfall Mark 9. ODAS 10. Recording Mark 11. Seaplane Anchorage Mark 12. Recreation Zone Mark 13. Private Mark 14. Mooring Mark 15. LANBY 16. Leading Mark 17. Measured Distance Mark 18. Notice Mark 19. TSS Mark	EN	1, *

20. Anchoring Prohibited Mark
21. Berthing Prohibited Mark
22. Overtaking Prohibited Mark
23. Two-Way Traffic Prohibited Mark
24. Reduced Wake Mark
25. Speed Limit Mark
26. Stop Mark
27. General Warning Mark
28. Sound Ship's Siren Mark
29. Restricted Vertical Clearance Mark
30. Maximum Vessel's Draught Mark
31. Restricted Horizontal Clearance Mark
32. Strong Current Warning Mark
33. Berthing Permitted Mark
34. Overhead Power Cable Mark
35. Channel Edge Gradient Mark
36. Telephone Mark
37. Ferry Crossing Mark
39. Pipeline Mark
40. Anchorage Mark
41. Clearing Mark
42. Control Mark
43. Diving Mark
44. Refuge Beacon
45. Foul Ground Mark
46. Yachting Mark
47. Heliport Mark
48. GNSS Mark
49. Seaplane Landing Mark
50. Entry Prohibited Mark

		51. Work in Progress Mark		
		52. Mark With Unknown Purpose		
		53. Wellhead Mark		
		54. Channel Separation Mark		
		55. Marine Farm Mark		
		56. Artificial Reef Mark		
		57. Ice Mark		
		58. Nature Reserve Mark		
		59. Fish Aggregating Device		
		60. Wreck Mark		
		61. Customs Mark		
		62. Causeway Mark		
		63. Wave Recorder		
		64. Jetski Prohibited		

1.54 Dangerous Feature

Type: FeatureType

Definition: -

CamelCase: DangerousFeature

Super type: AbstractFeatureType

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1

Feature bindings:

Type	Association Name	Association Ends

		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	DangerousFeatureAssociation	DangerousFeature	danger	0..*	AtonAssociation	markingAton	1..*
	--FeatureType-- Dangerous Feature 0..* DangerousFeatureAssociation danger markingAton --FeatureType-- AtonAssociation						

1.55 Aton Aggregation

Type: FeatureType

Definition: Used to identify an aggregation of two or more objects. This aggregation may be named content of categoryOfAggregation should be put in information attribute when converting to S-57.

CamelCase: AtonAggregation

Super type: AbstractFeatureType

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Aggregation		1. Leading Line 3. Measured Distance 2. Range System	CO	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	AtonAggregations	Aton Aggregation	peerAtonAggregation	0..*	AidsToNavigation	atonAggregationBy	0..*
	--FeatureType-- Aton Aggregation 0..* AtonAggregations peerAtonAggregation atonAggregationBy 0..* --FeatureType-- AidsToNavigation						

1.56 Aton Association

Type: FeatureType

Definition: Used to identify an association between two or more objects. The association may be named content of categoryOfAssociation should be put in information attribute when converting to S-57

CamelCase: AtonAssociation

Super type: AbstractFeatureType

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Association		1. Channel Markings 2. Danger Markings	CO	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	DangerousFeatureAssociation	Aton Association	markingAton	1..*	DangerousFeature	danger	0..*
	--FeatureType-- Aton Association 1..* DangerousFeatureAssociation markingAton danger --FeatureType-- DangerousFeature						
Association	AtonAssociations	Aton Association	peerAtonAssociation	0..*	AidsToNavigation	atonAssociationBy	0..*
	--FeatureType-- Aton Association 0..* AtonAssociations peerAtonAssociation atonAssociationBy 0..* AidsToNavigation						

1.57 Quality of Non-Bathymetric Data

Type: FeatureType

Definition: An area within which a uniform assessment of the quality of the non-bathymetric data exists.

CamelCase: QualityOfNonBathymetricData

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Temporal Variation		1. Extreme Event 2. Likely to Change and Significant Shoaling Expected 3. Likely to Change But Significant Shoaling Not Expected 4. Likely to Change 5. Unlikely to Change 6. Unassessed	EN	1, 1
Orientation Uncertainty			RE	0, 1

Horizontal Distance Uncertainty			RE	0, 1
Horizontal Position Uncertainty			C	1, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Information			C	0, 1
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Information in National Language			TE	0, 1
Textual Description			C	0, 1
File Reference			(S) TE	1, 1
Language			(S) TE	0, 1
Vertical Uncertainty			C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1

1.58 Data Coverage

Type: FeatureType

Definition: A geographical area that describes the coverage and extent of spatial objects.

CamelCase: DataCoverage

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Maximum Display Scale			IN	1, 1
Minimum Display Scale			IN	1, 1

1.59 Local Direction of Buoyage

Type: FeatureType

Definition: An area within which the navigational system of marks has been established in relation to a specific direction.

CamelCase: LocalDirectionOfBuoyage

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

1.60 Navigational System of Marks

Type: FeatureType

Definition: An area within which the navigational system of marks has been established in relation to a specific direction.

CamelCase: NavigationalSystemOfMarks

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. CEVNI 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulations - Two Sides 14. Brazilian National Inland Waterway Regulations - Side Independent 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	1, 1

1.61 Sounding Datum

Type: FeatureType

Definition: The horizontal plane or tidal datum to which soundings have been reduced. Also called datum for sounding reduction.

CamelCase: SoundingDatum

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water 17. Mean High Water Springs 18. High Water 19. Approximate Mean Sea Level 20. High Water Springs 21. Mean Higher High Water 22. Equinoctial Spring Low Water 23. Lowest Astronomical Tide 24. Local Datum 25. International Great Lakes Datum 1985	EN	1, 1

		26. Mean Water Level		
		27. Lower Low Water Large Tide		
		28. Higher High Water Large Tide		
		29. Nearly Highest High Water		
		30. Highest Astronomical Tide		
		31. Local Low Water Reference Level		
		32. Local High Water Reference Level		
		33. Local Mean Water Reference Level		
		34. Equivalent Height of Water (German GIW)		
		35. Highest Shipping Height of Water (German HSW)		
		36. Reference Low Water Level According to Danube Commission		
		37. Highest Shipping Height of Water According to Danube Commission		
		38. Dutch River Low Water Reference Level (OLR)		
		39. Russian Project Water Level		
		40. Russian Normal Backwater Level		
		41. Ohio River Datum		
		43. Dutch High Water Reference Level		
		44. Baltic Sea Chart Datum 2000		
		45. Dutch Estuary Low Water Reference Level (OLW)		

1.62 Vertical Datum of Data

Type: FeatureType

Definition: Any level surface (for example Mean Sea Level) taken as a surface of reference to which the elevations within a data set are reduced. Also called datum level, reference level, reference plane, levelling datum, datum for heights.

CamelCase: VerticalDatumOfData

Super type: AbstractFeatureType

Primitive: surface,

Feature use type: meta

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water 6. Lowest Low Water Springs 7. Approximate Mean Low Water Springs 8. Indian Spring Low Water 9. Low Water Springs 10. Approximate Lowest Astronomical Tide 11. Nearly Lowest Low Water 12. Mean Lower Low Water 13. Low Water 14. Approximate Mean Low Water 15. Approximate Mean Lower Low Water 16. Mean High Water 17. Mean High Water Springs 18. High Water 19. Approximate Mean Sea Level 20. High Water Springs 21. Mean Higher High Water 22. Equinoctial Spring Low Water 23. Lowest Astronomical Tide 24. Local Datum 25. International Great Lakes Datum 1985 26. Mean Water Level 27. Lower Low Water Large Tide 28. Higher High Water Large Tide	EN	1, 1

		29. Nearly Highest High Water		
		30. Highest Astronomical Tide		
		31. Local Low Water Reference Level		
		32. Local High Water Reference Level		
		33. Local Mean Water Reference Level		
		34. Equivalent Height of Water (German GIW)		
		35. Highest Shipping Height of Water (German HSW)		
		36. Reference Low Water Level According to Danube Commission		
		37. Highest Shipping Height of Water According to Danube Commission		
		38. Dutch River Low Water Reference Level (OLR)		
		39. Russian Project Water Level		
		40. Russian Normal Backwater Level		
		41. Ohio River Datum		
		43. Dutch High Water Reference Level		
		44. Baltic Sea Chart Datum 2000		
		45. Dutch Estuary Low Water Reference Level (OLW)		

4 Information Types

4.1 AtoN Fixing Method

Type: InformationType

Definition: Method used for fixing the position of an aid to navigation.

CamelCase: AtoNFixingMethod

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Reference Point			TE	0, 1
Horizontal Datum		1. WGS 72	EN	0, 1

2. WGS 84
3. European 1950
4. Potsdam Datum
5. Adindan
6. Afgooye
7. Ain el Abd 1970
8. Anna 1 Astro 1965
9. Antigua Island Astro 1943
10. Arc 1950
11. Arc 1960
12. Ascension Island 1958
13. Astro Beacon 'E' 1945
14. Astro DOS 71/4
15. Astro Tern Island (FRIG) 1961
16. Astronomical Station 1952
17. Australian Geodetic 1966
18. Australian Geodetic 1984
19. Ayabelle Lighthouse
20. Bellevue (IGN)
21. Bermuda 1957
22. Bissau
23. Bogota Observatory
24. Bukit Rimpah
25. Camp Area Astro
26. Campo Inchauspe 1969
27. Canton Astro 1966
28. Cape Datum
29. Cape Canaveral
30. Carthage
31. Chatam Island Astro 1971
32. Chua Astro

33. Corrego Alegre	
34. Dabola	
35. Djakarta (Batavia)	
36. DOS 1968	
37. Easter Island 1967	
38. European 1979	
39. Fort Thomas 1955	
40. Gan 1970	
41. Geodetic Datum 1949	
42. Graciosa Base SW 1948	
43. Guam 1963	
44. Gunung Segara	
45. GUX 1 Astro	
46. Herat North	
47. Hjorsey 1955	
48. Hong Kong 1963	
49. Hu-Tzu-Shan	
50. Indian	
51. Indian 1954	
52. Indian 1975	
53. Ireland 1965	
54. ISTS 061 Astro 1968	
55. ISTS 073 Astro 1969	
56. Johnston Island 1961	
57. Kandawala	
58. Kerguelen Island 1949	
59. Kertau 1968	
60. Kusaie Astro 1951	
61. L. C. 5 Astro 1961	
62. Leigon	
63. Liberia 1964	

64. Luzon
65. Mahe 1971
66. Massawa
67. Merchich
68. Midway Astro 1961
69. Minna
70. Montserrat Island Astro 1958
71. M'poraloko
72. Nahrwan
73. Naparima, BWI
74. North American 1927
75. North American 1983
76. Observatorio Meteorologico 1939
77. Old Egyptian 1907
78. Old Hawaiian
79. Oman
80. Ordnance Survey of Great Britain 1936
81. Pico de las Nieves
82. Pitcairn Astro 1967
83. Point 58
84. Pointe Noire 1948
85. Porto Santo 1936
86. Provisional South American 1956
87. Provisional South Chilean 1963
88. Puerto Rico
89. Qatar National
90. Qornoq
91. Reunion
92. Rome 1940
93. Santo (DOS) 1965

- | |
|--|
| 94. Sao Braz |
| 95. Sapper Hill 1943 |
| 96. Schwarzeck |
| 97. Selvagem Grande 1938 |
| 98. South American 1969 |
| 99. South Asia |
| 100. Tananarive Observatory
1925 |
| 101. Timbalai 1948 |
| 102. Tokyo |
| 103. Tristan Astro 1968 |
| 104. Viti Levu 1916 |
| 105. Wake-Eniwetok 1960 |
| 106. Wake Island Astro 1952 |
| 107. Yacare |
| 108. Zanderij |
| 109. American Samoa 1962 |
| 110. Deception Island |
| 111. Indian 1960 |
| 112. Indonesian 1974 |
| 113. North Sahara 1959 |
| 114. Pulkovo 1942 |
| 116. S-JTSK |
| 117. Voirol 1950 |
| 118. Average Terrestrial System
1977 |
| 119. Compensation Geodesique
du Quebec 1977 |
| 120. Finnish (KKJ) |
| 121. Ordnance Survey of Ireland |
| 122. Revised Kertau |
| 123. Revised Nahrwan |
| 124. GGRS 76 (Greece) |

		125. Nouvelle Triangulation de France 126. RT 90 (Sweden) 127. Geocentric Datum of Australia 128. BJZ54 (A954 Beijing Coordinates) 129. Modified BJZ54 130. GDZ80 131. Local Datum		
Source Date			DA	1, 1
Positioning Procedure			TE	1, 1

4.2 Aton Status Information

Type: InformationType

Definition: -

CamelCase: AtonStatusInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Change Details			C	1, 1
Aton Commissioning		1. Buoy Establishment 2. Light Establishment 3. Beacon Establishment 4. Audible Signal Establishment 5. Fog Signal Establishment 6. AIS Transmitter Establishment 7. V-AIS Establishment 8. RACON Establishment 9. DGPS Station Establishment 10. ELORAN Station Establishment 11. DGLONASS Station Establishment 12. E-Chayka Station Establishment	(S) EN	0, 1

		13. EGNOS establishment		
Aton Removal		1. Buoy Removal 2. Buoy Temporary Removal 3. Light Removal 4. Light Temporary Removal 5. Beacon Removal 6. Beacon Temporary Removal 7. Fog Signal Removal 8. Fog Signal Temporary Removal 9. Audible Signal Removal 10. Audible Signal Temporary Removal 11. V-AIS Removal 12. V-AIS Temporary Removal 13. RACON signal removal 14. RACON Temporary Removal 15. DGPS removal 16. DGPS temporary removal 17. EGNOS removal 18. EGNOS temporary removal 19. LORAN C Station Removal 20. LORAN C Station Temporary Removal 21. eLORAN removal 22. eLORAN temporary removal 23. Chayka Station Removal 24. Chayka Station Temporary Removal 25. E-Chayka Station Removal 26. E-Chayka Station Temporary Removal	(S) EN	0, 1
Aton Replacement		1. Buoy Change	(S) EN	0, 1

		2. Buoy Temporary Change 3. Light Change 4. Light Temporary Change 5. Sector Light Change 6. Sector Light Temporary Change 7. Beacon Change 8. Beacon Temporary Change 9. Fog Signal Change 10. Fog Signal Temporary Change 11. Audible Signal Change 12. Audible Signal Temporary Change 13. V-AIS Change 14. V-AIS Temporary Change 15. RACON signal change 16. RACON Temporary Change		
Fixed Aton Change		1. Beacon Missing 2. Beacon Damaged 3. Light beacon Unlit 4. Light beacon Unreliable 5. Light beacon Not synchronized 6. Light beacon damaged 7. Beacon Topmark Missing 8. Beacon Topmark Damaged 9. Beacon Daymark Unreliable 10. Floodlit beacon Unlit 11. Beacon Restored To Normal	(S) EN	0, 1
Floating Aton Change		1. Buoy Adrift 2. Buoy Damaged 3. Buoy Daymark Unreliable	(S) EN	0, 1

		4. Buoy Destroyed 5. Buoy Missing 6. Buoy Move 7. Buoy off Position 8. Buoy re-establishment 9. Buoy Restored to Normal 10. Buoy Topmark Damaged 11. Buoy Topmark Missing 12. Buoy Will Be Withdrawn 13. Buoy Withdrawn 14. Decommissioned for winter 15. Lifted for Winter 16. Light buoy Light damaged 17. Light buoy Light not synchronized 18. Light buoy Light unlit 19. Light buoy Light unreliable 20. Marine Aids to Navigation Unreliable 21. Recommissioned for navigation season 22. Replaced by Winter Spar 23. Seasonal Decommissioning Complete 24. Seasonal Decommissioning in Progress 25. Seasonal recommissioning complete 26. Seasonal recommissioning in progress		
Audible Signal Aton Change		1. Audible Signal Out Of Service 2. Fog Signal Out Of Service 3. Audible Signal Operating Properly 4. Fog Signal Operating Properly	(S) EN	0, 1

Lighted Aton Change		1. Light Unlit 2. Light Unreliable 3. Light Re-Establishment 4. Light Range Reduced 5. Light Without Rhythm 6. Light Out Of Synchronization 7. Light Daymark Unreliable 8. Light Operating Properly 9. Sector light Sector obscured 10. Front leading/range light Unlit 11. Rear leading/range light Unlit 12. Front leading/range light Unreliable 13. Rear leading/range light Unreliable 14. Front leading/range light Light range reduced 15. Rear leading/range light Light range reduced 16. Front leading/range light without rhythm 17. Rear leading/range light without rhythm 18. Leading/range lights out of synchronization 19. Front leading/range beacon Unreliable 20. Rear leading/range beacon Unreliable 21. Front leading/range light is operating properly 22. Rear leading/range light is operating properly 23. Front leading/range beacon restored to normal 24. Rear leading/range beacon restored to normal	(S) EN	0, 1
Electronic Aton Change		1. AIS Transmitter Out Of Service	(S) EN	0, 1

		2. AIS Transmitter Unreliable 3. AIS Transmitter Operating Properly 4. V-AIS Out Of Service 5. V-AIS Unreliable 6. V-AIS Operating Properly 7. RACON Out Of Service 8. RACON Unreliable 9. RACON Operating Properly 10. DGPS Out Of Service 11. DGPS Operating Properly 12. DGPS Unreliable 13. LORAN C operating properly 14. LORAN C unreliable 15. LORAN C out of service 16. ELORAN Operating Properly 17. ELORAN Unreliable 18. ELORAN Out Of Service 19. DGLOANSS operating properly 20. DGLOANSS unreliable 21. DGLOANSS out of service 22. Chayka Operating Properly 23. Chayka Unreliable 24. Chayka Out Of Service 25. E-Chayka Operating Properly 26. E-Chayka Unreliable 27. E-Chayka Out Of Service 28. EGNOS Operating Properly 29. EGNOS Unreliable 30. EGNOS Out Of Service		
Change Types		1. Advanced notice of changes	EN	0, 1

		2. Discrepancy		
		3. Proposed changes		
		4. Temporary changes		

4.3 Positioning Information

Type: InformationType

Definition: Information about how a position was obtained. (proposed by CCG)

CamelCase: PositioningInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Positioning Device			TE	1, 1
Positioning Method			C	0, 1
Positioning Equipment		1. DGPS Receiver 2. GLONASS Receiver 3. GPS Receiver 4. GPS/WAAS Receiver	(S) EN	1, 1
NMEAString			(S) TE	1, 1

4.4 Spatial Quality

Type: InformationType

Definition: The indication of the quality of the locational information for features in a dataset.

CamelCase: SpatialQuality

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Quality of Horizontal Measurement		1. Surveyed 2. Unsurveyed 3. Inadequately Surveyed 4. Approximate 5. Position Doubtful 6. Unreliable 7. Reported (Not Surveyed) 8. Reported (Not Confirmed)	EN	0, 1

		9. Estimated 10. Precisely Known 11. Calculated		
Spatial Accuracy			C	0, 1
Fixed Date Range			(S) C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Horizontal Position Uncertainty			(S) C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Vertical Uncertainty			(S) C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1

5 Simple attributes

5.1 AtoN Number

Type: SimpleAttributeType

Definition: Identifier from a list of Aids to Navigation publication, such as List of Lights.

CamelCase: AtoNNumber

Value type: URI

5.2 Category of Aggregation

Type: CodeListType

Definition: Named aggregations between two or more aids to navigation and/or navigationally relevant features.

CamelCase: CategoryOfAggregation

Value type: codelist

Listed values:

Code	Label	Definition
1	Leading Line	A line passing through two or more clearly defined charted objects, and along which a vessel can approach safely.
3	Measured Distance	A course at sea, whose ends are indicated by ranges ashore, and whose length has been accurately measured for determining the speed of vessels.
2	Range System	Navigation system limited in their positioning capability to coastal regions,

or those systems limited to making landfall

5.3 Category of Association

Type: CodeListType

Definition: Named associations between two or more aids to navigation and/or navigationally relevant features.

CamelCase: CategoryOfAssociation

Value type: codelist

Listed values:

Code	Label	Definition
1	Channel Markings	A group of channel marks which indicate channel limits.
2	Danger Markings	One of more aids to navigation and the danger(s) that are marked.

5.4 Category Of Physical AIS Aid To Navigation

Type: EnumerationType

Definition: A classification of AIS AtoNs that correspond to an actual, physical Aid to Navigation at a real-world location.

CamelCase: CategoryOfPhysicalAIS AidToNavigation

Value type: enumeration

Listed values:

Code	Label	Definition
1	Physical AIS Type 1	Simple transmission of static, pre-programmed information.
2	Physical AIS Type 2	Transmission of dynamic, real-time updated information via connected sensors.
3	Physical AIS Type 3	Full two-way communication: transmission + remote control / configuration.

5.5 Category Of Power Source

Type: EnumerationType

Definition: The classification of energy sources used to power an Aid to Navigation (AtoN).

CamelCase: CategoryOfPowerSource

Value type: enumeration

Listed values:

Code	Label	Definition
1	Battery	A fortified structure on which artillery is mounted.
2	generator	-
3	solar panel	-

4	electrical service	-
---	--------------------	---

5.6 Category Of Synthetic AIS Aid To Navigation

Type: EnumerationType

Definition: A classification of AIS AtoNs that are transmitted electronically and linked to a real-world object but do not physically exist at the broadcast location.

CamelCase: CategoryOfSyntheticAISaidtoNavigation

Value type: enumeration

Listed values:

Code	Label	Definition
1	predicted	-
2	monitored	-

5.7 Change Types

Type: EnumerationType

Definition: Different categories or kinds of modifications that can be made to data, positions, or objects. For example, changes may involve updates to position, orientation, or attributes of an object.

CamelCase: ChangeTypes

Value type: enumeration

Listed values:

Code	Label	Definition
1	Advanced notice of changes	-
2	Discrepancy	(1) A difference between results of duplicate or comparable measures of a quantity. (2) The difference in computed values of a quantity obtained by different processes using data from the same survey.
3	Proposed changes	-
4	Temporary changes	-

5.8 NMEAString

Type: SimpleAttributeType

Definition: NMEA string captured from the positioning device.(proposed by CCG)

CamelCase: NMEAString

Value type: text

5.9 Seasonal Action Required

Type: SimpleAttributeType

Definition: Indicates whether specific actions or maintenance must be performed based on the season (for example, repositioning, removal, or other adjustments).

CamelCase: SeasonalActionRequired

Value type: text

5.10 Shackle Type

Type: EnumerationType

Definition: Types of shackle.

CamelCase: ShackleType

Value type: enumeration

Listed values:

Code	Label	Definition
1	forelock shackles	-
2	clenching shackles	-
3	bolt shackles	-
4	screw pin shackles	-
5	kenter shackle	-
6	quick release link	-

5.11 AtoN Maintenance Record

Type: SimpleAttributeType

Definition: A reference following the Uniform Resource Identifier (URI) principles to a record of maintenance.

CamelCase: aToNMaintenanceRecord

Value type: text

5.12 Administrative Division

Type: SimpleAttributeType

Definition: A generic term for an administrative region within a country at a level below that of the sovereign state.

CamelCase: administrativeDivision

Value type: text

5.13 Aid Availability Category

Type: EnumerationType

Definition: A Category denoting the significance of an Aid to Navigation, expressed in terms of the probability that an AtoN or system of AtoN, as defined by the Competent Authority, is performing its specified function at any randomly chosen time. This is expressed as a percentage of total time that an AtoN or system of AtoN should be performing their specified function.

CamelCase: aidAvailabilityCategory

Value type: enumeration

Listed values:

Code	Label	Definition
1	Category 1	An AtoN or system of AtoN that is considered by the Competent Authority to be of vital navigational significance.
2	Category 2	An AtoN or system of AtoN that is considered by the Competent Authority to be of important navigational significance.

3	Category 3	An AtoN or system of AtoN that is considered by the Competent Authority to be of necessary navigational significance.
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5.14 Aton Commissioning

Type: EnumerationType

Definition: The process of deploying and activating a new Aid to Navigation (AtoN), ensuring that it is properly installed and operational.

CamelCase: atonCommissioning

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Establishment	A new buoy has been or will be established.
2	Light Establishment	A new light has been or will be established.
3	Beacon Establishment	A new beacon has been or will be established.
4	Audible Signal Establishment	A new audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be established.
5	Fog Signal Establishment	A new fog signal has been or will be established.
6	AIS Transmitter Establishment	A new AIS site has been or will be established.
7	V-AIS Establishment	A new V-AIS has been or will be established.
8	RACON Establishment	A new RACON has been or will be established.
9	DGPS Station Establishment	A new DGPS station has been or will be established.
10	ELORAN Station Establishment	A new eLORAN station has been or will be established.
11	DGLONASS Station Establishment	A new DGLONASS station has been or will be established.
12	E-Chayka Station Establishment	A new e-Chayka station has been or will be established.
13	EGNOS establishment	.

5.15 Aton Removal

Type: EnumerationType

Definition: The process of decommissioning and physically removing an AtoN from its designated location, either temporarily or permanently.

CamelCase: atonRemoval

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Removal	Buoy has been or will be permanently removed from service.
2	Buoy Temporary Removal	Buoy has been or will be temporarily removed from service.
3	Light Removal	Light has been or will be permanently removed from service.
4	Light Temporary Removal	Light has been or will be temporarily removed from service.
5	Beacon Removal	Beacon has been or will be permanently removed from service.
6	Beacon Temporary Removal	Beacon has been or will be temporarily removed from service.
7	Fog Signal Removal	Fog signal has been or will be permanently removed from service.
8	Fog Signal Temporary Removal	Fog signal has been or will be temporarily removed from service.
9	Audible Signal Removal	Audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be permanently removed from service.
10	Audible Signal Temporary Removal	Audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be temporarily removed from service.
11	V-AIS Removal	V-AIS has been or will be permanently removed from service.
12	V-AIS Temporary Removal	V-AIS has been or will be temporarily removed from service.
13	RACON signal removal	.
14	RACON Temporary Removal	RACON has been or will be temporarily removed from service.
15	DGPS removal	.
16	DGPS temporary removal	.
17	EGNOS removal	.
18	EGNOS temporary removal	.
19	LORAN C Station Removal	LORAN C station has been or will be permanently removed from service.
20	LORAN C Station Temporary Removal	LORAN C station has been or will be temporarily removed from service.

21	eLORAN removal	.
22	eLORAN temporary removal	.
23	Chayka Station Removal	Chayka station has been or will be permanently removed from service.
24	Chayka Station Temporary Removal	Chayka station has been or will be temporarily removed from service.
25	E-Chayka Station Removal	The e-Chayka station has been or will be permanently removed from service.
26	E-Chayka Station Temporary Removal	The e-Chayka station has been or will be temporarily removed from service

5.16 Aton Replacement

Type: EnumerationType

Definition: The act of swapping an existing AtoN with a new or upgraded unit, either due to maintenance needs or technological improvements.

CamelCase: atonReplacement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Change	The characteristics of the buoy have been or will be changed.
2	Buoy Temporary Change	The characteristics of the buoy have been or will be temporarily changed.
3	Light Change	The characteristics of the light have been or will be changed.
4	Light Temporary Change	The characteristics of the light have been or will be temporarily changed.
5	Sector Light Change	The characteristics of the sector light have been or will be changed.
6	Sector Light Temporary Change	The characteristics of the sector light have been or will be temporarily changed.
7	Beacon Change	The characteristics of the beacon have been or will be changed.
8	Beacon Temporary Change	The characteristics of the beacon have been or will be temporarily changed.
9	Fog Signal Change	The characteristics of the fog signal have been or will be changed.
10	Fog Signal Temporary Change	The characteristics of the fog signal have been or will be temporarily changed.
11	Audible Signal Change	The characteristics of the audible signal (device activated by e.g. sea state or

		wind, irrespective of visibility) have been or will be changed.
12	Audible Signal Temporary Change	The characteristics of the audible signal (device activated by e.g. sea state or wind, irrespective of visibility) have been or will be temporarily changed.
13	V-AIS Change	The characteristics of the V-AIS have been or will be changed.
14	V-AIS Temporary Change	The characteristics of the V-AIS have been or will be temporarily changed.
15	RACON signal change	.
16	RACON Temporary Change	The characteristics of the RACON have been or will be temporarily changed.

5.17 Audible Signal Aton Change

Type: EnumerationType

Definition: Any modification to an AtoN that uses sound signals, such as foghorns or bells, to assist in navigation under low visibility conditions.

CamelCase: audibleSignalAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Audible Signal Out Of Service	The audible signal (device activated by e.g. sea state or wind, irrespective of visibility) is inoperative.
2	Fog Signal Out Of Service	The fog signal is inoperative.
3	Audible Signal Operating Properly	The audible signal (device activated by e.g. sea state or wind, irrespective of visibility) is operating as advertised.
4	Fog Signal Operating Properly	The fog signal is operating as advertised.

5.18 Based On Fixed Marks

Type: SimpleAttributeType

Definition: A straight route (known as a recommended track, range or leading line), which comprises: a. at least two structures (usually beacons or daymarks) and/or natural features, which may carry lights and/or top-marks. The structures/features are positioned so that when observed to be in line, a vessel can follow a known bearing with safety. (Adapted from International Association of Lighthouse Authorities - IALA Aids to Navigation Guide, 1990); or b. a single structure or natural feature, which may carry lights and/or a topmark, and a specified bearing which can be followed with safety. (S-57 Edition 3.1, Appendix A Chapter 2, Page 2.72, November 2000, as amended).

CamelCase: basedOnFixedMarks

Value type: boolean

5.19 Beacon Shape

Type: EnumerationType

Definition: Describes the characteristic geometric form of the beacon.

CamelCase: beaconShape

Value type: enumeration

Listed values:

Code	Label	Definition
1	Stake, Pole, Perch, Post	An elongated wood or metal pole, driven into the ground or seabed, which serves as a navigational aid or a support for a navigational aid.
2	Withy	A tree without roots stuck or spoiled into the bottom of the sea to serve as a navigational aid.
3	Beacon Tower	A solid structure of the order of 10 metres in height used as a navigational aid.
4	Lattice Beacon	A structure consisting of strips of metal or wood crossed or interlaced to form a structure to serve as an aid to navigation or as a support for an aid to navigation.
5	Pile Beacon	A long heavy timber(s) or section(s) of steel, wood, concrete, etc., forced into the seabed to serve as an aid to navigation or as a support for an aid to navigation.
6	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
7	Buoyant Beacon	A tall spar-like beacon fitted with a permanently submerged buoyancy chamber, the lower end of the body is secured to seabed sinker either by a flexible joint or by a cable under tension.

5.20 BridleLink Type

Type: SimpleAttributeType

Definition: The type of bridle link, which could refer to a structural or mechanical connection (like in a cable, chain, or rigging system).

CamelCase: bridleLinkType

Value type: text

5.21 Building Shape

Type: EnumerationType

Definition: The specific shape of the building.

CamelCase: buildingShape

Value type: enumeration

Listed values:

Code	Label	Definition
5	High-Rise Building	A building having many storeys.
6	Pyramid	A polyhedron of which one face is a polygon of any number of sides, and the

		other faces are triangles with a common vertex.
7	Cylindrical	Shaped like a cylinder, which is a solid geometrical figure generated by straight lines fixed in direction and describing with one of its points a closed curve, especially a circle.
8	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
9	Cubic	A shape the sides of which are six equal squares; a regular hexahedron.

5.22 Buoy Shape

Type: EnumerationType

Definition: The principal shape and/or design of a buoy.

CamelCase: buoyShape

Value type: enumeration

Listed values:

Code	Label	Definition
1	Conical	The upper part of the body above the water-line, or the greater part of the superstructure, has approximately the shape or the appearance of a pointed cone with the point upwards.
2	Can	The upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a cylinder, or a truncated cone that approximates to a cylinder, with a flat end uppermost.
3	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
4	Pillar	The upper part of the body above the water-line, or the greater part of the superstructure is a narrow vertical structure, pillar or lattice tower.
5	Spar	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a pole, or of a very long cylinder, floating upright.
6	Barrel	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a barrel or cylinder floating horizontally.
7	Superbuoy	A very large buoy designed to carry a signal light of high luminous intensity at a high elevation.
8	Ice Buoy	A specially constructed shuttle shaped buoy which is used in ice conditions.

5.23 Cable Length

Type: SimpleAttributeType

Definition: A unit of distance originally equal to the length of a ship's anchor cable, but now generally considered to be about 600 feet. In the British Navy it is 608 feet, or exactly one-tenth of a nautical mile. In the United States Navy it is 720 feet but is infrequently used.

CamelCase: cableLength

Value type: real

5.24 Candela

Type: SimpleAttributeType

Definition: The unit of luminous intensity in the SI system.

CamelCase: candela

Value type: real

5.25 Category of Cable

Type: EnumerationType

Definition: Classification of the cable based on the services provided.

CamelCase: categoryOfCable

Value type: enumeration

Listed values:

Code	Label	Definition
1	Power Line	A cable that transmits or distributes electrical power.
3	Transmission Line	Multiple un-insulated cables usually supported by steel lattice towers. Such features are generally more prominent than normal power lines.
4	Telephone	A cable that transmits telephone signals.
5	Telegraph	An apparatus, system or process for communication at a distance by electric transmission over wire.
6	Mooring Cable	A chain or very strong fibre or wire rope used to anchor or moor vessels or buoys.
7	Ferry	A vessel for transporting passengers, vehicles, and/or goods across a stretch of water, especially as a regular service.
8	Fibre Optic Cable	A cable made of glass or plastic fiber designed to guide light along its length, fibre optic cables are widely used in fiber-optic communication, which permits transmission over longer distances and at higher data rates than other forms of communication.

5.26 Category of Cardinal Mark

Type: EnumerationType

Definition: The four quadrants (north, east, south and west) are bounded by the true bearings NW-NE, NE-SE, SE-SW and SW-NW taken from the point of interest. A cardinal mark is named after the quadrant in which it is placed. The name of the cardinal mark

indicates that it should be passed to the named side of the mark.

CamelCase: categoryOfCardinalMark

Value type: enumeration

Listed values:

Code	Label	Definition
1	North Cardinal Mark	Quadrant bounded by the true bearing NW-NE taken from the point of interest; it should be passed to the north side of the mark.
2	East Cardinal Mark	Quadrant bounded by the true bearing NE-SE taken from the point of interest. It should be passed to the east side of the mark.
3	South Cardinal Mark	Quadrant bounded by the true bearing SE-SW taken from the point of interest; it should be passed to the south side of the mark.
4	West Cardinal Mark	Quadrant bounded by the true bearing SW-NW taken from the point of interest; it should be passed to the west side of the mark.

5.27 Category of Fog Signal

Type: EnumerationType

Definition: Classification of the various means of generating the fog signal.

CamelCase: categoryOfFogSignal

Value type: enumeration

Listed values:

Code	Label	Definition
1	Explosive	A signal produced by the firing of explosive charges.
2	Diaphone	A diaphone uses compressed air and generally emits a powerful low-pitched sound, which often concludes with a brief sound of suddenly lowered pitch, termed the 'grunt'.
3	Siren	A type of fog signal apparatus which produces sound by virtue of the passage of air through slots or holes in a revolving disk.
4	Nautophone	A horn having a diaphragm oscillated by electricity.
5	Reed	[1] A reed uses compressed air and emits a weak, high pitched sound. [2] Any of various water or marsh plants with a firm stem. (Concise Oxford English Dictionary)
6	Tyfon	A diaphragm horn which operates under the influence of compressed air or steam.

7	Bell	A ringing sound with a short range.
8	Whistle	A distinctive sound made by a jet of air passing through an orifice. The apparatus may be operated automatically, by hand or by air being forced up a tube by waves acting on a buoy.
9	Gong	A sound produced by vibration of a disc when struck.
10	Horn	A horn uses compressed air or electricity to vibrate a diaphragm and exists in a variety of types which differ greatly in their sound and power.

5.28 Category of Installation Buoy

Type: EnumerationType

Definition: Classification of fixed installation buoy.

CamelCase: categoryOfInstallationBuoy

Value type: enumeration

Listed values:

Code	Label	Definition
1	Catenary Anchor Leg Mooring	Incorporates a large buoy which remains on the surface at all times and is moored by 4 or more anchors. Mooring hawsers and cargo hoses lead from a turntable on top of the buoy, so that the buoy does not turn as the ship swings to wind and stream.
2	Single Buoy Mooring	A large mooring buoy used by tankers to load and unload in port approaches or in offshore oil and gas fields.

5.29 Category of Landmark

Type: EnumerationType

Definition: Classification of prominent cultural and natural features in the landscape.

CamelCase: categoryOfLandmark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
2	Cemetery	A site and associated structures devoted to the burial of the dead.
3	Chimney	A vertical structure containing a passage or flue for discharging smoke and gases of combustion.

4	Dish Aerial	A parabolic aerial for the receipt and transmission of high frequency radio signals.
5	Flagstaff	A staff or pole on which flags are raised.
6	Flare Stack	A tall structure used for burning-off waste oil or gas.
7	Mast	A relatively tall structure usually held vertical by guy lines.
8	Windsock	A tapered fabric sleeve mounted so as to catch and swing with the wind, thus indicating the wind direction.
9	Monument	A structure erected and/or maintained as a memorial to a person and/or event.
10	Column/Pillar	A cylindrical or slightly tapering body of considerably greater length than diameter erected vertically.
11	Memorial Plaque	A slab of metal, usually ornamented, erected as a memorial to a person or event.
12	Obelisk	A tapering shaft usually of stone or concrete, square or rectangular in section, with a pyramidal apex.
13	Statue	A representation of a living being, sculptured, moulded, or cast in a variety of materials (for example: marble, metal, or plaster).
14	Cross	A monument, or other structure in form of a cross.
15	Dome	A landmark comprising a hemispherical or spheroidal shaped structure.
16	Radar Scanner	A device used for directing a radar beam through a search pattern.
17	Tower	A relatively tall, narrow structure that may either stand alone or may form part of another structure.
18	Windmill	A system of vanes attached to a tower and driven by wind (excluding wind turbines).
19	Windmotor	A modern structure for the use of wind power.
20	Spire/Minaret	A tall conical or pyramid-shaped structure often built on the roof or tower of a building, especially a church or mosque.
21	Large Rock or Boulder on Land	An isolated rocky formation or a single large stone.
22	Triangulation Mark	A recoverable point on the earth, whose geographic position has been determined by angular methods with

		geodetic instruments. A triangulation point is a selected point, which has been marked with a station mark, or it is a conspicuous natural or artificial feature.
23	Boundary Mark	A marker identifying the location of a surveyed boundary line.
24	Observation Wheel	Wheels with passenger cars mounted external to the rim and independently rotated by electric motors.
25	Torii	A form of decorative gateway or portal, consisting of two upright wooden posts connected at the top by two horizontal crosspieces, commonly found at the entrance to Shinto temples.
26	Bridge	A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.
27	Dam	A barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding.

5.30 Category of Lateral Mark

Type: EnumerationType

Definition: Classification of lateral marks in the IALA Buoyage System.

CamelCase: categoryOfLateralMark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Port-Hand Lateral Mark	Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
2	Starboard-Hand Lateral Mark	Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
3	Preferred Channel to Starboard Lateral Mark	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
4	Preferred Channel to Port Lateral Mark	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
5	Right-Hand Side of the Waterway	Indicates the right-hand side of the inland waterway.
6	Left-Hand Side of the Waterway	Indicates the left-hand side of the inland waterway.

7	Right-Hand Side of the Channel	Indicates the right-hand side of a channel of an inland waterway.
8	Left-Hand Side of the Channel	Indicates the left-hand side of a channel of an inland waterway.
9	Bifurcation of the Waterway	Indicates a bifurcation of the inland waterway.
10	Bifurcation of the Channel	Indicates a bifurcation of a channel of an inland waterway.
11	Channel Near the Right Bank	Indicates that the channel is near the right bank.
12	Channel Near the Left Bank	Indicates that the channel is near the left bank.
13	Channel Cross-Over to the Right Bank	Indicates that the channel crosses from the left to the right bank.
14	Channel Cross-Over to the Left Bank	Indicates that the channel crosses from the right to the left bank.
15	Danger Point or Obstacles at the Right-Hand Side	Indicates a danger point or obstacles at the right-hand side.
16	Danger Point or Obstacles at the Left-Hand Side	Indicates a danger point or obstacles at the left-hand side.
17	Turn Off at the Right-Hand Side	Indicates a turn off at the right-hand side.
18	Turn Off at the Left-Hand Side	Indicates a turn off at the left-hand side.
19	Junction at the Right-Hand Side	Indicates a junction at the right-hand side.
20	Junction at the Left-Hand Side	Indicates a junction at the left-hand side.
21	Harbour Entry at the Right-Hand Side	Indicates a harbour entry at the right-hand side.
22	Harbour Entry at the Left-Hand Side	Indicates a harbour entry at the left-hand side.
23	Bridge Pier Mark	Indicates a bridge pier in an inland waterway.
24	Entry From a Lake to a Narrower Waterway, Right Bank	Indicates the right bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
25	Entry From a Lake to a Narrower Waterway, Left Bank	Indicates the left bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
26	Change Bank	Change bank.
27	Continue Along Bank	Continue along bank.

5.31 Category of Light

Type: EnumerationType

Definition: Classification of different light types.

CamelCase: categoryOfLight

Value type: enumeration

Listed values:

Code	Label	Definition
1	Directional Function	A light illuminating a sector of very narrow angle and intended to mark a direction to follow.
4	Leading Light	A light associated with other lights so as to form a leading line to be followed.
5	Aero Light	An aero light is established for aeronautical navigation and may be of higher power than marine lights and visible from well offshore.
6	Air Obstruction Light	A light marking an obstacle which constitutes a danger to air navigation.
8	Flood Light	A broad beam light used to illuminate a structure or area.
9	Strip Light	A light whose source has a linear form generally horizontal, which can reach a length of several metres.
10	Subsidiary Light	A light placed on or near the support of a main light and having a special use in navigation.
11	Spotlight	A powerful light focused so as to illuminate a small area.
12	Front	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
13	Rear	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
14	Lower	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
15	Upper	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
17	Emergency	A light available as a backup to a main light which will be illuminated should the main light fail.
18	Bearing Light	A light which enables its approximate bearing to be obtained without the use of a compass.
19	Horizontally Disposed	A group of lights of identical character and almost identical position, that are disposed horizontally.
20	Vertically Disposed	A group of lights of identical character and almost identical position, that are

disposed vertically.

5.32 Category of Navigation Line

Type: EnumerationType

Definition: Classification of route guidance given to vessels.

CamelCase: categoryOfNavigationLine

Value type: enumeration

Listed values:

Code	Label	Definition
1	Clearing Line	A straight line that marks the boundary between a safe and a dangerous area or that passes clear of a navigational danger.
2	Transit Line	A line passing through one or more fixed marks.
3	Leading Line Bearing a Recommended Track	A line passing through one or more clearly defined objects, along the path of which a vessel can approach safely up to a certain distance off.

5.33 Category of Offshore Platform

Type: EnumerationType

Definition: Classification of an offshore raised structure.

CamelCase: categoryOfOffshorePlatform

Value type: enumeration

Listed values:

Code	Label	Definition
1	Oil Rig	A temporary mobile structure, either fixed or floating, used in the exploration stages of oil and gas fields.
2	Production Platform	A term used to indicate a permanent offshore structure equipped to control the flow of oil or gas. It does not include entirely submarine structures.
3	Observation/Research Platform	A platform from which one's surroundings or events can be observed, noted or recorded such as for scientific study.
4	Articulated Loading Platform	A metal lattice tower, buoyant at one end and attached at the other by a universal joint to a concrete filled base on the sea bed. The platform may be fitted with a helicopter platform, emergency accommodation and hawser/hose retrieval.
5	Single Anchor Leg Mooring	A rigid frame or tube with a buoyancy device at its upper end , secured at its lower end to a universal joint on a large steel or concrete base resting on the sea

		bed, and at its upper end to a mooring buoy by a chain or wire.
6	Mooring Tower	A platform secured to the sea bed and surmounted by a turntable to which ships moor.
7	Artificial Island	A man-made structure usually built for the exploration or exploitation of marine resources, marine scientific research, tidal observations, etc.
8	Floating Production, Storage and Off-Loading Vessel	An offshore oil/gas facility consisting of a moored tanker/barge by which the product is extracted, stored and exported.
9	Accommodation Platform	A platform used primarily for eating, sleeping and recreation purposes.
10	Navigation, Communication and Control Buoy	A floating structure with control room, power and storage facilities, attached to the sea bed by a flexible pipeline and cables.
11	Floating Oil Tank	A floating structure, anchored to the seabed, for storing oil.

5.34 Category of Pile

Type: EnumerationType

Definition: Classification of pile, driven into the earth as a foundation or support for a structure.

CamelCase: categoryOfPile

Value type: enumeration

Listed values:

Code	Label	Definition
1	Stake	An elongated wood or metal pole embedded in the seabed to serve as a marker or support.
3	Post	A vertical piece of timber, metal or concrete forced into the earth or sea bed.
4	Tripodal	A single structure comprising 3 or more piles held together (sections of heavy timber, steel or concrete), and forced into the earth or sea bed.
5	Piling	A number of piles, usually in a straight line, and usually connected or bolted together.
6	Area of Piles	A number of piles, usually in a straight line, but not connected by structural members.
7	Pipe	A vertical hollow cylinder of metal, wood, or other material forced into the earth or seabed.

5.35 Category of Radar Transponder Beacon

Type: EnumerationType

Definition: Classification of radar transponder beacon based on functionality.

CamelCase: categoryOfRadarTransponderBeacon

Value type: enumeration

Listed values:

Code	Label	Definition
1	Ramark, Radar Beacon Transmitting Continuously	A radar marker beacon which continuously transmits a signal appearing as a radial line on a radar screen, the line indicating the direction of the beacon. Ramarks are intended primarily for marine use. The name 'ramark' is derived from the words radar marker.
2	Racon, Radar Transponder Beacon	A radar beacon which returns a coded signal which provides identification of the beacon, as well as range and bearing. The range and bearing are indicated by the location of the first character received on the radar screen. The name 'racon' is derived from the words radar beacon.
3	Leading Racon/Radar Transponder Beacon	A radar beacon that may be used (in conjunction with at least one other radar beacon) to indicate a leading line.

5.36 Category of Radio Station

Type: EnumerationType

Definition: Classification of radio services offered by a radio station.

CamelCase: categoryOfRadioStation

Value type: enumeration

Listed values:

Code	Label	Definition
1	Circular (Non-Directional) Marine or Aero-Marine Radiobeacon	A radio station which need not necessarily be manned, the emissions of which, radiated around the horizon, enable its bearing to be determined by means of the radio direction finder of a ship.
2	Directional Radiobeacon	A special type of radiobeacon station the emissions of which are intended to provide a definite track for guidance.
3	Rotating Pattern Radiobeacon	A special type of radiobeacon station emitting a beam of waves to which a uniform turning movement is given, the bearing of the station being determined by means of an ordinary listening receiver and a stop watch. Also referred to as a rotating loop radiobeacon.
4	Consol Beacon	A type of long range position fixing beacon.

5	Radio Direction-Finding Station	A radio station intended to determine only the direction of other stations by means of transmission from the latter.
6	Coast Radio Station Providing QTG Service	A radio station which is prepared to provide QTG service; that is to say, to transmit upon request from a ship a radio signal, the bearing of which can be taken by that ship.
7	Aeronautical Radiobeacon	A radio beacon designed for aeronautical use.
8	Decca	The Decca Navigator System is a high accuracy, short to medium range radio navigational aid intended for coastal and landfall navigation.
9	Loran C	A low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
10	Differential GNSS	A radiobeacon transmitting DGPS correction signals.
11	Toran	An electronic position fixing system used mainly by aircraft.
12	Omega	A long-range radio navigational aid which operates within the VLF frequency band. The system comprises eight land based stations.
13	Syledis	A ranging position fixing system operating at 420-450 MHz over a range of up to 400 Km.
14	Chaika	Chaika is a low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
19	Radio Telephone Station	The equipment needed at one station to carry on two way voice communication by radio waves only.
20	AIS Base Station	An onshore AIS unit that monitors traffic in the waterways.

5.37 Category of Silo/Tank

Type: EnumerationType

Definition: Classification based on the product for which a silo or tank is used.

CamelCase: categoryOfSiloTank

Value type: enumeration

Listed values:

Code	Label	Definition
1	Silo in General	A large storage structure used for storing loose materials.
2	Tank in General	A fixed structure for storing liquids.

3	Grain Elevator	A storage building for grain. Usually a tall frame, metal or concrete structure with an especially compartmented interior.
4	Water Tower	A tower supporting an elevated storage tank of water.

5.38 Category of Special Purpose Mark

Type: EnumerationType

Definition: Classification of an aid to navigation which signifies some special purpose.

CamelCase: categoryOfSpecialPurposeMark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Firing Danger Mark	A mark used to indicate a firing danger area, usually at sea.
2	Target Mark	Any object toward which something is directed. The distinctive marking or instrumentation of a ground point to aid its identification on a photograph.
3	Marker Ship Mark	A mark marking the position of a ship which is used as a target during some military exercise.
4	Degaussing Range Mark	A mark used to indicate a degaussing range.
5	Barge Mark	A mark of relevance to barges.
6	Cable Mark	A mark used to indicate the position of submarine cables or the point at which they run on to the land.
7	Spoil Ground Mark	A mark used to indicate the limit of a spoil ground.
8	Outfall Mark	A mark used to indicate the position of an outfall or the point at which it leaves the land.
9	ODAS	Ocean Data Acquisition System.
10	Recording Mark	A mark used to record data for scientific purposes.
11	Seaplane Anchorage Mark	A mark used to indicate a seaplane anchorage.
12	Recreation Zone Mark	A mark used to indicate a recreation zone.
13	Private Mark	A privately maintained mark.
14	Mooring Mark	A mark indicating a mooring or moorings.
15	LANBY	A large buoy designed to take the place of a lightship where construction of an

		offshore light station is not feasible.
16	Leading Mark	Aids to navigation or other indicators so located as to indicate the path to be followed. Leading marks identify a leading line when they are in transit.
17	Measured Distance Mark	A mark forming part of a transit indicating one end of a measured distance.
18	Notice Mark	A notice board or sign indicating information to the mariner.
19	TSS Mark	A mark indicating a Traffic Separation Scheme.
20	Anchoring Prohibited Mark	A mark indicating an anchoring prohibited area.
21	Berthing Prohibited Mark	A mark indicating that berthing is prohibited.
22	Overtaking Prohibited Mark	A mark indicating that overtaking is prohibited.
23	Two-Way Traffic Prohibited Mark	A mark indicating a one-way route.
24	Reduced Wake Mark	A mark indicating that vessels must not generate excessive wake.
25	Speed Limit Mark	A mark indicating that a speed limit applies.
26	Stop Mark	A mark indicating the place where the bow of a ship must stop when traffic lights show red.
27	General Warning Mark	A mark indicating that special caution must be exercised in the vicinity of the mark.
28	Sound Ship's Siren Mark	A mark indicating that a ship should sound its siren or horn.
29	Restricted Vertical Clearance Mark	A mark indicating the minimum vertical space available for passage.
30	Maximum Vessel's Draught Mark	A mark indicating the maximum draught of vessel permitted.
31	Restricted Horizontal Clearance Mark	A mark indicating the minimum horizontal space available for passage.
32	Strong Current Warning Mark	A mark warning of strong currents.
33	Berthing Permitted Mark	A mark indicating that berthing is allowed.
34	Overhead Power Cable Mark	A mark indicating an overhead power cable.
35	Channel Edge Gradient Mark	A mark indicating the gradient of the slope of a dredge channel edge.
36	Telephone Mark	A mark indicating the presence of a

		telephone.
37	Ferry Crossing Mark	A mark indicating that a ferry route crosses the ship route; often used with a 'sound ship's siren' mark.
39	Pipeline Mark	A mark used to indicate the position of submarine pipelines or the point at which they run on to the land.
40	Anchorage Mark	A mark indicating an anchorage area.
41	Clearing Mark	A mark used to indicate a clearing line.
42	Control Mark	A mark indicating the location at which a restriction or requirement exists.
43	Diving Mark	A mark indicating that diving may take place in the vicinity.
44	Refuge Beacon	A mark providing or indicating a place of safety.
45	Foul Ground Mark	A mark indicating a foul ground.
46	Yachting Mark	A mark installed for use by yachtsmen.
47	Heliport Mark	A mark indicating an area where helicopters may land.
48	GNSS Mark	A mark indicating a location at which a GNSS position has been accurately determined.
49	Seaplane Landing Mark	A mark indicating an area where sea-planes land.
50	Entry Prohibited Mark	A mark indicating that entry is prohibited.
51	Work in Progress Mark	A mark indicating that work (generally construction) is in progress.
52	Mark With Unknown Purpose	A mark whose detailed characteristics are unknown.
53	Wellhead Mark	A mark indicating a borehole that produces or is capable of producing oil or natural gas.
54	Channel Separation Mark	A mark indicating the point at which a channel divides separately into two channels.
55	Marine Farm Mark	A mark indicating the existence of a fish, mussel, oyster or pearl farm/culture.
56	Artificial Reef Mark	A mark indicating the existence or the extent of an artificial reef.
57	Ice Mark	A mark, used year round, that may be submerged when ice passes through the area.
58	Nature Reserve Mark	A mark used to define the boundary of a nature reserve.

59	Fish Aggregating Device	A fish aggregating (or aggregation) device (FAD) is a man-made object used to attract ocean going pelagic fish such as marlin, tuna and mahi-mahi (dolphin fish). They usually consist of buoys or floats tethered to the ocean floor with concrete blocks.
60	Wreck Mark	A mark used to indicate the existence of a wreck.
61	Customs Mark	A mark used to indicate the existence of a customs checkpoint.
62	Causeway Mark	A mark used to indicate the existence of a causeway.
63	Wave Recorder	A surface following buoy used to measure wave activity.
64	Jetski Prohibited	A mark indicating a jetski prohibited area.

5.39 Category of Temporal Variation

Type: EnumerationType

Definition: An assessment of the likelihood of change over time.

CamelCase: categoryOfTemporalVariation

Value type: enumeration

Listed values:

Code	Label	Definition
1	Extreme Event	Indication of the possible impact of a significant event (for example hurricane, earthquake, volcanic eruption, landslide, etc), which is considered likely to have changed the seafloor or landscape significantly.
2	Likely to Change and Significant Shoaling Expected	Continuous or frequent change (for example river siltation, sand waves, seasonal storms, ice bergs, etc) that is likely to result in new significant shoaling.
3	Likely to Change But Significant Shoaling Not Expected	Continuous or frequent change (for example sand wave shift, seasonal storms, ice bergs, etc) that is not likely to result in new significant shoaling.
4	Likely to Change	Continuous or frequent change to non-bathymetric features (for example river siltation, glacier creep/recession, sand dunes, buoys, marine farms, etc).
5	Unlikely to Change	Significant change to the seafloor is not expected.
6	Unassessed	Not having been assessed.

5.40 City Name

Type: SimpleAttributeType

Definition: The name of a town or city.

CamelCase: cityName

Value type: text

5.41 Colour

Type: EnumerationType

Definition: The property possessed by an object of producing different sensations on the eye as a result of the way it reflects or emits light.

CamelCase: colour

Value type: enumeration

Listed values:

Code	Label	Definition
1	White	The achromatic object colour of greatest lightness characteristically perceived to belong to objects that reflect diffusely nearly all incident energy throughout the visible spectrum.
2	Black	The achromatic color of least lightness characteristically perceived to belong to objects that neither reflect nor transmit light.
3	Red	A color whose hue resembles that of blood or of the ruby or is that of the long-wave extreme of the visible spectrum.
4	Green	Of the color green.
5	Blue	A color whose hue is that of the clear sky or that of the portion of the color spectrum lying between green and violet.
6	Yellow	A color whose hue resembles that of ripe lemons or sunflowers or is that of the portion of the spectrum lying between green and orange.
7	Grey	Of the color grey.
8	Brown	Any of a group of colors between red and yellow in hue, of medium to low lightness, and of moderate to low saturation.
9	Amber	A variable color averaging a dark orange yellow.
10	Violet	Any of a group of colors of reddish-blue hue, low lightness, and medium saturation.
11	Orange	Any of a group of colors that are between red and yellow in hue.
12	Magenta	A deep purplish red.
13	Pink	Any of a group of colors bluish red to red in hue, of medium to high lightness, and

		of low to moderate saturation.
14	Green A	
15	Green B	
16	White Temporary	
17	Red Temporary	
18	Yellow Temporary	
19	Green Preferred	
20	Green Temporary	

5.42 Colour Pattern

Type: EnumerationType

Definition: A regular repeated design containing more than one colour.

CamelCase: colourPattern

Value type: enumeration

Listed values:

Code	Label	Definition
1	Horizontal Stripes	Straight bands or stripes of differing colours oriented horizontally.
2	Vertical Stripes	Straight bands or stripes of differing colours oriented vertically.
3	Diagonal Stripes	Straight bands or stripes of differing colours oriented diagonally (that is, not horizontally or vertically).
4	Squared	Often referred to as checker plate, where alternate colours are used to create squares similar to a chess or draught board. The pattern may be straight or diagonal.
5	Stripes (Direction Unknown)	Straight bands or stripes of differing colours oriented in an unknown direction.
6	Border Stripe	A band or stripe of colour which is displayed around the outer edge of the object, which may also form a border to an inner pattern or plain colour.
7	Single Colour	One solid colour of uniform coverage.
8	Rectangle	A four-sided shape that is made up of two pairs of parallel lines and that has four right angles, on a different coloured background.
9	Triangle	A shape that is made up of three lines and three angles, on a different coloured background.

5.43 Condition

Type: EnumerationType

Definition: The various conditions of buildings and other constructions.

CamelCase: condition

Value type: enumeration

Listed values:

Code	Label	Definition
1	Under Construction	Being built but not yet capable of function.
2	Ruined	A structure in a decayed or deteriorated condition resulting from neglect or disuse, or a damaged structure in need of repair.
3	Under Reclamation	An area of the sea, a lake or the navigable part of a river that is being reclaimed as land, usually by the dumping of earth and other material.
4	Wingless	A windmill or wind turbine from which the vanes or turbine blades are missing.
5	Planned Construction	Detailed planning has been completed but construction has not been initiated.

5.44 Counter Weight Type

Type: SimpleAttributeType

Definition: The type of counterweight used to balance or stabilize a structure, commonly found in lifting systems, buoys, or aids to navigation.

CamelCase: counterWeightType

Value type: text

5.45 Country Name

Type: SimpleAttributeType

Definition: The name of a nation.

CamelCase: countryName

Value type: text

5.46 Date End

Type: SimpleAttributeType

Definition: The latest date on which an object (for example a buoy) will be present.

CamelCase: dateEnd

Value type: S100_TruncatedDate

5.47 Date Start

Type: SimpleAttributeType

Definition: The earliest date on which an object (for example a buoy) will be present.

CamelCase: dateStart

Value type: S100_TruncatedDate

5.48 Delivery Point

Type: SimpleAttributeType

Definition: Details of where post can be delivered such as the apartment, name and/or number of a street, building or PO Box.

CamelCase: deliveryPoint

Value type: text

5.49 Depth Range Minimum Value

Type: SimpleAttributeType

Definition: The minimum (shoalest) value of a depth range.

CamelCase: depthRangeMinimumValue

Value type: real

5.50 Diameter

Type: SimpleAttributeType

Definition: The measurement of the width of a circular or cylindrical Aids to Navigation component, typically referring to buoys, beacons, or light structures.

CamelCase: diameter

Value type: real

5.51 Display Name

Type: SimpleAttributeType

Definition: A statement expressing if a feature name is to be displayed in certain system display settings or not.

CamelCase: displayName

Value type: boolean

5.52 Effective Intensity

Type: SimpleAttributeType

Definition: The luminous intensity of a fictitious juxtaposed steady-burning point light source that would appear to exhibit a luminosity equal to that of the rhythmic point light source it describes. The apparent reduction in intensity of the rhythmic light is subjective and is due to the nature of the response of the eye of the observer.

CamelCase: effectiveIntensity

Value type: real

5.53 Electronic Aton Change

Type: EnumerationType

Definition: Modifications to electronic or digital AtoNs, such as AIS (Automatic Identification System) AtoNs, virtual AtoNs, or remote-controlled systems.

CamelCase: electronicAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	AIS Transmitter Out Of Service	The terrestrial AIS transmitter is inoperative due to a technical issue.

2	AIS Transmitter Unreliable	The terrestrial AIS transmitter is unreliable due to a technical issue or maintenance.
3	AIS Transmitter Operating Properly	The terrestrial AIS transmitter is operating as advertised.
4	V-AIS Out Of Service	Virtual AIS aid to navigation is extinguished.
5	V-AIS Unreliable	Virtual AIS aid is unreliable due to a technical issue or maintenance.
6	V-AIS Operating Properly	Virtual AIS aid to navigation is operating as advertised.
7	RACON Out Of Service	The RACON is inoperative.
8	RACON Unreliable	The RACON is unreliable due to a technical issue or maintenance.
9	RACON Operating Properly	The RACON is operating as advertised.
10	DGPS Out Of Service	The DGPS station is inoperative due to a technical issue.
11	DGPS Operating Properly	The DGPS station is operating as advertised.
12	DGPS Unreliable	The DGPS station is unreliable due to a technical issue or maintenance.
13	LORAN C operating properly	.
14	LORAN C unreliable	.
15	LORAN C out of service	.
16	ELORAN Operating Properly	The eLORAN station is operating as advertised.
17	ELORAN Unreliable	The eLORAN station is unreliable due to a technical issue or maintenance.
18	ELORAN Out Of Service	The eLORAN station is inoperative due to a technical issue.
19	DGLOANSS operating properly	.
20	DGLOANSS unreliable	.
21	DGLOANSS out of service	.
22	Chayka Operating Properly	The Chayka station is operating as advertised.
23	Chayka Unreliable	The Chayka station is unreliable due to a technical issue or maintenance.
24	Chayka Out Of Service	The Chayka station is inoperative due to a technical issue.
25	E-Chayka Operating Properly	The e-Chayka station is operating as advertised.

26	E-Chayka Unreliable	The e-Chayka station is unreliable due to a technical issue or maintenance.
27	E-Chayka Out Of Service	The e-Chayka station is inoperative due to a technical issue.
28	EGNOS Operating Properly	The EGNOS station is operating as advertised.
29	EGNOS Unreliable	The EGNOS station is unreliable due to a technical issue or maintenance.
30	EGNOS Out Of Service	The EGNOS station is inoperative due to a technical issue.

5.54 Elevation

Type: SimpleAttributeType

Definition: The altitude of the ground level of an object, measured from a specified vertical datum.

CamelCase: elevation

Value type: real

5.55 Estimated Range of Transmission

Type: SimpleAttributeType

Definition: The estimated range of a non-optical electromagnetic transmission.

CamelCase: estimatedRangeOfTransmission

Value type: real

5.56 Exhibition Condition of Light

Type: EnumerationType

Definition: The outward display of the light.

CamelCase: exhibitionConditionOfLight

Value type: enumeration

Listed values:

Code	Label	Definition
1	Light Shown Without Change of Character	A light shown throughout the 24 hours without change of character.
2	Daytime Light	A light which is only exhibited by day.
3	Fog Light	A light which is exhibited in fog or conditions of reduced visibility.
4	Night Light	A light which is only exhibited at night.

5.57 File Locator

Type: SimpleAttributeType

Definition: The location of a fragment of text or other information in a support file.

CamelCase: fileLocator

Value type: text

5.58 File Reference

Type: SimpleAttributeType

Definition: The file name of an externally referenced text file.

CamelCase: fileReference

Value type: text

5.59 Fixed Aton Change

Type: EnumerationType

Definition: Modifications or updates to fixed AtoNs, such as lighthouses or beacons, which are permanently positioned.

CamelCase: fixedAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Beacon Missing	No beacon at the advertised position.
2	Beacon Damaged	The beacon has sustained damage due to external factors (wind, sea state, collision with a vessel).
3	Light beacon Unlit	.
4	Light beacon Unreliable	.
5	Light beacon Not synchronized	.
6	Light beacon damaged	.
7	Beacon Topmark Missing	The topmark of the beacon is missing.
8	Beacon Topmark Damaged	The topmark of the beacon is damaged due to external factors (wind, sea state, collision with a vessel).
9	Beacon Daymark Unreliable	Colour of the beacon daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
10	Floodlit beacon Unlit	.
11	Beacon Restored To Normal	The beacon has been restored to normal condition.

5.60 Flare Bearing

Type: SimpleAttributeType

Definition: The bearing about which the light flare symbol is rotated to be displayed in ECDIS.

CamelCase: flareBearing

Value type: integer

5.61 Floating Aton Change

Type: EnumerationType

Definition: Adjustments or replacements related to floating AtoNs, such as buoys, which are anchored but can move with water currents.

CamelCase: floatingAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Adrift	The buoy is no longer secured to its moorings and is adrift.
2	Buoy Damaged	The buoy has been damaged due to external factors (wind, sea state, collision with a vessel).
3	Buoy Daymark Unreliable	Colour of the buoy daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
4	Buoy Destroyed	The buoy has suffered extensive damage and is not useable.
5	Buoy Missing	No buoy at its advertised/charted position or in the vicinity.
6	Buoy Move	The buoy has been or will be moved intentionally.
7	Buoy off Position	The buoy has been dragged off its advertised position due to wind or current affecting the mooring system.
8	Buoy re-establishment	.
9	Buoy Restored to Normal	The buoy has been restored to normal condition.
10	Buoy Topmark Damaged	The topmark of the buoy is damaged due to external factors (wind, sea state, collision with a vessel).
11	Buoy Topmark Missing	The topmark of the buoy is missing.
12	Buoy Will Be Withdrawn	The buoy has been scheduled for removal from service for a fixed term.
13	Buoy Withdrawn	The buoy has been removed from service for a fixed term.
14	Decommissioned for winter	.
15	Lifted for Winter	.
16	Light buoy Light damaged	.
17	Light buoy Light not synchronized	.
18	Light buoy Light unlit	.
19	Light buoy Light unreliable	.
20	Marine Aids to Navigation Unreliable	The position or status of Marine Aids to Navigation, over an extensive area, is

		unreliable due to a natural event (freshet, storm surge, flooding).
21	Recommissioned for navigation season	.
22	Replaced by Winter Spar	.
23	Seasonal Decommissioning Complete	The completion of the process to remove summer buoys (and possibly replace some with winter spar buoys).
24	Seasonal Decommissioning in Progress	The commencement of the process to remove summer buoys (and possibly replace some with winter spar buoys).
25	Seasonal recommissioning complete	.
26	Seasonal recommissioning in progress	.

5.62 Function

Type: EnumerationType

Definition: A specific role that describes a feature.

CamelCase: functionFihoXyz

Value type: enumeration

Listed values:

Code	Label	Definition
2	Harbour-Masters Office	A local official who has charge of mooring and berthing of vessels, collecting harbour fees, etc.
3	Customs Office	Serves as a government office where customs duties are collected, the flow of goods are regulated and restrictions enforced, and shipments or vehicles are cleared for entering or leaving a country.
4	Health Office	The office which is charged with the administration of health laws and sanitary inspections.
5	Hospital	An institution or establishment providing medical or surgical treatment for the ill or wounded.
6	Post Office	The public department, agency or organisation responsible primarily for the collection, transmission and distribution of mail.
7	Hotel	An establishment, especially of a comfortable or luxurious kind, where paying visitors are provided with accommodation, meals and other services.
8	Railway Station	A building with platforms where trains arrive, load, discharge and depart.
9	Police Station	The headquarters of a local police force and that is where those under arrest are first charged.

10	Water-Police Station	The headquarters of a local water-police force.
11	Pilot Office	The office or headquarters of pilots; the place where the services of a pilot may be obtained.
12	Pilot Lookout	A distinctive structure or place on shore from which personnel keep watch upon events at sea or along the coast.
13	Bank Office	An office for custody, deposit, loan, exchange or issue of money.
14	Headquarters for District Control	The quarters of an executive officer (director, manager, etc.) with responsibility for an administrative area.
15	Transit Shed/Warehouse	A building or part of a building for storage of wares or goods.
16	Factory	A building or buildings with equipment for manufacturing; a workshop.
17	Power Station	A stationary plant containing apparatus for large scale conversion of some form of energy (such as hydraulic, steam, chemical or nuclear energy) into electrical energy.
18	Administrative	A building for the management of affairs.
19	Educational Facility	A building concerned with education (for example school, college, university, etc).
20	Church	A building for public Christian worship.
21	Chapel	A place for Christian worship other than a parish, cathedral or church, especially one attached to a private house or institution.
22	Temple	A building for public Jewish worship.
23	Pagoda	A Hindu or Buddhist temple or sacred building.
24	Shinto Shrine	A building for public Shinto worship.
25	Buddhist Temple	A building for public Buddhist worship.
26	Mosque	A Muslim place of worship.
27	Marabout	A shrine marking the burial place of a Muslim holy man.
28	Lookout	Keeping a watch upon events at sea or along the coast.
29	Communication	Transmitting and/or receiving electronic communication signals.
30	Television	A system for reproducing on a screen visual images transmitted (usually with sound) by radio signals.

31	Radio	Transmitting and/or receiving radio-frequency electromagnetic waves as a means of communication.
32	Radar	A method, system or technique of using beamed, reflected, and timed radio waves for detecting, locating, or tracking objects, and for measuring altitudes.
33	Light Support	A structure serving as a support for one or more lights.
34	Microwave	Broadcasting and receiving signals using microwaves.
35	Cooling	Generation of chilled liquid and/or gas for cooling purposes.
36	Observation	A place from which the surroundings can be observed but at which a watch is not habitually maintained.
37	Timeball	A visual time signal in the form of a ball.
38	Clock	Instrument for measuring time and recording hours.
39	Control	Used to control the flow of traffic within a specified range of an installation.
40	Airship Mooring	Equipment or structure to secure an airship.
41	Stadium	An arena for holding and viewing events.
42	Bus Station	A building where buses and coaches regularly stop to take on and/or let off passengers, especially for long-distance travel.
43	Passenger Terminal Building	A building within a terminal for the loading and unloading of passengers.
44	Sea Rescue Control	A unit responsible for promoting efficient organization of search and rescue services and for coordinating the conduct of search and rescue operations within a search and rescue region.
45	Observatory	A building designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.
46	Ore Crusher	A building or structure used to crush ore.
47	Boathouse	A building or shed, usually built partly over water, for sheltering a boat or boats.
48	Pumping Station	A facility to move solids, liquids or gases by means of pressure or suction.

5.63 Headline

Type: SimpleAttributeType

Definition: Words set at the head of a passage or page to introduce or categorize.

CamelCase: headline

Value type: text

5.64 Height

Type: SimpleAttributeType

Definition: The value of the vertical distance to the highest point of the object, measured from a specified vertical datum.

CamelCase: height

Value type: real

5.65 Height/Length Units

Type: EnumerationType

Definition: Units of measure of waterway distances.

CamelCase: heightLengthUnits

Value type: enumeration

Listed values:

Code	Label	Definition
1	Metres	The basic unit of length in the International System of Units (SI) system.
2	Feet	A unit of length equal to 12 inches, 1/6 of a fathom, or 30.480 centimetres.
3	Kilometres	A unit of length, the common measure of distances equal to 1000 metres, and equivalent to 3280.8 feet or 0.621 mile.
4	Hectometres	Heights/lengths are specified in hectometres (100 metres).
5	Statute Miles	A unit equal to 5280 feet.
6	Nautical Miles	A unit of length equal to 1,852 metres. This value was approved by the International Hydrographic Conference of 1929 and has been adopted by nearly all maritime states.

5.66 Horizontal Accuracy

Type: SimpleAttributeType

Definition: The degree to which a horizontal measurement is accurate, typically referring to the accuracy of an object's position in the horizontal plane (latitude and longitude).

CamelCase: horizontalAccuracy

Value type: real

5.67 Horizontal Datum

Type: EnumerationType

Definition: Horizontal reference surface or the reference coordinate system used for geodetic control in the calculation of coordinates of points on the earth.

CamelCase: horizontalDatum

Value type: enumeration

Listed values:

Code	Label	Definition
1	WGS 72	A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system's fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. The WGS 72 is based on selected satellite, surface gravity and astrogeodetic data available through 1972.
2	WGS 84	A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system's fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. WGS 84 is the reference coordinate system used by the Global Positioning System.
3	European 1950	A geodetic datum first defined in 1950 suitable for use in Europe - west: Andorra; Cyprus; Denmark - onshore and offshore; Faroe Islands - onshore; France - offshore; Germany - offshore North Sea; Gibraltar; Greece - offshore; Israel - offshore; Italy including San Marino and Vatican City State; Ireland offshore; Malta; Netherlands - offshore; North Sea; Norway including Svalbard - onshore and offshore; Portugal - mainland - offshore; Spain - onshore; Turkey - onshore and offshore; United Kingdom UKCS offshore east of 6W including Channel Islands (Guernsey and Jersey). Egypt - Western Desert; Iraq - onshore; Jordan. European Datum 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. European Datum 1950 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222'51.4456N, longitude: 1303'58.9283E (of Greenwich). European Datum 1950 is a geodetic datum for Topographic mapping, geodetic survey.
4	Potsdam Datum	A geodetic datum first defined in 1990 suitable for use in Germany - Thuringen. Potsdam Datum/83 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Potsdam Datum/83 origin is Fundamental point: Rauenberg. Latitude: 5227'12.021N, longitude: 1322'04.928E (of Greenwich). This station was destroyed in 1910 and the station at Potsdam substituted as the fundamental point. Potsdam Datum/83 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from BKG via

		EuroGeographics. http://crs.bkg.bund.de/PD/83 is the realisation of DHDN in Thuringen. It is the resultant of applying a transformation derived at 13 points on the border between East and West Germany to Pulkovo 1942/83 points in Thuringen.
5	Adindan	A geodetic datum first defined in 1958 suitable for use in Eritrea; Ethiopia; South Sudan; Sudan. Adindan references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Adindan origin is Fundamental point: Station 15; Adindan. Latitude: 2210'07.110N, longitude: 3129'21.608E (of Greenwich). Adindan is a geodetic datum for Topographic mapping. It was defined by information from US Coast and Geodetic Survey via Geophysical Research vol 67 #11, October 1962. The 12th parallel traverse of 1966-70 (Point 58 datum, code 6620) is connected to the Blue Nile 1958 network in western Sudan. This has given rise to misconceptions that the Blue Nile network is used in west Africa.
6	Afgooye	A geodetic datum first defined in and suitable for use in Somalia - onshore. Afgooye references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Afgooye is a geodetic datum for Topographic mapping.
7	Ain el Abd 1970	A geodetic datum first defined in 1970 and suitable for use in Bahrain, Kuwait and Saudi Arabia - onshore. Ain el Abd 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Ain el Abd 1970 origin is Fundamental point: Ain El Abd. Latitude: 2814'06.171N, longitude: 4816'20.906E (of Greenwich). Ain el Abd 1970 is a geodetic datum for Topographic mapping.
8	Anna 1 Astro 1965	A geodetic datum first defined in 1965 suitable for use in Cocos (Keeling) Islands - onshore. Cocos Islands 1965 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Cocos Islands 1965 origin is Fundamental point: Anna 1. Cocos Islands 1965 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
9	Antigua Island Astro 1943	A geodetic datum first defined in 1943 suitable for use in Antigua island - onshore. Antigua 1943 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Antigua 1943 origin is Fundamental point: station A14. Antigua 1943 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.
10	Arc 1950	A geodetic datum first defined in 1950 suitable for use in Botswana; Malawi; Zambia; Zimbabwe. Arc 1950 references

		the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Arc 1950 origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000S, longitude: 2530'44.622E (of Greenwich). Arc 1950 is a geodetic datum for Topographic mapping, geodetic survey.
11	Arc 1960	A geodetic datum first defined in 1960 suitable for use in Kenya; Tanzania; Uganda. Arc 1960 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Arc 1960 origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000S, longitude: 2530'44.622E (of Greenwich). Arc 1960 is a geodetic datum for Topographic mapping, geodetic survey.
12	Ascension Island 1958	A geodetic datum first defined in 1958 suitable for use in St Helena, Ascension and Tristan da Cunha - Ascension Island - onshore. Ascension Island 1958 references the International 1924 ellipsoid and the Greenwich prime meridian. Ascension Island 1958 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
13	Astro Beacon 'E' 1945	Astro beacon 'E' 1945
14	Astro DOS 71/4	A geodetic datum first defined in 1971 suitable for use in St Helena, Ascension and Tristan da Cunha - St Helena Island - onshore. Astro DOS 71 references the International 1924 ellipsoid and the Greenwich prime meridian. Astro DOS 71 origin is Fundamental point: DOS 71/4, Ladder Hill Fort, latitude: 1555'30S, longitude: 543'25W (of Greenwich). Astro DOS 71 is a geodetic datum for Geodetic control, military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000) and St. Helena Government, Environment and Natural Resources Directorate (ENRD).
15	Astro Tern Island (FRIG) 1961	A geodetic datum first defined in 1961 suitable for use in United States (USA) - Hawaii - Tern Island and Sorel Atoll. Tern Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Tern Island 1961 origin is Fundamental point: station FRIG on tern island, station B4 on Sorol Atoll. Tern Island 1961 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (original 1987 first edition and 3rd edition, Amendment 1, 3 January 2000). Two independent astronomic determinations considered to be consistent through adoption of common transformation to WGS 84 (see tfm code 15795).
16	Astronomical Station 1952	Astronomical station 1952.

17	Australian Geodetic 1966	A geodetic datum first defined in 1966 suitable for use in Australia - onshore and offshore. Papua New Guinea - onshore. Australian Geodetic Datum 1966 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1966 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515S, longitude: 13312'30.0771E (of Greenwich). Australian Geodetic Datum 1966 is a geodetic datum for Topographic mapping. It was defined by information from Australian Map Grid Technical Manual. National Mapping Council of Australia Technical Publication 7; 1972.
18	Australian Geodetic 1984	A geodetic datum first defined in 1984 suitable for use in Australia - Queensland, South Australia, Western Australia, federal areas offshore west of 129E. Australian Geodetic Datum 1984 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1984 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515S, longitude: 13312'30.0771E (of Greenwich). Australian Geodetic Datum 1984 is a geodetic datum for Topographic mapping. It was defined by information from GDA technical manual v2_2, Intergovernmental Committee on Surveying and Mapping. www.anzlic.org.au/icsm/gdtm/ Uses all data from 1966 adjustment with additional observations, improved software and a geoid model.
19	Ayabelle Lighthouse	A geodetic datum suitable for use in Djibouti - onshore and offshore. Ayabelle Lighthouse references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Ayabelle Lighthouse origin is Fundamental point: Ayabelle Lighthouse. Ayabelle Lighthouse is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
20	Bellevue (IGN)	A geodetic datum first defined in 1960 suitable for use in Vanuatu - southern islands - Aneityum, Efate, Erromango and Tanna. Bellevue references the International 1924 ellipsoid and the Greenwich prime meridian. Bellevue is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.
21	Bermuda 1957	A geodetic datum first defined in 1957 suitable for use in Bermuda - onshore. Bermuda 1957 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Bermuda 1957 origin is

		Fundamental point: Fort George base. Latitude 3222'44.36N, longitude 6440'58.11W (of Greenwich). Bermuda 1957 is a geodetic datum for Topographic mapping. It was defined by information from Various oil industry sources.
22	Bissau	A geodetic datum first defined in and is suitable for use in Guinea-Bissau - onshore. Bissau references the International 1924 ellipsoid and the Greenwich prime meridian. Bissau origin is Bissau is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 ftp://164.214.2.65/pub/gig/tr8350.2/changes.pdf .
23	Bogota Observatory	A geodetic datum first defined in 1975 suitable for use in Colombia - mainland and offshore Caribbean. Bogota 1975 references the International 1924 ellipsoid and the Greenwich prime meridian. Bogota 1975 origin is Fundamental point: Bogota observatory. Latitude: 435'56.570N, longitude: 7404'51.300W (of Greenwich). Bogota 1975 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico Agustin Codazzi (IGAC) special publication no. 1, 4th edition (1975) Geodesia: Resultados Definitivos de Parte de las Redes Geodesicas Establecidas en el Pais. Replaces 1951 adjustment. Replaced by MAGNA-SIRGAS (datum code 6685).
24	Bukit Rimpah	A geodetic datum suitable for use in Indonesia - Banga and Belitung Islands. Bukit Rimpah references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Bukit Rimpah origin is 200'40.16S, 10551'39.76E (of Greenwich). Bukit Rimpah is a geodetic datum for Topographic mapping.
25	Camp Area Astro	A geodetic datum suitable for use in Antarctica - McMurdo Sound, Camp McMurdo area. Camp Area Astro references the International 1924 ellipsoid and the Greenwich prime meridian. Camp Area Astro is a geodetic datum for Geodetic and topographic survey. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
26	Campo Inchauspe 1969	A geodetic datum suitable for use in Argentina - mainland onshore and Atlantic offshore Tierra del Fuego. Campo Inchauspe references the International 1924 ellipsoid and the Greenwich prime meridian. Campo Inchauspe origin is Fundamental point: Campo Inchauspe. Latitude: 3558'16.56S, longitude: 6210'12.03W (of Greenwich). Campo Inchauspe is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/

27	Canton Astro 1966	A geodetic datum first defined in 1966 suitable for use in Kiribati - Phoenix Islands: Kanton, Orona, McKean Atoll, Birnie Atoll, Phoenix Seamounts. Phoenix Islands 1966 references the International 1924 ellipsoid and the Greenwich prime meridian. Phoenix Islands 1966 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
28	Cape Datum	A geodetic datum suitable for use in Botswana; Lesotho; South Africa - mainland; Swaziland. Cape references the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Cape origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000S, longitude: 2530'44.622E (of Greenwich). Cape is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.
29	Cape Canaveral	A geodetic datum first defined in 1963 suitable for use in North America - onshore - Bahamas and USA - Florida (east). Cape Canaveral references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Cape Canaveral origin is Fundamental point: Central 1950. Latitude: 28 29'32.36555N, longitude 80 34'38.77362W (of Greenwich). Cape Canaveral is a geodetic datum for US space and military operations. It was defined by information from US NGS and DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
30	Carthage	A geodetic datum first defined in 1925 suitable for use in Tunisia - onshore and offshore. Carthage references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Carthage origin is Fundamental point: Carthage. Latitude: 40.9464506g = 3651'06.50N, longitude: 8.8724368g E of Paris = 1019'20.72E (of Greenwich). Carthage is a geodetic datum for Topographic mapping. Fundamental point astronomic coordinates determined in 1878.
31	Chatam Island Astro 1971	A geodetic datum first defined in 1971 suitable for use in New Zealand - Chatham Islands group - onshore. Chatham Islands Datum 1971 references the International 1924 ellipsoid and the Greenwich prime meridian. Chatham Islands Datum 1971 is a geodetic datum for Geodetic survey, topographic mapping, engineering survey. It was defined by information from Office of Surveyor General (OSG) Technical Report 14, June 2001. Replaced by Chatham Islands Datum 1979 (code 6673).
32	Chua Astro	A geodetic datum suitable for use in Brazil - south of 18S and west of 54W, plus Distrito Federal. Paraguay - north.

		Chua references the International 1924 ellipsoid and the Greenwich prime meridian. Chua origin is Fundamental point: Chua. Latitude: 1945'41.160S, longitude: 4806'07.560W (of Greenwich). Chua is a geodetic datum for Geodetic survey. It was defined by information from NIMA http://earth-info.nima.mil/ . The Chua origin and associated network is in Brazil with a connecting traverse through northern Paraguay. It was used in Brazil only as input into the Corrego Alegre adjustment and for government work in Distrito Federal.
33	Corrego Alegre	A geodetic datum first defined in 1972 suitable for use in Brazil - onshore - west of 54W and south of 18S; also south of 15S between 54W and 42W; also east of 42W. Corrego Alegre 1970-72 references the International 1924 ellipsoid and the Greenwich prime meridian. Corrego Alegre 1970-72 origin is Fundamental point: Corrego Alegre. Latitude: 1950'14.91S, longitude: 4857'41.98W (of Greenwich). Corrego Alegre 1970-72 is a geodetic datum for Topographic mapping, geodetic survey. Superseded by SAD69. It was defined by information from IBGE. Replaces 1961 adjustment (datum code 1074). NIMA gives coordinates of origin as latitude: 1950'15.14S, longitude: 4857'42.75W; these may refer to 1961 adjustment.
34	Dabola	A geodetic datum first defined in 1981 suitable for use in Guinea - onshore. Dabola 1981 references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Dabola 1981 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.
35	Djakarta (Batavia)	A geodetic datum suitable for use in Indonesia - onshore Java and Bali. Batavia (Jakarta) references the Bessel 1841 ellipsoid and the Jakarta prime meridian. Batavia (Jakarta) origin is Fundamental point: Longitude at Batavia astronomical station. Latitude: 607'39.522S, longitude: 000'00.0E (of Jakarta). Latitude and azimuth at Genuk. Batavia (Jakarta) is a geodetic datum for Topographic mapping.
36	DOS 1968	DOS 1968.
37	Easter Island 1967	A geodetic datum first defined in 1967 suitable for use in Chile - Easter Island onshore. Easter Island 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Easter Island 1967 is a geodetic datum for Military and topographic mapping, +/- 25 meters in each component. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
38	European 1979	A geodetic datum first defined in 1979 suitable for use in Europe - west. European Datum 1979 references the International 1924 ellipsoid and the

		Greenwich prime meridian. European Datum 1979 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222'51.4456N, longitude: 1303'58.9283E (of Greenwich). European Datum 1979 is a geodetic datum for Scientific network. Replaced by 1987 adjustment.
39	Fort Thomas 1955	Fort Thomas 1955 datum.
40	Gan 1970	A geodetic datum first defined in 1970 suitable for use in Maldives - onshore. Gan 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Gan 1970 is a geodetic datum for Topographic mapping. It was defined by information from Various industry sources. In some references incorrectly named Gandajika 1970.
41	Geodetic Datum 1949	A geodetic datum first defined in 1949 suitable for use in New Zealand - North Island, South Island, Stewart Island - onshore and nearshore. New Zealand Geodetic Datum 1949 references the International 1924 ellipsoid and the Greenwich prime meridian. New Zealand Geodetic Datum 1949 origin is Fundamental point: Papatahi. Latitude: 4119' 8.900S, longitude: 17502'51.000E (of Greenwich). New Zealand Geodetic Datum 1949 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Land Information New Zealand. http://www.linz.govt.nz/rcs/linz/pub/web/root/core/SurveySystem/GeodeticInfo/GeodeticDatums/nzgd2000factsheet/index.jsp . Replaced by New Zealand Geodetic Datum 2000 (code 6167) from March 2000.
42	Graciosa Base SW 1948	Graciosa Base SW 1948 datum.
43	Guam 1963	A geodetic datum first defined in 1963 suitable for use in Guam - onshore. Guam 1963 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Guam 1963 origin is Fundamental point: Tagcha. Latitude: 1322'38.49N, longitude: 14445'51.56E (of Greenwich). Guam 1963 is a geodetic datum for Topographic mapping. It was defined by information from US National Geospatial Intelligence Agency (NGA). http://earth-info.nga.mil/ Replaced by NAD83(HARN)
44	Gunung Segara	A geodetic datum suitable for use in Indonesia - Kalimantan - onshore east coastal area including Mahakam delta coastal and offshore shelf areas. Gunung Segara references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Gunung Segara origin is Station P5 (Gunung Segara). Latitude 032'12.83S, longitude 11708'48.47E (of Greenwich). Gunung Segara is a geodetic datum for Topographic mapping. It was defined by information from TotalFinaElf.

45	GUX 1 Astro	GUX 1 Astro datum.
46	Herat North	A geodetic datum suitable for use in Afghanistan. Herat North references the International 1924 ellipsoid and the Greenwich prime meridian. Herat North origin is Fundamental point: Herat North. Latitude: 3423'09.08N, longitude: 6410'58.94E (of Greenwich). Herat North is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
47	Hjorsey 1955	A geodetic datum first defined in 1955 suitable for use in Iceland - onshore. Hjorsey 1955 references the International 1924 ellipsoid and the Greenwich prime meridian. Hjorsey 1955 origin is Fundamental point: Latitude: 6431'29.26N, longitude: 2222'05.84W (of Greenwich). Hjorsey 1955 is a geodetic datum for 1/50,000 scale topographic mapping. It was defined by information from Landmaelingar Islands (National Survey of Iceland).
48	Hong Kong 1963	A geodetic datum first defined in 1963 suitable for use in China - Hong Kong - onshore and offshore. Hong Kong 1963 references the Clarke 1858 ellipsoid and the Greenwich prime meridian. Hong Kong 1963 origin is Fundamental point: Trig Zero, 38.4 feet south along the transit circle of the Kowloon Observatory. Latitude 2218'12.82N, longitude 11410'18.75E (of Greenwich). Hong Kong 1963 is a geodetic datum for Topographic mapping and hydrographic charting. It was defined by information from Survey and Mapping Office, Lands Department. http://www.info.gov.hk/landsd/ . Replaced by Hong Kong 1963(67) for military purposes only in 1967. Replaced by Hong Kong 1980.
49	Hu-Tzu-Shan	A geodetic datum first defined in 1950 suitable for use in Taiwan, Republic of China - onshore - Taiwan Island, Penghu (Pescadores) Islands. Hu Tzu Shan 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. Hu Tzu Shan 1950 origin is Fundamental point: Hu Tzu Shan. Latitude: 2358'32.34N, longitude: 12058'25.975E (of Greenwich). Hu Tzu Shan 1950 is a geodetic datum for Topographic mapping. It was defined by information from NIMA US NGA, http://earth-info.nga.mil/GandG/index.html
50	Indian	Indian datum.
51	Indian 1954	A geodetic datum first defined in 1954 suitable for use in Myanmar (Burma) - onshore; Thailand - onshore. Indian 1954 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1954 origin is Extension of Kalianpur 1937 over Myanmar and Thailand. Indian 1954 is a

		geodetic datum for Topographic mapping.
52	Indian 1975	A geodetic datum first defined in 1975 suitable for use in Thailand - onshore plus offshore Gulf of Thailand. Indian 1975 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1975 origin is Fundamental point: Khau Sakaerang. Indian 1975 is a geodetic datum for Topographic mapping.
53	Ireland 1965	A geodetic datum first defined in 1975 suitable for use in Ireland - onshore. United Kingdom (UK) - Northern Ireland (Ulster) - onshore. Ireland 1965 references the Airy Modified 1849 ellipsoid and the Greenwich prime meridian. Ireland 1965 origin is Adjusted to best mean fit 9 stations of the OSNI 1952 primary adjustment in Northern Ireland plus the 1965 values of 3 stations in the Republic of Ireland. Ireland 1965 is a geodetic datum for Geodetic survey, topographic mapping and engineering survey. It was defined by information from The Irish Grid - A Description of the Co-ordinate Reference System published by Ordnance Survey of Ireland, Dublin and Ordnance Survey of Northern Ireland, Belfast. Differences from the 1965 adjustment (datum code 6299) are: average difference in Eastings 0.092m; average difference in Northings 0.108m; maximum vector difference 0.548m.
54	ISTS 061 Astro 1968	A geodetic datum first defined in 1968 suitable for use in South Georgia and the South Sandwich Islands - South Georgia onshore. ISTS 061 Astro 1968 references the International 1924 ellipsoid and the Greenwich prime meridian. ISTS 061 Astro 1968 origin is Fundamental point: ISTS 061. ISTS 061 Astro 1968 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
55	ISTS 073 Astro 1969	A geodetic datum first defined in 1969 suitable for use in British Indian Ocean Territory - Chagos Archipelago - Diego Garcia. ISTS 073 Astro 1969 references the International 1924 ellipsoid and the Greenwich prime meridian. ISTS 073 Astro 1969 origin is Fundamental point: ISTS 073. ISTS 073 Astro 1969 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
56	Johnston Island 1961	A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands - Johnston Island. Johnston Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Johnston Island 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA

		/ NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
57	Kandawala	A geodetic datum first defined in 1930 suitable for use in Sri Lanka - onshore. Kandawala references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Kandawala origin is Fundamental point: Kandawala. Latitude: 714'06.838N, longitude: 7952'36.670E. Kandawala is a geodetic datum for Topographic mapping. It was defined by information from Abeyratne, Featherstone and Tantrigoda in Survey Review vol. 42 no. 317 (July 2010).
58	Kerguelen Island 1949	A geodetic datum first defined in 1949 suitable for use in French Southern Territories - Kerguelen onshore. References the International 1924 ellipsoid and the Greenwich prime meridian. Origin is K0 1949. Is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris.
59	Kertau 1968	A geodetic datum first defined in 1968 suitable for use in Malaysia - West Malaysia onshore and offshore east coast; Singapore - onshore and offshore. Kertau 1968 references the Everest 1830 Modified ellipsoid and the Greenwich prime meridian. Kertau 1968 origin is Fundamental point: Kertau. Latitude: 327'50.710N, longitude: 10237'24.550E (of Greenwich). Kertau 1968 is a geodetic datum for Geodetic survey, cadastre. It was defined by information from Defence Geographic Centre. Replaces MRT48 and earlier adjustments. Adopts metric conversion of 39.370113 inches per metre. Not used for 1969 metrication of RSO grid - see Kertau (RSO) (code 6751).
60	Kusaie Astro 1951	A geodetic datum first defined in 1951 suitable for use in Federated States of Micronesia - Kosrae (Kusaie). Kusaie 1951 references the International 1924 ellipsoid and the Greenwich prime meridian. Kusaie 1951 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
61	L. C. 5 Astro 1961	L. C. 5 Astro 1961 datum.
62	Leigon	A geodetic datum suitable for use in Ghana - onshore and offshore. Leigon references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Leigon origin is Fundamental point: GCS Station 121, Leigon. Latitude: 538'52.27N, longitude: 011'46.08W (of Greenwich). Leigon is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Replaced Accra datum (code 6168) from 1978. Coordinates at Leigon

		fundamental point defined as Accra datum values for that point.
63	Liberia 1964	A geodetic datum first defined in 1964 suitable for use in Liberia - onshore. Liberia 1964 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Liberia 1964 origin is Fundamental point: Robertsfield. Latitude: 613'53.02N, longitude: 1021'35.44W (of Greenwich). Liberia 1964 is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
64	Luzon	A geodetic datum first defined in 1911 suitable for use in Philippines - onshore. Luzon references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Luzon origin is Fundamental point: Balacan. Latitude: 1333'41.000N, longitude: 12152'03.000E (of Greenwich). Luzon is a geodetic datum for Topographic mapping. It was defined by information from Coast and Geodetic Survey Replaced by Philippine Reference system of 1992 (datum code 6683).
65	Mahe 1971	A geodetic datum first defined in 1971 suitable for use in Seychelles - Mahe Island. Mahe 1971 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Mahe 1971 origin is Fundamental point: Station SITE. Latitude: 440'14.644S, longitude: 5528'44.488E (of Greenwich). Mahe 1971 is a geodetic datum for US military survey. It was defined by information from Clifford Mugnier's September 2007 PE,RS Grids and Datums article on Seychelles (www.asprs.org/resources/grids/). South East Island 1943 (datum code 1138) used for topographic mapping, cadastral and hydrographic survey.
66	Massawa	A geodetic datum suitable for use in Eritrea - onshore and offshore. Massawa references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Massawa is a geodetic datum for Topographic mapping.
67	Merchich	A geodetic datum first defined in 1922 suitable for use in Morocco - onshore. Merchich references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Merchich origin is Fundamental point: Merchich. Latitude: 3326'59.672N, longitude: 733'27.295W (of Greenwich). Merchich is a geodetic datum for Topographic mapping.
68	Midway Astro 1961	A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands - Midway Islands - Sand Island and Eastern Island. Midway 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Midway 1961 is a geodetic datum for Military and topographic mapping. It was defined by information

		from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
69	Minna	A geodetic datum suitable for use in Nigeria - onshore and offshore. Minna references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Minna origin is Fundamental point: Minna base station L40. Latitude: 938'08.87N, longitude: 630'58.76E (of Greenwich). Minna is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
70	Montserrat Island Astro 1958	A geodetic datum first defined in 1958 suitable for use in Montserrat - onshore. Montserrat 1958 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Montserrat 1958 origin is Fundamental point: station M36. Montserrat 1958 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.
71	M'poraloko	A geodetic datum suitable for use in Gabon - onshore and offshore. M'poraloko references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. M'poraloko is a geodetic datum for Topographic mapping.
72	Nahrwan	A geodetic datum first defined in 1934 suitable for use in Iraq - onshore; Iran - onshore northern Gulf coast and west bordering southeast Iraq. Nahrwan 1934 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1934 origin is Fundamental point: Nahrwan south base. Latitude: 3319'10.87N, longitude: 4443'25.54E (of Greenwich). Nahrwan 1934 is a geodetic datum for Oil exploration and production. It was defined by information from Various industry sources. This adjustment later discovered to have a significant orientation error. In Iran replaced by FD58. In Iraq, replaced by Karbala 1979.
73	Naparima, BWI	A geodetic datum first defined in 1972 suitable for use in Trinidad and Tobago - Tobago - onshore. Naparima 1972 references the International 1924 ellipsoid and the Greenwich prime meridian. Naparima 1972 origin is Fundamental point: Naparima. Latitude: 1016'44.860N, longitude: 6127'34.620W (of Greenwich). Naparima 1972 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Naparima 1972 is an extension of the Naparima 1955 network of Trinidad to include Tobago.
74	North American 1927	A geodetic datum first defined in 1927 suitable for use in North and central America; Antigua and Barbuda; Bahamas; Belize; British Virgin Islandss. Usage shall be onshore only except that onshore and offshore shall apply to Canada east coast (New Brunswick;

		Newfoundland and Labrador; Prince Edward Island; Quebec). Cuba. Mexico (Gulf of Mexico and Caribbean coasts only). USA Alaska. USA Gulf of Mexico (Alabama; Florida; Louisiana; Mississippi; Texas). USA East Coast. Bahamas onshore plus offshore over internal continental shelf only. North American Datum 1927 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. North American Datum 1927 origin is Fundamental point: Meade's Ranch. Latitude: 3913'26.686N, longitude: 9832'30.506W (of Greenwich). North American Datum 1927 is a geodetic datum for Topographic mapping. In United States (USA) and Canada, replaced by North American Datum 1983 (NAD83) (code 6269) ; in Mexico, replaced by Mexican Datum of 1993 (code 1042).
75	North American 1983	A geodetic datum first defined in 1986 suitable for use in North America - onshore and offshore: Canada; Puerto Rico; United States (USA); US Virgin Islands; British Virgin Islands. North American Datum 1983 references the GRS 1980 ellipsoid and the Greenwich prime meridian. North American Datum 1983 origin is Origin at geocentre. North American Datum 1983 is a geodetic datum for Topographic mapping. Although the 1986 adjustment included connections to Greenland and Mexico, it has not been adopted there. In Canada and US, replaced NAD27.
76	Observatorio Meteorologico 1939	A geodetic datum first defined in 1939 suitable for use in Portugal - western Azores onshore - Flores, Corvo. Azores Occidental Islands 1939 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Occidental Islands 1939 origin is Fundamental point: Observatorio Meteorologico Flores. Azores Occidental Islands 1939 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon via EuroGeographics; http://crs.bkg.bund.de/crs-eu/.
77	Old Egyptian 1907	A geodetic datum first defined in 1907 suitable for use in Egypt - onshore and offshore. Egypt 1907 references the Helmert 1906 ellipsoid and the Greenwich prime meridian. Egypt 1907 origin is Fundamental point: Station F1 (Venus). Latitude: 3001'42.86N, longitude: 3116'33.60E (of Greenwich). Egypt 1907 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey.
78	Old Hawaiian	A geodetic datum suitable for use in United States (USA) - Hawaii - main islands onshore. Old Hawaiian references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Old Hawaiian origin is Fundamental point: Oahu West Base Astro. Latitude: 2118'13.89N, longitude 15750'55.79W (of Greenwich). Old Hawaiian is a geodetic datum for Topographic

		mapping. It was defined by information from http://www.ngs.noaa.gov/ (NADCON readme file). Hawaiian Islands were never on NAD27 but rather on Old Hawaiian Datum. NADCON conversion program provides transformation from Old Hawaiian Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.
79	Oman	A geodetic datum first defined in 2013 suitable for use in Oman - onshore and offshore. Oman National Geodetic Datum 2014 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Oman National Geodetic Datum 2014 origin is 20 stations of the Oman primary network tied to ITRF2008 at epoch 2013.15. Oman National Geodetic Datum 2014 is a geodetic datum for Geodetic Survey. It was defined by information from National Survey Authority, Sultanate of Oman. Replaces WGS 84 (G874).
80	Ordnance Survey of Great Britain 1936	A geodetic datum first defined in 1936 suitable for use in United Kingdom (UK) - offshore to boundary of UKCS within 4946'N to 6101'N and 733'W to 333'E; onshore Great Britain (England, Wales and Scotland). Isle of Man onshore. OSGB 1936 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSGB 1936 origin is Prior to 2002, fundamental point: Herstmonceux, Latitude: 5051'55.271N, longitude: 020'45.882E (of Greenwich). From April 2002 the datum is defined through the application of the OSTN transformation from ETRS89. OSGB 1936 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain. The average accuracy of OSTN compared to the old triangulation network (down to 3rd order) is 0.1m. With the introduction of OSTN15, the area for OSGB 1936 has effectively been extended from Britain to cover the adjacent UK Continental Shelf.
81	Pico de las Nieves	A geodetic datum suitable for use in Spain - Canary Islands onshore. Pico de las Nieves 1984 references the International 1924 ellipsoid and the Greenwich prime meridian. Pico de las Nieves 1984 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaces Pico de las Nieves 1968 (PN68). Replaced by REGCAN95.
82	Pitcairn Astro 1967	A geodetic datum first defined in 1967 suitable for use in Pitcairn - Pitcairn Island. Pitcairn 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Pitcairn 1967 origin is Fundamental point: Pitcairn Astro. Latitude: 2504'06.87S, longitude: 13006'47.83W (of Greenwich). Pitcairn 1967 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA

		TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaced by Pitcairn 2006.
83	Point 58	A geodetic datum first defined in 1969 suitable for use in Senegal - central, Mali - southwest, Burkina Faso - central, Niger - southwest, Nigeria - north, Chad - central. All in proximity to the parallel of latitude of 12N. Point 58 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Point 58 origin is Fundamental point: Point 58. Latitude: 1252'44.045N, longitude: 358'37.040E (of Greenwich). Point 58 is a geodetic datum for Geodetic survey. It was defined by information from IGN Paris. Used as the basis for computation of the 12th Parallel traverse conducted 1966-70 from Senegal to Chad and connecting to the Blue Nile 1958 (Adindan) triangulation in Sudan.
84	Pointe Noire 1948	Pointe Noire 1948 datum.
85	Porto Santo 1936	A geodetic datum first defined in 1936 suitable for use in Portugal - Madeira, Porto Santo and Desertas islands - onshore. Porto Santo 1936 references the International 1924 ellipsoid and the Greenwich prime meridian. Porto Santo 1936 origin is SE Base on Porto Santo island. Porto Santo 1936 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt Replaced by 1995 adjustment (datum code 6663). For Selvagens see Selvagem Grande (code 6616).
86	Provisional South American 1956	A geodetic datum first defined in 1956 suitable for use in Aruba - onshore; Bolivia; Bonaire - onshore; Brazil - offshore - Amazon Cone shelf; Chile - onshore north of 4330'S; Curacao - onshore; Ecuador - mainland onshore; Guyana - onshore; Peru - onshore; Venezuela - onshore. Provisional South American Datum 1956 references the International 1924 ellipsoid and the Greenwich prime meridian. Provisional South American Datum 1956 origin is Fundamental point: La Canoa. Latitude: 834'17.170N, longitude: 6351'34.880W (of Greenwich). Provisional South American Datum 1956 is a geodetic datum for Topographic mapping. Same origin as La Canoa datum.
87	Provisional South Chilean 1963	A geodetic datum first defined in 1963 suitable for use in Argentina and Chile - Tierra del Fuego, onshore. Hito XVIII 1963 references the International 1924 ellipsoid and the Greenwich prime meridian. Hito XVIII 1963 origin is Chile-Argentina boundary survey. Hito XVIII 1963 is a geodetic datum for Geodetic survey. It was defined by information from Various oil company records. Used in Tierra del Fuego.
88	Puerto Rico	A geodetic datum first defined in 1901 suitable for use in Puerto Rico, US Virgin

		Islands and British Virgin Islands - onshore. Puerto Rico references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Puerto Rico origin is Fundamental point: Cardona Island Lighthouse. Latitude:1757'31.40N, longitude: 6638'07.53W (of Greenwich). Puerto Rico is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain and http://www.ngs.noaa.gov/ (NADCON readme file). NADCON conversion program provides transformation from Puerto Rico Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.
89	Qatar National	A geodetic datum first defined in 1995 suitable for use in Qatar - onshore. Qatar National Datum 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Qatar National Datum 1995 origin is defined by transformation from WGS 84 - see coordinate operation code 1840. Qatar National Datum 1995 is a geodetic datum for Topographic mapping. It was defined by information from Qatar Centre for Geographic Information.
90	Qornoq	A geodetic datum first defined in 1927 suitable for use in Greenland - west coast onshore. Qornoq 1927 references the International 1924 ellipsoid and the Greenwich prime meridian. Qornoq 1927 origin is Fundamental point: Station 7008. Latitude: 6431'06.27N, longitude: 5112'24.86W (of Greenwich). Qornoq 1927 is a geodetic datum for Topographic mapping. It was defined by information from Kort, Matrikelstyrelsen, Copenhagen. Origin coordinates from NIMA http://earth-info.nima.mil/.
91	Reunion	A geodetic datum first defined in 1947 suitable for use in Reunion - onshore. Reunion 1947 references the International 1924 ellipsoid and the Greenwich prime meridian. Reunion 1947 origin is Fundamental point: Piton des Neiges (Borne). Latitude: 2105'13.119S, longitude: 5529'09.193E (of Greenwich). Reunion 1947 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris. Replaced by RGR92 (datum code 6627).
92	Rome 1940	A geodetic datum first defined in and is suitable for use in Italy - onshore and offshore; San Marino, Vatican City State. Monte Mario (Rome) references the International 1924 ellipsoid and the Rome prime meridian. Monte Mario (Rome) origin is Fundamental point: Monte Mario. Latitude: 4155'25.51N, longitude: 000' 00.00E (of Rome). Monte Mario (Rome) is a geodetic datum for Topographic mapping. Replaced Genova datum, Bessel 1841 ellipsoid, from 1940.

93	Santo (DOS) 1965	A geodetic datum first defined in 1965 suitable for use in Vanuatu - northern islands - Aese, Ambrym, Aoba, Epi, Espiritu Santo, Maewo, Malo, Malkula, Paama, Pentecost, Shepherd and Tutuba. Santo 1965 references the International 1924 ellipsoid and the Greenwich prime meridian. Santo 1965 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.
94	Sao Braz	A geodetic datum first defined in 1995 suitable for use in Portugal - eastern Azores onshore - Sao Miguel, Santa Maria, Formigas. Azores Oriental Islands 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Oriental Islands 1995 origin is Fundamental point: Forte de So Bras. Origin and orientation constrained to those of the 1940 adjustment. Azores Oriental Islands 1995 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon; http://www.igeo.pt/ Classical and GPS observations. Replaces 1940 adjustment (datum code 6184).
95	Sapper Hill 1943	A geodetic datum first defined in 1943 suitable for use in Falkland Islands (Malvinas) - onshore. Sapper Hill 1943 references the International 1924 ellipsoid and the Greenwich prime meridian. Sapper Hill 1943 is a geodetic datum for Topographic mapping.
96	Schwarzeck	A geodetic datum suitable for use in Namibia - onshore and offshore. Schwarzeck references the Bessel Namibia (GLM) ellipsoid and the Greenwich prime meridian. Schwarzeck origin is Fundamental point: Schwarzeck. Latitude: 2245'35.820S, longitude: 1840'34.549E (of Greenwich). Fixed during German South West Africa-British Bechuanaland boundary survey of 1898-1903. Schwarzeck is a geodetic datum for Topographic mapping. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.
97	Selvagem Grande 1938	A geodetic datum suitable for use in Portugal - Selvagens islands (Madeira province) - onshore. Selvagem Grande references the International 1924 ellipsoid and the Greenwich prime meridian. Selvagem Grande is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt .
98	South American 1969	A geodetic datum first defined in 1969 suitable for use in Brazil - onshore and offshore. In rest of South America - onshore north of approximately 45S and

		Tierra del Fuego. South American Datum 1969 references the GRS 1967 Modified ellipsoid and the Greenwich prime meridian. South American Datum 1969 origin is Fundamental point: Chua. Geodetic latitude: 1945'41.6527S; geodetic longitude: 4806'04.0639W (of Greenwich). (Astronomic coordinates: Latitude 1945'41.34S +/- 0.05, longitude 4806'07.80W +/- 0.08). South American Datum 1969 is a geodetic datum for Topographic mapping. It was defined by information from DMA 1974. SAD69 uses GRS 1967 ellipsoid but with 1/f to exactly 2 decimal places. In Brazil only, replaced by SAD69(96) (datum code 1075).
99	South Asia	South Asia datum.
100	Tananarive Observatory 1925	A geodetic datum first defined in 1925 suitable for use in Madagascar - onshore and nearshore. Tananarive 1925 references the International 1924 ellipsoid and the Greenwich prime meridian. Tananarive 1925 origin is Fundamental point: Tananarive observatory. Latitude: 1855'02.10S, longitude: 4733'06.75E (of Greenwich). Tananarive 1925 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.
101	Timbalai 1948	A geodetic datum first defined in 1948 suitable for use in Brunei - onshore and offshore; Malaysia - East Malaysia (Sabah; Sarawak) - onshore and offshore. Timbalai 1948 references the Everest 1830 (1967 Definition) ellipsoid and the Greenwich prime meridian. Timbalai 1948 origin is Fundamental point: Station P85 at Timbalai. Latitude: 517' 3.548N, longitude: 11510'56.409E (of Greenwich). Timbalai 1948 is a geodetic datum for Topographic mapping. It was defined by information from Defence Geographic Centre. In 1968, the original adjustment was densified in Sarawak and extended to Sabah.
102	Tokyo	A geodetic datum first defined in 1918 suitable for use in Japan - onshore; North Korea - onshore; South Korea - onshore. Tokyo references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Tokyo origin is Fundamental point: Nikon-Keido-Genten. Latitude: 3539'17.5148N, longitude: 13944'40.5020E (of Greenwich). Longitude derived in 1918. Tokyo is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Geographic Survey Institute; Japan; Bulletin 40 (March 1994). Also http://vldb.gsi.go.jp/sokuchi/datum/tokyo datum.html. In Japan, replaces Tokyo 1892 (code 1048) and replaced by Japanese Geodetic Datum 2000 (code 6611). In Korea used only for geodetic applications before being replaced by Korean 1985 (code 6162).

103	Tristan Astro 1968	A geodetic datum first defined in 1968 suitable for use in St Helena, Ascension and Tristan da Cunha - Tristan da Cunha island group including Tristan, Inaccessible, Nightingale, Middle and Stoltenhoff Islands. Tristan 1968 references the International 1924 ellipsoid and the Greenwich prime meridian. Tristan 1968 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
104	Viti Levu 1916	A geodetic datum first defined in 1912 suitable for use in Fiji - Viti Levu island. Viti Levu 1912 references the Clarke 1880 (international foot) ellipsoid and the Greenwich prime meridian. Viti Levu 1912 Latitude origin was obtained astronomically at station Monavatu = 1753'28.285S, longitude origin was obtained astronomically at station Suva = 17825'35.835E. Viti Levu 1912 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Clifford J. Mugnier in Photogrammetric Engineering and Remote Sensing, October 2000, www.asprs.org . For topographic mapping, replaced by Fiji 1956. For other purposes, replaced by Fiji 1986.
105	Wake-Eniwetok 1960	A geodetic datum first defined in 1960 suitable for use in Marshall Islands - onshore. Wake atoll onshore. Marshall Islands 1960 references the Hough 1960 ellipsoid and the Greenwich prime meridian. Marshall Islands 1960 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
106	Wake Island Astro 1952	A geodetic datum first defined in 1952 suitable for use in Wake atoll - onshore. Wake Island 1952 references the International 1924 ellipsoid and the Greenwich prime meridian. Wake Island 1952 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
107	Yacare	A geodetic datum suitable for use in Uruguay - onshore. Yacare references the International 1924 ellipsoid and the Greenwich prime meridian. Yacare origin is Fundamental point: Yacare. Latitude: 3035'53.68S, longitude: 5725'01.30W (of Greenwich). Yacare is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/
108	Zanderij	A geodetic datum suitable for use in Suriname - onshore and offshore. Zanderij references the International 1924 ellipsoid and the Greenwich prime

		meridian. Zanderij is a geodetic datum for Topographic mapping.
109	American Samoa 1962	A geodetic datum first defined in 1962 suitable for use in American Samoa - Tutuila, Aunu'u, Ofu, Olesega and Ta'u islands. American Samoa 1962 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. American Samoa 1962 origin is Fundamental point: Betty 13 eccentric. Latitude: 1420'08.34S, longitude: 17042'52.25W (of Greenwich). American Samoa 1962 is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 revision of January 2000. Oil industry sources for origin description details.
110	Deception Island	A geodetic datum suitable for use in Antarctica - South Shetland Islands - Deception Island. Deception Island references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Deception Island is a geodetic datum for Military and scientific mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
111	Indian 1960	A geodetic datum suitable for use in Cambodia - onshore; Vietnam - onshore and offshore Cuu Long basin. Indian 1960 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1960 origin is DMA extension over IndoChina of the Indian 1954 network adjusted to better fit local geoid. Indian 1960 is a geodetic datum for Topographic mapping. Also known as Indian (DMA Reduced).
112	Indonesian 1974	A geodetic datum first defined in 1974 suitable for use in Indonesia - onshore. Indonesian Datum 1974 references the Indonesian National Spheroid ellipsoid and the Greenwich prime meridian. Indonesian Datum 1974 origin is Fundamental point: Padang. Latitude: 056'38.414S, longitude: 10022' 8.804E (of Greenwich). Ellipsoidal height 3.190m, gravity-related height 14.0m above mean sea level. Indonesian Datum 1974 is a geodetic datum for Topographic mapping. It was defined by information from Bakosurtanal 1979 paper by Jacob Rais. Replaced by DGN95.
113	North Sahara 1959	A geodetic datum first defined in 1959 suitable for use in Algeria - onshore and offshore. Nord Sahara 1959 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nord Sahara 1959 origin is Coordinates of primary network readjusted on ED50 datum and then transformed conformally to Clarke 1880 (RGS) ellipsoid. Nord Sahara 1959 is a geodetic datum for Topographic mapping. It was defined by information from Le System Geodesique Nord-Sahara; IGN Paris Adjustment includes Morocco and Tunisia but use only in Algeria. Within Algeria the adjustment is north of 32N but use has been extended

		southwards in many disconnected projects, some based on independent astro stations rather than the geodetic network.
114	Pulkovo 1942	A geodetic datum first defined in 1942 suitable for use in Armenia; Azerbaijan; Belarus; Estonia - onshore; Georgia - onshore; Kazakhstan; Kyrgyzstan; Latvia - onshore; Lithuania - onshore; Moldova; Russian Federation - onshore; Tajikistan; Turkmenistan; Ukraine - onshore; Uzbekistan. Pulkovo 1942 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Pulkovo 1942 origin is Fundamental point: Pulkovo observatory. Latitude: 5946'18.550N, longitude: 3019'42.090E (of Greenwich). Pulkovo 1942 is a geodetic datum for Topographic mapping.
116	S-JTSK	A geodetic datum suitable for use in Czech Republic; Slovakia. System Jednotne Trigonometricke Site Katastralni references the Bessel 1841 ellipsoid and the Greenwich prime meridian. System Jednotne Trigonometricke Site Katastralni origin is Modification of Austrian MGI datum, code 6312. System Jednotne Trigonometricke Site Katastralni is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Research Institute for Geodesy Topography and Cartography (VUGTK); Prague. S-JTSK = System of the Unified Trigonometrical Cadastral Network.
117	Voirol 1950	Voirol 1950 datum.
118	Average Terrestrial System 1977	A geodetic datum first defined in 1977 suitable for use in Canada - New Brunswick; Nova Scotia; Prince Edward Island. Average Terrestrial System 1977 references the Average Terrestrial System 1977 ellipsoid and the Greenwich prime meridian. Average Terrestrial System 1977 is a geodetic datum for Topographic mapping. It was defined by information from New Brunswick Geographic Information Corporation land and water information standards manual. In use from 1979.
119	Compensation Geodesique du Quebec 1977	Compensation Geodesique du Quebec 1977.
120	Finnish (KKJ)	A geodetic datum first defined in 1966 suitable for use in Finland - onshore. Kartastokoordinaattijarjestelma (1966) references the International 1924 ellipsoid and the Greenwich prime meridian. Kartastokoordinaattijarjestelma (1966) origin is Adjustment with fundamental point SF31 based on ED50 transformed to best fit the older VVJ adjustment. Kartastokoordinaattijarjestelma (1966) is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by

		information from National Land Survey of Finland; http://www.maanmittauslaitos.fi . Adopted in 1970.
121	Ordnance Survey of Ireland	A geodetic datum first defined in 1952 suitable for use in United Kingdom (UK) - Northern Ireland (Ulster) - onshore. OSNI 1952 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSNI 1952 origin is Position fixed to the coordinates from the 19th century Principle Triangulation of station Divis. Scale and orientation controlled by position of Principle Triangulation stations Knocklayd and Trostan. OSNI 1952 is a geodetic datum for Geodetic survey and topographic mapping. It was defined by information from Ordnance Survey of Northern Ireland. Replaced by Geodetic Datum of 1965 alias 1975 Mapping Adjustment or TM75 (datum code 6300).
122	Revised Kertau	A geodetic datum first defined in 1969 suitable for use in Malaysia - West Malaysia; Singapore. Kertau (RSO) references the Everest 1830 (RSO 1969) ellipsoid and the Greenwich prime meridian. Kertau (RSO) is a geodetic datum for Metrication of RSO grid. It was defined by information from Defence Geographic Centre. Adopts metric conversion of 0.914398 metres per yard exactly. This is a truncation of the Sears 1922 ratio.
123	Revised Nahrwan	A geodetic datum first defined in 1967 suitable for use in Arabian Gulf; Qatar - offshore; United Arab Emirates (UAE) - Abu Dhabi; Dubai; Sharjah; Ajman; Fujairah; Ras Al Kaimah; Umm Al Qaiwain - onshore and offshore. Nahrwan 1967 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1967 origin is Fundamental point: Nahrwan south base. Latitude: 3319'10.87N, longitude: 4443'25.54E (of Greenwich). Nahrwan 1967 is a geodetic datum for Topographic mapping. In Iraq, replaces Nahrwan 1934.
124	GGRS 76 (Greece)	A geodetic datum first defined in 1987 suitable for use in Greece - onshore. Greek Geodetic Reference System 1987 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Greek Geodetic Reference System 1987 origin is Fundamental point: Dionysos. Latitude 3804'33.8N, longitude 2355'51.0E of Greenwich; geoid height 7.0 m. Greek Geodetic Reference System 1987 is a geodetic datum for Topographic mapping. It was defined by information from L. Portokalakis; Public Petroleum Corporation of Greece. Replaced (old) Greek datum. Oil industry work based on ED50.
125	Nouvelle Triangulation de France	A geodetic datum first defined in 1895 suitable for use in France - onshore - mainland and Corsica. Nouvelle Triangulation Francaise references the Clarke 1880 (IGN) ellipsoid and the

		Greenwich prime meridian. Nouvelle Triangulation Francaise origin is Fundamental point: Pantheon. Latitude: 4850'46.522N, longitude: 220'48.667E (of Greenwich). Nouvelle Triangulation Francaise is a geodetic datum for Topographic mapping.
126	RT 90 (Sweden)	A geodetic datum first defined in 1982 suitable for use in Sweden - onshore and offshore. Rikets koordinatsystem 1990 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Rikets koordinatsystem 1990 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Survey of Sweden Replaces RT38 adjustment (datum code 6308).
127	Geocentric Datum of Australia	A geodetic datum first defined in 1994 suitable for use in Australia including Lord Howe Island, Macquarie Islands, Ashmore and Cartier Islands, Christmas Island, Cocos (Keeling) Islands, Norfolk Island. All onshore and offshore. Geocentric Datum of Australia 1994 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Geocentric Datum of Australia 1994 origin is ITRF92 at epoch 1994.0. Geocentric Datum of Australia 1994 is a geodetic datum for Topographic mapping, geodetic survey. It was defined by information from Australian Surveying and Land Information Group Internet WWW page. http://www.auslig.gov.au/geodesy/datum/s/gda.htm#specs Coincident with WGS84 to within 1 metre.
128	BJZ54 (A954 Beijing Coordinates)	A geodetic datum first defined in 1954 suitable for use in China - onshore. Beijing 1954 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Beijing 1954 origin is Pulkovo, transferred through Russian triangulation. Beijing 1954 is a geodetic datum for Topographic mapping. It was defined by information from Chinese Science Bulletin, 2009, 54:2714-2721 Scale determined through three baselines in northeast China. Discontinuities at boundaries of adjustment blocks. From 1982 replaced by Xian 1980 and New Beijing.
129	Modified BJZ54	Modified BJZ54 datum.
130	GDZ80	GDZ80 datum.
131	Local Datum	An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.

5.68 Horizontal Distance Uncertainty

Type: SimpleAttributeType

Definition: The best estimate of the horizontal accuracy of horizontal clearances and distances.

CamelCase: horizontalDistanceUncertainty

Value type: real

5.69 Horizontal Length

Type: SimpleAttributeType

Definition: A measurement of the longer of two linear axis.

CamelCase: horizontalLength

Value type: real

5.70 Horizontal Width

Type: SimpleAttributeType

Definition: A measurement of the shorter of two linear axis.

CamelCase: horizontalWidth

Value type: real

5.71 ID Code

Type: SimpleAttributeType

Definition: Identification code as specified in predefined system. Also called identification number.

CamelCase: iDCode

Value type: text

5.72 Information in National Language

Type: SimpleAttributeType

Definition: Textual information in national language characters.

CamelCase: informationInNationalLanguage

Value type: text

5.73 Inspection Frequency

Type: SimpleAttributeType

Definition: A statement of how frequently an item is inspected.

CamelCase: inspectionFrequency

Value type: text

5.74 Inspection Requirements

Type: SimpleAttributeType

Definition: A statement of what requirements are in place for how inspection of an item is carried out.

CamelCase: inspectionRequirements

Value type: text

5.75 Installation Date

Type: SimpleAttributeType

Definition: The date when an item was installed.

CamelCase: installationDate

Value type: date

5.76 IsSlatted

Type: SimpleAttributeType

Definition: -

CamelCase: isSlatted

Value type: boolean

5.77 Language

Type: SimpleAttributeType

Definition: The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way.

CamelCase: language

Value type: text

5.78 Legs Details

Type: SimpleAttributeType

Definition: -

CamelCase: legsDetails

Value type: text

5.79 Light Characteristic

Type: EnumerationType

Definition: The distinct character, such as fixed, flashing, or occulting, which is given to each light to avoid confusion with neighbouring ones.

CamelCase: lightCharacteristic

Value type: enumeration

Listed values:

Code	Label	Definition
1	Fixed	A signal light that shows continuously, in any given direction, with constant luminous intensity and colour.
2	Flashing	A rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
3	Long-Flashing	A single-flashing light in which a single flash of not less than two seconds duration is regularly repeated.
4	Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 50 flashes per minutes but less than 80 flashes per minutes. It may be: - Continuous quick-flashing: A quick-flashing light in which a flash is regularly repeated. - Group quick-flashing: A quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
5	Very Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 80 flashes per minute but less than 160 flashes per minute. It may be:- Continuous very quick-flashing: A very

		quick-flashing light in which a flash is regularly repeated.- Group very quick-flashing: A very quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
6	Continuous Ultra Quick-Flashing	A rhythmic light in which flashes are regularly repeated at a rate of not less than 160 flashes per minute.
7	Isophased	A light with all durations of light and darkness equal.
8	Occulting	A rhythmic light in which the total duration of light in a period is clearly longer than the total duration of darkness and all the eclipses are of equal duration. It may be: - Single-occulting: An occulting light in which an eclipse is regularly repeated. - Group-occulting: An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated. - Composite group-occulting: An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
12	Morse	A rhythmic light in which appearances of light of two clearly different durations are grouped to represent a character or characters in the Morse code.
13	Fixed and Flash	A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity.
14	Flash and Long-Flash	A rhythmic light in which a flashing light is combined with a long-flashing light of higher luminous intensity.
15	Occulting and Flash	A rhythmic light in which an occulting light is combined with a flashing light of higher luminous intensity.
16	Fixed and Long-Flash	A rhythmic light in which a fixed light is combined with a long-flashing light of higher luminous intensity.
17	Occulting Alternating	An alternating light in which the total duration of light in each period is clearly longer than the total duration of darkness and in which the intervals of darkness (occultations) are all of equal duration.
18	Long-Flash Alternating	An alternating single-flashing light in which an appearance of light of not less than two seconds duration is regularly repeated.
19	Flash Alternating	An alternating rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.

20	Group Alternating	Occulting light in which the occultations are combined in groups, each group including the same number of occultations, and in which the groups are repeated at regular intervals.
25	Quick-Flash Plus Long-Flash	A rhythmic light in which a group of quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
26	Very Quick-Flash Plus Long-Flash	A rhythmic light in which a group of very quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
27	Ultra Quick-Flash Plus Long-Flash	A rhythmic light in which a group of ultra quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
28	Alternating	A signal light that shows, in any given direction, two or more colours in a regularly repeated sequence with a regular periodicity.
29	Fixed and Alternating Flashing	A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity and different colour.
30	Group-Occulting Light	An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated.
31	Composite Group-Occulting Light	An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
32	Group flashing light	A flashing light in which a group of flashes, specified in number, is regularly repeated.
33	Composite Group-Flashing Light	A flashing light in which a sequence of groups of one or more flashes, which are specified in number, is regularly repeated, and the groups comprise different numbers of flashes.
34	Group Quick Light	A light in which flashes are combined in groups including the same number of quick flashes (repetition rate : 50-79 per minute) and in which groups are repeated at regular intervals.
35	Group Very Quick Light	A light in which very quick flashes are combined in groups including the same number of flashes (repetition rate : 80-159 per minute) and in which groups are repeated at regular intervals.

5.80 Light Visibility

Type: EnumerationType

Definition: The specific visibility of a light, with respect to the light's intensity and ease of recognition.

CamelCase: lightVisibility

Value type: enumeration

Listed values:

Code	Label	Definition
1	High Intensity	Non-marine lights with a higher power than marine lights and visible from well off shore (often 'Aero' lights).
2	Low Intensity	Non-marine lights with lower power than marine lights.
3	Faint	A decrease in the apparent intensity of a light which may occur in the case of partial obstructions.
4	Intensified	A light in a sector is intensified (that is, has longer range than other sectors).
5	Unintensified	A light in a sector is unintensified (that is, has shorter range than other sectors).
6	Visibility Deliberately Restricted	A light sector is deliberately reduced in intensity, for example to reduce its effect on a built-up area.
7	Obscured	Said of the arc of a light sector designated by its limiting bearings in which the light is not visible from seaward.
8	Partially Obscured	This value specifies that parts of the sector are obscured.
9	Visible in Line of Range	Lights that must in line to be visible.

5.81 Lighted Aton Change

Type: EnumerationType

Definition: Updates or modifications to light-emitting AtoNs, including changing light characteristics, intensity, or operational status.

CamelCase: lightedAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Light Unlit	The light is extinguished.
2	Light Unreliable	The light is unreliable due to technical problems.
3	Light Re-Establishment	The re-establishment of a light which was previously announced as either destroyed or temporarily removed.
4	Light Range Reduced	The nominal range of the light is less than the advertised range.
5	Light Without Rhythm	Due to technical problems the light has no more rhythm and is in fixed light mode.

6	Light Out Of Synchronization	The light is no longer synchronized with another light or group of lights.
7	Light Daymark Unreliable	The light daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
8	Light Operating Properly	The light is operating as advertised
9	Sector light Sector obscured	.
10	Front leading/range light Unlit	.
11	Rear leading/range light Unlit	.
12	Front leading/range light Unreliable	.
13	Rear leading/range light Unreliable	.
14	Front leading/range light Light range reduced	.
15	Rear leading/range light Light range reduced	.
16	Front leading/range light without rhythm	.
17	Rear leading/range light without rhythm	.
18	Leading/range lights out of synchronization	.
19	Front leading/range beacon Unreliable	.
20	Rear leading/range beacon Unreliable	.
21	Front leading/range light is operating properly	.
22	Rear leading/range light is operating properly	.
23	Front leading/range beacon restored to normal	.
24	Rear leading/range beacon restored to normal	.

5.82 MMSI Code

Type: SimpleAttributeType

Definition: The Maritime Mobile Service Identity (MMSI) Code is formed of a series of nine digits which are transmitted over the radio path in order to uniquely identify ship stations, ship earth stations, coast stations, coast earth stations, and group calls. These identities are formed in such a way that the identity or part thereof can be used by telephone and telex subscribers connected to the general telecommunications network principally to call ships automatically.

CamelCase: mmsiCode

Value type: text

5.83 Major Light

Type: SimpleAttributeType

Definition: A statement expressing if a light is considered to be a major light in terms of ECDIS display in a particular area.

CamelCase: majorLight

Value type: boolean

5.84 Manned Structure

Type: SimpleAttributeType

Definition: An expression of the feature being permanently manned or not.

CamelCase: mannedStructure

Value type: boolean

5.85 Manufacturer

Type: SimpleAttributeType

Definition: -

CamelCase: manufacturer

Value type: text

5.86 Marks Navigational - System Of

Type: EnumerationType

Definition: The system of navigational buoyage a region complies with.

CamelCase: marksNavigationalSystemOf

Value type: enumeration

Listed values:

Code	Label	Definition
1	IALA A	Navigational aids conform to the International Association of Lighthouse Authorities - IALA A system.
2	IALA B	Navigational aids conform to the International Association of Lighthouse Authorities - IALA B system.
9	No System	Navigational aids do not conform to any defined system.
10	Other System	Navigational aids conform to a defined system other than International Association of Lighthouse Authorities - IALA.
11	CEVNI	CEVNI (European Code for Navigation on Inland Waterways) is the European code for rivers, canals land lakes in most of Europe.
12	Russian Inland Waterway Regulations	Navigational aids conform to the Russian inland waterway regulations.
13	Brazilian National Inland Waterway Regulations - Two Sides	Navigational aids conform to the Brazilian national inland waterway regulations for two sides.
14	Brazilian National Inland Waterway Regulations - Side Independent	Navigational aids conform to the Brazilian national inland waterway regulations - side independent.

15	Paraguay-Parana Waterway - Brazilian Complementary Aids	Navigational aids conform to the Brazilian complementary aids on the Paraguay-Parana waterway.
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5.87 Maximal Permitted Draught

Type: SimpleAttributeType

Definition: The maximal permitted draught of a vessel or convoy according to the particular article/clause of the applicable law/regulation.

CamelCase: maximalPermittedDraught

Value type: real

5.88 Maximum Display Scale

Type: SimpleAttributeType

Definition: The largest intended viewing scale for the data.

CamelCase: maximumDisplayScale

Value type: integer

5.89 Minimum Display Scale

Type: SimpleAttributeType

Definition: The smallest intended viewing scale for the data.

CamelCase: minimumDisplayScale

Value type: integer

5.90 Moire Effect

Type: SimpleAttributeType

Definition: A short range (up to 2km) type of directional light. Sodium lighting gives a yellow background to a screen on which a vertical black line will be seen by an observer on the centre line.

CamelCase: moireEffect

Value type: boolean

5.91 Multiplicity Known

Type: SimpleAttributeType

Definition: The number of features of identical character that exist as a co-located group is or is not known.

CamelCase: multiplicityKnown

Value type: boolean

5.92 Name

Type: SimpleAttributeType

Definition: The individual name of a feature.

CamelCase: name

Value type: text

5.93 Nature of Construction

Type: EnumerationType

Definition: The building's primary construction material.

CamelCase: natureOfConstruction

Value type: enumeration

Listed values:

Code	Label	Definition
1	Masonry	Constructed of stones or bricks, usually quarried, shaped, and mortared.
2	Concreted	Constructed of concrete, a material made of sand and gravel that is united by cement into a hardened mass used for roads, foundations, etc.
3	Loose Boulders	Constructed from large stones or blocks of concrete, often placed loosely for protection against waves or water turbulence.
4	Hard Surfaced	Constructed with a surface of hard material, usually a term applied to roads surfaced with asphalt or concrete.
5	Unsurfaced	Constructed with no extra protection, usually a term applied to roads not surfaced with a hard material.
6	Wooden	Constructed from wood.
7	Metal	Constructed from metal.
8	Glass Reinforced Plastic	Constructed from a plastic material strengthened with fibres of glass.
9	Painted	The application of paint to some other construction or natural feature.
10	Framework	Constructed from a lattice framework of, often diagonal, intersecting struts.
11	Latticed	A structure of crossed wooden or metal strips usually arranged to form a diagonal pattern of open spaces between the strips.
12	Glass	[1] Any artificial or natural substance having similar properties and composition, as fused borax, obsidian, or the like. [2] Something made of such a substance, as a windowpane.
13	Fiberglass	Constructed from fiberglass.
14	Plastic	Constructed from plastic.

5.94 Number of Features

Type: SimpleAttributeType

Definition: The number of features of identical character that exist as a co-located group.

CamelCase: numberOfFeatures

Value type: integer

5.95 Orientation Uncertainty

Type: SimpleAttributeType

Definition: The best estimate of the accuracy of a bearing.

CamelCase: orientationUncertainty

Value type: real

5.96 Orientation Value

Type: SimpleAttributeType

Definition: The angular distance measured from true north to the major axis of the feature.

CamelCase: orientationValue

Value type: real

5.97 Peak Intensity

Type: SimpleAttributeType

Definition: The maximum luminous intensity of a light during its flash cycle.

CamelCase: peakIntensity

Value type: real

5.98 Pictorial Representation

Type: SimpleAttributeType

Definition: Indicates whether a pictorial representation of the feature is available.

CamelCase: pictorialRepresentation

Value type: text

5.99 Positioning Device

Type: SimpleAttributeType

Definition: A tool or instrument used to determine the geographic location of an object or point, such as a GPS receiver, radar, or other location-determining technologies.

CamelCase: positioningDevice

Value type: text

5.100 Positioning Equipment

Type: EnumerationType

Definition: Devices and systems used to determine and maintain the correct position of an AtoN, including GPS, GNSS, and mooring systems.

CamelCase: positioningEquipment

Value type: enumeration

Listed values:

Code	Label	Definition
1	DGPS Receiver	.
2	GLONASS Receiver	.
3	GPS Receiver	.
4	GPS/WAAS Receiver	.

5.101 Positioning Procedure

Type: SimpleAttributeType

Definition: The method or process used to determine the location of an object or point in space. This may involve the use of technologies like GPS or triangulation methods.

CamelCase: positioningProcedure

Value type: text

5.102 Postal Code

Type: SimpleAttributeType

Definition: Known in various countries as a postcode, or ZIP code, the postal code is a series of letters and/or digits that identifies each postal delivery area.

CamelCase: postalCode

Value type: text

5.103 Product

Type: EnumerationType

Definition: The various substances which are transported, stored or exploited.

CamelCase: product

Value type: enumeration

Listed values:

Code	Label	Definition
1	Oil	A thick, slippery liquid that will not dissolve in water, usually petroleum based in the context of storage tanks.
2	Gas	A substance with particles that can move freely, usually a fuel substance in the context of storage tanks.
3	Water	A colourless, odourless, tasteless liquid that is a compound of hydrogen and oxygen.
4	Stone	A general term for rock and rock fragments ranging in size from pebbles and gravel to boulders or large rock masses.
5	Coal	A hard black mineral that is burned as fuel.
6	Ore	A solid rock or mineral from which metal is obtained.
7	Chemicals	Any substance obtained by or used in a chemical process.
8	Drinking Water	Water that is suitable for human consumption.
9	Milk	A white fluid secreted by female mammals as food for their young.
10	Bauxite	A mineral from which aluminum is obtained.
11	Coke	A solid substance obtained after gas and tar have been extracted from coal, used

		as a fuel.
12	Iron Ingots	An oblong lump of cast iron metal.
13	Salt	Sodium chloride obtained from mines or by the evaporation of sea water.
14	Sand	Loose material consisting of small but easily distinguishable, separate grains, between 0.0625 and 2.000 millimetres in diameter.
15	Timber	Wood prepared for use in building or carpentry.
16	Sawdust/Wood Chips	Powdery fragments of wood made in sawing timber or coarse chips produced for use in manufacturing pressed board.
17	Scrap Metal	Discarded metal suitable for being reprocessed.
18	Liquefied Natural Gas	Natural gas that has been liquefied for ease of transport by cooling the gas to -162 Celsius.
19	Liquefied Petroleum Gas	A compressed gas consisting of flammable light hydrocarbons and derived from petroleum.
20	Wine	The fermented juice of grapes.
21	Cement	A substance made of powdered lime and clay, mixed with water.
22	Grain	A small hard seed, especially that of any cereal plant such as wheat, rice, corn, rye etc.
23	Electricity	Electric charge or current.
24	Ice	Frozen water.
25	Clay	(Particles of less than 0.002mm); stiff, sticky earth that becomes hard when baked.

5.104 Quality of Horizontal Measurement

Type: EnumerationType

Definition: The degree of reliability attributed to a position.

CamelCase: qualityOfHorizontalMeasurement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Surveyed	The position(s) was(were) determined by the operation of making measurements for determining the relative position of points on, above or beneath the earth's surface. Survey implies a regular, controlled survey of any date.

2	Unsurveyed	Survey data is does not exist or is very poor.
3	Inadequately Surveyed	Position data is of a very poor quality.
4	Approximate	A position that is considered to be less than third-order accuracy, but is generally considered to be within 30.5 metres of its correct geographic location. Also may apply to an object whose position does not remain fixed.
5	Position Doubtful	Of uncertain position. The expression is used principally on charts to indicate that a wreck, shoal, etc., has been reported in various positions and not definitely determined in any.
6	Unreliable	A feature's position has been obtained from questionable or unreliable data.
7	Reported (Not Surveyed)	An object whose position has been reported and its position confirmed by some means other than a formal survey such as an independent report of the same object.
8	Reported (Not Confirmed)	An object whose position has been reported and its position has not been confirmed.
9	Estimated	The most probable position of an object determined from incomplete data or data of questionable accuracy.
10	Precisely Known	A position that is of a known value, such as the position of an anchor berth or other defined object.
11	Calculated	A position that is computed from data.

5.105 Quality of Vertical Measurement

Type: EnumerationType

Definition: The reliability of the value of a sounding.

CamelCase: qualityOfVerticalMeasurement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Depth Known	The depth from the chart datum to the seabed (or to the top of a drying feature) is known.
2	Depth or Least Depth Unknown	The depth from chart datum to the seabed, or the shoalest depth of the feature is unknown.
3	Doubtful Sounding	A depth that may be less than indicated.
4	Unreliable Sounding	A depth that is considered to be an unreliable value.

5	No Bottom Found at Value Shown	Upon investigation the bottom was not found at this depth.
6	Least Depth Known	The shoalest depth over a feature is of known value.
7	Least Depth Unknown, Safe Clearance at Value Shown	The least depth over a feature is unknown, but there is considered to be safe clearance at this depth.
8	Value Reported (Not Surveyed)	Depth value obtained from a report, but not fully surveyed.
9	Value Reported (Not Confirmed)	Depth value obtained from a report, which it has not been possible to confirm.
10	Maintained Depth	The depth at which a channel is kept by human influence, usually by dredging.
11	Not Regularly Maintained	Depths may be altered by human influence, but will not be routinely maintained.

5.106 Radar Band

Type: SimpleAttributeType

Definition: The band code character of the electromagnetic spectrum within which radar wave lengths lie.

CamelCase: radarBand

Value type: text

5.107 Radar Conspicuous

Type: SimpleAttributeType

Definition: A feature which returns a strong radar echo.

CamelCase: radarConspicuous

Value type: boolean

5.108 Reference Point

Type: SimpleAttributeType

Definition: Any point which can serve as a reference or base for the measurement of other quantities. Also called datum point.

CamelCase: referencePoint

Value type: text

5.109 Remote Monitoring System

Type: SimpleAttributeType

Definition: Classification or name of system used to remotely monitor a feature.

CamelCase: remoteMonitoringSystem

Value type: text

5.110 Scale Minimum

Type: SimpleAttributeType

Definition: The minimum scale at which the feature may be used for example for ECDIS presentation.

CamelCase: scaleMinimum

Value type: integer

5.111 Sector Bearing

Type: SimpleAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector bearing specifies the limit of the sector.

CamelCase: sectorBearing

Value type: real

5.112 Sector Extension

Type: SimpleAttributeType

Definition: Distance in screen millimetres (mm) by which a sector is extended from its origin.

CamelCase: sectorExtension

Value type: boolean

5.113 Sector Line Length

Type: SimpleAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector line length specifies the displayed length of the line, in ground units, defining the limit of the sector.

CamelCase: sectorLineLength

Value type: integer

5.114 Signal Duration

Type: SimpleAttributeType

Definition: The time occupied by a single instance of light/sound or eclipse/silence in a signal sequence.

CamelCase: signalDuration

Value type: real

5.115 Signal Frequency

Type: SimpleAttributeType

Definition: The frequency of a signal.

CamelCase: signalFrequency

Value type: integer

5.116 Signal Generation

Type: EnumerationType

Definition: The mechanism used to generate a fog or light signal.

CamelCase: signalGeneration

Value type: enumeration

Listed values:

Code	Label	Definition
1	Automatically	Signal generation is initiated by a self regulating mechanism such as a timer or light sensor.
2	By Wave Action	The signal is generated by the motion of the sea surface such as a bell in a buoy.

3	By Hand	The signal is generated by a manually operated mechanism such as a hand cranked siren.
4	By Wind	The signal is generated by the motion of air such as a wind driven whistle.
5	Radio Activated	Activated by radio signal.
6	Call Activated	Activated by making a call to a manned station.

5.117 Signal Group

Type: SimpleAttributeType

Definition: The number of signals, the combination of signals or the morse character(s) within one period of full sequence.

CamelCase: signalGroup

Value type: text

5.118 Signal Output

Type: SimpleAttributeType

Definition: Strength of signal output.

CamelCase: signalOutput

Value type: real

5.119 Signal Period

Type: SimpleAttributeType

Definition: The time occupied by an entire cycle of intervals of light and eclipse.

CamelCase: signalPeriod

Value type: real

5.120 Signal Status

Type: EnumerationType

Definition: The indication of an element of a signal sequence being a period of light/sound or eclipse/silence.

CamelCase: signalStatus

Value type: enumeration

Listed values:

Code	Label	Definition
1	Lit/Sound	The indication of an element of a signal sequence being a period of light or sound.
2	Eclipsed/Silent	The indication of an element of a signal sequence being a period of eclipse or silence.

5.121 Sinker Type

Type: SimpleAttributeType

Definition: Refers to the type of sinker or weight used to anchor an object (such as a buoy) to the seafloor or a fixed position.

CamelCase: sinkerType

Value type: text

5.122 Source

Type: SimpleAttributeType

Definition: The publication, document, or reference work from which information comes or is acquired.

CamelCase: source

Value type: text

5.123 Source Date

Type: SimpleAttributeType

Definition: The production date of the source; for example the date of measurement.

CamelCase: sourceDate

Value type: date

5.124 Status

Type: EnumerationType

Definition: The condition of an object at a given instant in time.

CamelCase: status

Value type: enumeration

Listed values:

Code	Label	Definition
1	Permanent	Intended to last or function indefinitely.
2	Occasional	Acting on special occasions; happening irregularly.
3	Recommended	Presented as worthy of confidence, acceptance, use, etc.
4	Not in Use	Use has ceased, but the facility still exists intact; disused.
5	Periodic/Intermittent	Recurring at intervals.
6	Reserved	Set apart for some specific use.
7	Temporary	Meant to last only for a time.
8	Private	Administered by an individual or corporation, rather than a State or a public body.
9	Mandatory	Compulsory; enforced.
11	Extinguished	No longer lit.
12	Illuminated	Lit by floodlights, strip lights, etc.
13	Historic	Famous in history; of historical interest.
14	Public	Belonging to, available to, used or shared by, the community as a whole and not restricted to private use.

15	Synchronized	Occur at a time, coincide in point of time, be contemporary or simultaneous.
16	Watched	Looked at or observed over a period of time especially so as to be aware of any movement or change.
17	Unwatched	Usually automatic in operation, without any permanently-stationed personnel to superintend it.
18	Existence Doubtful	A feature that has been reported but has not been definitely determined to exist.
19	On Request	When you ask for it.
20	Drop Away	To become lower in level.
21	Rising	To become higher in level.
22	Increasing	Becoming larger in magnitude.
23	Decreasing	Becoming smaller in magnitude.
24	Strong	Not easily broken or destroyed.
25	Good	In a satisfactory condition to use.
26	Moderately	Fairly but not very.
27	Poor	Not as good as it could be or should.
28	Buoyed	Marked by buoys.
29	Fully Operational	Entire observation platform is operating in accordance with, or exceeding, manufacturer specifications.
30	Partially Operational	At least one instrument that is part of an observation platform is not operating to manufacturer specification.
31	Drifting	Floating platform at the mercy of environmental elements, whether intentional or not.
32	Broken	Fractured or in pieces.
33	Offline	Observation platform is intentionally not reporting an environmental observation.
34	Discontinued	Observation station, suite of instruments, or an individual instrument, for a particular location, has been removed and is no longer at the particular location.
35	Manual Observation	Observations made by a human observer.
36	Unknown Status	Status of an observation platform, suite of instruments, or individual instrument is not known or unspecified.

37	Confirmed	Made certain as to truth, accuracy, validity, availability, etc.
38	Candidate	Item selected for an action.
39	Under Modification	Item that is in the process of being modified.
41	Under Removal / Deletion	Item in the process of being removed or deleted.
42	Removed / Deleted	Item that has been removed or deleted.
43	Candidate for Modification	Item selected for modification.

5.125 Swivel Type

Type: SimpleAttributeType

Definition: This refers to the type of swivel mechanism that allows rotation between two connected parts and is often used to reduce twisting in cables or ropes.

CamelCase: swivelType

Value type: text

5.126 Technique of Vertical Measurement

Type: EnumerationType

Definition: Survey method used to obtain depth information.

CamelCase: techniqueOfVerticalMeasurement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Found by Echo Sounder	The depth was determined by using an instrument that determines depth of water by measuring the time interval between emission of a sonic or ultrasonic signal and return of its echo from the bottom.
2	Found by Side Scan Sonar	The depth was computed from a record produced by active sonar in which fixed acoustic beams are directed into the water perpendicularly to the direction of travel to scan the seabed and generate a record of the seabed configuration.
3	Found by Multi Beam	The depth was determined by using a wide swath echo sounder that uses multiple beams to measure depths directly below and transverse to the ship's track.
4	Found by Diver	The depth was determined by a person skilled in the practice of diving.
5	Found by Lead Line	The depth was determined by using a line, graduated with attached marks and fastened to a sounding lead.
6	Swept by Wire-Drag	The given area was determined to be free from navigational dangers to a

		certain depth by towing a buoyed wire at the desired depth by two launches, or a least depth was identified using the same technique.
7	Found by Laser	The depth was determined by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
8	Swept by Vertical Acoustic System	The given area has been swept using a system comprised of multiple echo sounder transducers attached to booms deployed from the survey vessel.
9	Found by Electromagnetic Sensor	The depth was determined by using an instrument that compares electromagnetic signals.
10	Photogrammetry	The science or art of obtaining reliable measurements from photographs.
11	Satellite Imagery	The depth was determined by using instruments placed aboard an artificial satellite.
12	Found by Levelling	The depth was determined by using levelling techniques to find the elevation of the point relative to a datum.
13	Swept by Side Scan Sonar	The given area was determined to be free from navigational dangers to a certain depth by towing a side scan sonar.
14	Computer Generated	The sounding was determined from a bottom model constructed using a computer.
15	Found by LIDAR	The depth was measured by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
16	Synthetic Aperture Radar	A radar with a synthetic aperture antenna which is composed of a large number of elementary transducing elements. The signals are electronically combined into a resulting signal equivalent to that of a single antenna of a given aperture in a given direction.
17	Hyperspectral Imagery	Term used to describe the imagery derived from subdividing the electromagnetic spectrum into very narrow bandwidths. These narrow bandwidths may be combined with or subtracted from each other in various ways to form images useful in precise terrain or target analysis.

5.127 Text

Type: SimpleAttributeType

Definition: A non-formatted digital text string.

CamelCase: text

Value type: text

5.128 Topmark/Daymark Shape

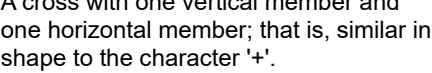
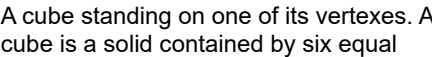
Type: EnumerationType

Definition: The shape a topmark or daymark exhibits.

CamelCase: topmarkDaymarkShape

Value type: enumeration

Listed values:

Code	Label	Definition
1	Cone (Point Up)	Is where the vertex points up. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities - IALA topmarks, lateral.
2	Cone (Point Down)	Is where the vertex points down. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities - IALA topmarks, lateral.
3	Sphere	A curved surface all points of which are equidistant from a fixed point within, called the centre.
4	2 Spheres	Two spheres, one above the other. Two black spheres are commonly used as an International Association of Lighthouse Authorities - IALA topmark (isolated danger).
5	Cylinder	A solid geometrical figure generated by straight lines fixed in direction and describing with one of point a closed curve, especially a circle (in which case the figure is circular cylinder, it's ends being parallel circles). Cylinders are commonly used as International Association of Lighthouse Authorities - IALA topmarks lateral.
6	Board	Usually of rectangular shape, made from timber or metal and used to provide a contrast with the natural background of a daymark. The actual daymark is often painted on to this board.
7	X-Shaped	Having a shape or a cross-section like the capital letter X. An x-shape as an International Association of Lighthouse Authorities - IALA topmark should be 3 dimensional in shape. It is made of at least three crossed bars.
8	Upright Cross	A cross with one vertical member and one horizontal member; that is, similar in shape to the character '+'. 
9	Cube (Point Up)	A cube standing on one of its vertexes. A cube is a solid contained by six equal 

		squares, a regular hexahedron.
10	2 Cones (Point to Point)	2 cones, one above the other, with their vertices together in the centre.
11	2 Cones (Base to Base)	2 cones, one above the other, with their bases together in the centre and their vertices pointing up and down.
12	Rhombus	A plane figure having four equal sides and equal opposite angles (two acute and two obtuse); an oblique equilateral parallelogram.
13	2 Cones (Points Upward)	2 cones, one above the other, with their vertices pointing up.
14	2 Cones (Points Downward)	2 cones, one above the other, with their vertices pointing down.
15	Besom (Point Up)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point up is where the thicker (untied) end of the besom is at the bottom.
16	Besom (Point Down)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point down is where the thinner (tied) end of the besom is at the bottom.
17	Flag	A flag mounted on a short pole.
18	Sphere Over a Rhombus	A sphere located above a rhombus.
19	Square	A plane figure with four right angles and four equal straight sides.
20	Rectangle (Horizontal)	Where the two longer opposite sides are standing horizontally. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
21	Rectangle (Vertical)	Where the two longer opposite sides are standing vertically. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
22	Trapezium (Up)	A quadrilateral having one pair of opposite sides parallel, and which stands on its longer parallel side.
23	Trapezium (Down)	A quadrilateral having one pair of opposite sides parallel, and which stands on its shorter parallel side.
24	Triangle (Point Up)	A figure having three angles and three sides, and which has a vertex at the top.
25	Triangle (Point Down)	A figure having three angles and three sides, and which has a side at the top.
26	Circle	A perfectly round plane figure whose circumference is everywhere equidistant from its centre.

27	Two Upright Crosses (One Over the Other)	Two upright crosses, generally vertically disposed one above the other.
28	T-Shape	Having a shape like the capital letter T.
29	Triangle Pointing Up Over a Circle	A triangle, vertex uppermost, located above a circle.
30	Upright Cross Over a Circle	An upright cross located above a circle.
31	Rhombus Over a Circle	A rhombus located above a circle.
32	Circle Over a Triangle Pointing Up	A circle located over a triangle, vertex uppermost.
33	Other Shape (See Shape Information)	An uncommon and/or non-standardized shape as textually described using an associated attribute.
34	Tubular	Having the form of or consisting of a tube.

5.129 Traffic Flow

Type: EnumerationType

Definition: Direction of vessels passing a reference point.

CamelCase: trafficFlow

Value type: enumeration

Listed values:

Code	Label	Definition
1	Inbound	Traffic flow in a general direction toward a port or similar destination.
2	Outbound	Traffic flow in a general direction away from a port or similar point of origin.
3	One-Way	Traffic flow in one general direction only.
4	Two-Way	Traffic flow in two generally opposite directions.

5.130 Type of Buoy

Type: SimpleAttributeType

Definition: Type equipment used as a buoy in a particular installation.

CamelCase: typeOfBuoy

Value type: text

5.131 Type of Environmental Observation Equipment

Type: SimpleAttributeType

Definition: Type of sensor used to observe the environment. For example Anemometer, fog monitor, etc.

CamelCase: typeOfEnvironmentalObservationEquipment

Value type: text

5.132 Type of Light

Type: SimpleAttributeType

Definition: Type of light equipment.

CamelCase: typeOfLight

Value type: text

5.133 Uncertainty Fixed

Type: SimpleAttributeType

Definition: The best estimate of the fixed horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyFixed

Value type: real

5.134 Uncertainty Variable Factor

Type: SimpleAttributeType

Definition: The factor to be applied to the variable component of an uncertainty equation so as to provide the best estimate of the variable horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyVariableFactor

Value type: real

5.135 Value of Geographic Range

Type: SimpleAttributeType

Definition: The greatest distance at which a light can be seen as a function of the curvature of the earth and the heights of the light source and the observer.

CamelCase: valueOfGeographicRange

Value type: real

5.136 Value of Luminous Range

Type: SimpleAttributeType

Definition: The greatest distance at which a light can be seen merely as a function of its luminous intensity, the meteorological visibility, and the sensitivity of the observer's eye.

CamelCase: valueOfLuminousRange

Value type: real

5.137 Value of Maximum Range

Type: SimpleAttributeType

Definition: The extreme distance at which an feature can be seen or a signal detected.

CamelCase: valueOfMaximumRange

Value type: real

5.138 Value of Nominal Range

Type: SimpleAttributeType

Definition: The luminous range of a light in a homogenous atmosphere in which the meteorological visibility is 10 sea miles.

CamelCase: valueOfNominalRange

Value type: real

5.139 Vertical Accuracy

Type: SimpleAttributeType

Definition: The degree to which a vertical measurement is accurate, typically referring to the accuracy of an object's position in the vertical plane (height or depth)

CamelCase: verticalAccuracy

Value type: real

5.140 Vertical Datum

Type: EnumerationType

Definition: The reference level used for expressing the vertical measurements of points on the earth's surface. Also called datum level, reference plane, levelling datum, datum for sounding reduction, datum for heights.

CamelCase: verticalDatum

Value type: enumeration

Listed values:

Code	Label	Definition
1	Mean Low Water Springs	The average height of the low waters of spring tides. This level is used as a tidal datum in some areas. Also called spring low water.
2	Mean Lower Low Water Springs	The average height of lower low water springs at a place.
3	Mean Sea Level	The average height of the surface of the sea at a tide station for all stages of the tide over a 19-year period, usually determined from hourly height readings measured from a fixed predetermined reference level.
4	Lowest Low Water	An arbitrary level conforming to the lowest tide observed at a place, or some what lower.
5	Mean Low Water	The average height of all low waters at a place over a 19-year period.
6	Lowest Low Water Springs	An arbitrary level conforming to the lowest water level observed at a place at spring tides during a period of time shorter than 19 years.
7	Approximate Mean Low Water Springs	An arbitrary level, usually within 0.3m from that of Mean Low Water Springs (MLWS).
8	Indian Spring Low Water	An arbitrary tidal datum approximating the level of the mean of the lower low water at spring tides. It was first used in waters surrounding India.
9	Low Water Springs	An arbitrary level, approximating that of mean low water springs (MLWS).
10	Approximate Lowest Astronomical Tide	An arbitrary level, usually within 0.3m from that of Lowest Astronomical Tide (LAT).
11	Nearly Lowest Low Water	An arbitrary level approximating the lowest water level observed at a place,

		usually equivalent to the Indian Spring Low Water (ISLW).
12	Mean Lower Low Water	The average height of the lower low waters at a place over a 19-year period.
13	Low Water	The lowest level reached at a place by the water surface in one oscillation. Also called low tide.
14	Approximate Mean Low Water	An arbitrary level, usually within 0.3m from that of Mean Low Water (MLW).
15	Approximate Mean Lower Low Water	An arbitrary level, usually within 0.3m from that of Mean Lower Low Water (MLLW).
16	Mean High Water	The average height of all high waters at a place over a 19-year period.
17	Mean High Water Springs	The average height of the high waters of spring tides. Also called spring high water.
18	High Water	The highest level reached at a place by the water surface in one oscillation.
19	Approximate Mean Sea Level	An arbitrary level, usually within 0.3m from that of Mean Sea Level (MSL).
20	High Water Springs	An arbitrary level, approximating that of mean high water springs (MHWS).
21	Mean Higher High Water	The average height of higher high waters at a place over a 19-year period.
22	Equinoctial Spring Low Water	The level of low water springs near the time of an equinox.
23	Lowest Astronomical Tide	The lowest tide level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
24	Local Datum	An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
25	International Great Lakes Datum 1985	A vertical reference system with its zero based on the mean water level at Rimouski/Pointe-au-Pere, Quebec, over the period 1970 to 1988.
26	Mean Water Level	The average of all hourly water levels over the available period of record.
27	Lower Low Water Large Tide	The average of the lowest low waters, one from each of 19 years of observations.
28	Higher High Water Large Tide	The average of the highest high waters, one from each of 19 years of observations.
29	Nearly Highest High Water	An arbitrary level approximating the highest water level observed at a place,

		usually equivalent to the high water springs.
30	Highest Astronomical Tide	The highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
31	Local Low Water Reference Level	Low water reference level of the local area.
32	Local High Water Reference Level	High water reference level of the local area.
33	Local Mean Water Reference Level	Mean water reference level of the local area.
34	Equivalent Height of Water (German GIW)	A low water level which is the result of a defined low water discharge - called equivalent discharge.
35	Highest Shipping Height of Water (German HSW)	Upper limit of water levels where navigation is allowed.
36	Reference Low Water Level According to Danube Commission	The water level at a discharge, which is exceeded 94 % of the year within a period of 30 years.
37	Highest Shipping Height of Water According to Danube Commission	The water level at a discharge, which is exceeded 1% of the year within a period of 30 years.
38	Dutch River Low Water Reference Level (OLR)	The water level at a discharge, which is exceeded 95% of the year within a period of 20 years.
39	Russian Project Water Level	Conditional low water level with established probability.
40	Russian Normal Backwater Level	Highest water level derived from the upper backwater stream in watercourse or reservoir under the normal operational conditions.
41	Ohio River Datum	The Ohio River datum.
43	Dutch High Water Reference Level	Dutch High Water Reference Level.
44	Baltic Sea Chart Datum 2000	The datum refers to each Baltic country's realization of the European Vertical Reference System (EVRS) with land-uplift epoch 2000, which is connected to the Normaal Amsterdams Peil (NAP).
45	Dutch Estuary Low Water Reference Level (OLW)	Dutch Estuary Low Water Reference Level (OLW)

5.141 Vertical Length

Type: SimpleAttributeType

Definition: The total vertical length of a feature.

CamelCase: verticalLength

Value type: real

5.142 Virtual AIS Aid to Navigation Type

Type: EnumerationType

Definition: A purpose of a virtual AIS Aid to Navigation.

CamelCase: virtualAISaidToNavigationType

Value type: enumeration

Listed values:

Code	Label	Definition
1	North Cardinal	Indicates that it should be passed to the north side of the aid.
2	East Cardinal	Indicates that it should be passed to the east side of the aid.
3	South Cardinal	Indicates that it should be passed to the south side of the aid.
4	West Cardinal	Indicates that it should be passed to the west side of the aid.
5	Port Lateral	Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
6	Starboard Lateral	Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
7	Preferred Channel to Port	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
8	Preferred Channel to Starboard	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
9	Isolated Danger	A mark used alone to indicate a dangerous reef or shoal. The mark may be passed on either hand.
10	Safe Water	Indicates that there is navigable water around the mark.
11	Special Purpose	A special purpose aid is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notice to Mariners
12	New Danger Marking	A mark used to indicate the existence of a recently identified new danger, such as a wreck.

5.143 Visual Prominence

Type: EnumerationType

Definition: The extent to which a feature, either natural or artificial, is visible from seaward.

CamelCase: visualProminence

Value type: enumeration

Listed values:

Code	Label	Definition
1	Visually Conspicuous	Term applied to a feature either natural or artificial which is distinctly and notably visible from seaward.
2	Not Visually Conspicuous	An object that may be visible from seaward, but cannot be used as a fixing mark and is not conspicuous.
3	Prominent	Objects which are easily identifiable, but do not justify being classed as conspicuous.

5.144 Wave Length Value

Type: SimpleAttributeType

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave.

CamelCase: waveLengthValue

Value type: real

5.145 Weight

Type: SimpleAttributeType

Definition: (1) The force with which a body is attracted by gravity. (2) Any block or mass of material used for its heaviness. (3) The relative value of an observation, source, or quantity when compared with other observations, sources, or quantities of the same or related quantities. The value determined by the most reliable method is assigned the greatest weight.

CamelCase: weight

Value type: real

6 Complex attributes

6.1 Cable Dimensions

Type: ComplexAttributeType

Definition: The dimensions of a cable to give its length and diameter.

CamelCase: CableDimensions

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Cable Length			RE	1, 1
Height/Length Units		1. Metres 2. Feet 3. Kilometres 4. Hectometres 5. Statute Miles	EN	1, 1

		6. Nautical Miles		
Diameter			RE	1, 1

6.2 Change Details

Type: ComplexAttributeType

Definition: -

CamelCase: ChangeDetails

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Aton Commissioning		1. Buoy Establishment 2. Light Establishment 3. Beacon Establishment 4. Audible Signal Establishment 5. Fog Signal Establishment 6. AIS Transmitter Establishment 7. V-AIS Establishment 8. RACON Establishment 9. DGPS Station Establishment 10. ELORAN Station Establishment 11. DGLONASS Station Establishment 12. E-Chayka Station Establishment 13. EGNOS establishment	EN	0, 1
Aton Removal		1. Buoy Removal 2. Buoy Temporary Removal 3. Light Removal 4. Light Temporary Removal 5. Beacon Removal 6. Beacon Temporary Removal 7. Fog Signal Removal 8. Fog Signal Temporary	EN	0, 1

		Removal 9. Audible Signal Removal 10. Audible Signal Temporary Removal 11. V-AIS Removal 12. V-AIS Temporary Removal 13. RACON signal removal 14. RACON Temporary Removal 15. DGPS removal 16. DGPS temporary removal 17. EGNOS removal 18. EGNOS temporary removal 19. LORAN C Station Removal 20. LORAN C Station Temporary Removal 21. eLORAN removal 22. eLORAN temporary removal 23. Chayka Station Removal 24. Chayka Station Temporary Removal 25. E-Chayka Station Removal 26. E-Chayka Station Temporary Removal		
Aton Replacement		1. Buoy Change 2. Buoy Temporary Change 3. Light Change 4. Light Temporary Change 5. Sector Light Change 6. Sector Light Temporary Change 7. Beacon Change 8. Beacon Temporary Change 9. Fog Signal Change 10. Fog Signal Temporary Change	EN	0, 1

		11. Audible Signal Change 12. Audible Signal Temporary Change 13. V-AIS Change 14. V-AIS Temporary Change 15. RACON signal change 16. RACON Temporary Change		
Fixed Aton Change		1. Beacon Missing 2. Beacon Damaged 3. Light beacon Unlit 4. Light beacon Unreliable 5. Light beacon Not synchronized 6. Light beacon damaged 7. Beacon Topmark Missing 8. Beacon Topmark Damaged 9. Beacon Daymark Unreliable 10. Floodlit beacon Unlit 11. Beacon Restored To Normal	EN	0, 1
Floating Aton Change		1. Buoy Adrift 2. Buoy Damaged 3. Buoy Daymark Unreliable 4. Buoy Destroyed 5. Buoy Missing 6. Buoy Move 7. Buoy off Position 8. Buoy re-establishment 9. Buoy Restored to Normal 10. Buoy Topmark Damaged 11. Buoy Topmark Missing 12. Buoy Will Be Withdrawn 13. Buoy Withdrawn	EN	0, 1

		14. Decommissioned for winter 15. Lifted for Winter 16. Light buoy Light damaged 17. Light buoy Light not synchronized 18. Light buoy Light unlit 19. Light buoy Light unreliable 20. Marine Aids to Navigation Unreliable 21. Recommissioned for navigation season 22. Replaced by Winter Spar 23. Seasonal Decommissioning Complete 24. Seasonal Decommissioning in Progress 25. Seasonal recommissioning complete 26. Seasonal recommissioning in progress		
Audible Signal Aton Change		1. Audible Signal Out Of Service 2. Fog Signal Out Of Service 3. Audible Signal Operating Properly 4. Fog Signal Operating Properly	EN	0, 1
Lighted Aton Change		1. Light Unlit 2. Light Unreliable 3. Light Re-Establishment 4. Light Range Reduced 5. Light Without Rhythm 6. Light Out Of Synchronization 7. Light Daymark Unreliable 8. Light Operating Properly 9. Sector light Sector obscured 10. Front leading/range light Unlit	EN	0, 1

		11. Rear leading/range light Unlit 12. Front leading/range light Unreliable 13. Rear leading/range light Unreliable 14. Front leading/range light Light range reduced 15. Rear leading/range light Light range reduced 16. Front leading/range light without rhythm 17. Rear leading/range light without rhythm 18. Leading/range lights out of synchronization 19. Front leading/range beacon Unreliable 20. Rear leading/range beacon Unreliable 21. Front leading/range light is operating properly 22. Rear leading/range light is operating properly 23. Front leading/range beacon restored to normal 24. Rear leading/range beacon restored to normal		
Electronic Aton Change		1. AIS Transmitter Out Of Service 2. AIS Transmitter Unreliable 3. AIS Transmitter Operating Properly 4. V-AIS Out Of Service 5. V-AIS Unreliable 6. V-AIS Operating Properly 7. RACON Out Of Service 8. RACON Unreliable 9. RACON Operating Properly 10. DGPS Out Of Service 11. DGPS Operating Properly	EN	0, 1

		12. DGPS Unreliable		
		13. LORAN C operating properly		
		14. LORAN C unreliable		
		15. LORAN C out of service		
		16. ELORAN Operating Properly		
		17. ELORAN Unreliable		
		18. ELORAN Out Of Service		
		19. DGLOANSS operating properly		
		20. DGLOANSS unreliable		
		21. DGLOANSS out of service		
		22. Chayka Operating Properly		
		23. Chayka Unreliable		
		24. Chayka Out Of Service		
		25. E-Chayka Operating Properly		
		26. E-Chayka Unreliable		
		27. E-Chayka Out Of Service		
		28. EGNOS Operating Properly		
		29. EGNOS Unreliable		
		30. EGNOS Out Of Service		

6.3 Obscured Sector

Type: ComplexAttributeType

Definition: -

CamelCase: ObscuredSector

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Limit			C	1, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			(S) C	1, 1

Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Information			C	0, 1
Language			(S) TE	0, 1
Text			(S) TE	1, 1

6.4 Contact Address

Type: ComplexAttributeType

Definition: Direction or superscription of a letter, package, etc., specifying the name of the place to which it is directed, and optionally a contact person or organisation who should receive it.

CamelCase: contactAddress

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Delivery Point			TE	0, 1
City Name			TE	0, 1
Administrative Division			TE	0, 1
Country Name			TE	0, 1
Postal Code			TE	0, 1

6.5 Directional Character

Type: ComplexAttributeType

Definition: A directional light is a light illuminating a sector of very narrow angle and intended to mark a direction to follow.

CamelCase: directionalCharacter

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Moire Effect			BO	0, 1
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

6.6 Feature Name

Type: ComplexAttributeType

Definition: Provides the name of an entity, defines the national language of the name, and provides the option to display the name at various system display settings.

CamelCase: featureName

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Display Name			BO	0, 1
Language			TE	0, 1
Name			TE	1, 1

6.7 Fixed Date Range

Type: ComplexAttributeType

Definition: An active period of a single fixed event or occurrence, as the date range between discrete start and end dates.

CamelCase: fixedDateRange

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Date End			TD	0, 1
Date Start			TD	0, 1

6.8 Horizontal Position Uncertainty

Type: ComplexAttributeType

Definition: The best estimate of the accuracy of a position.

CamelCase: horizontalPositionUncertainty

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed			RE	1, 1
Uncertainty Variable Factor			RE	0, 1

6.9 Information

Type: ComplexAttributeType

Definition: Textual information about the feature. The information may be provided as a string of text or as a file name of a single external text file that contains the text.

CamelCase: information

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
File Locator			TE	0, 1
File Reference			TE	0, 1
Headline			TE	0, 1
Language			TE	1, 1

Text			TE	0, 1
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6.10 Light Sector

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.

CamelCase: lightSector

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary	EN	1, *
Directional Character			C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1

Orientation Value			(S) RE	1, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	EN	0, *
Sector Limit			C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Value of Nominal Range			RE	0, 1
Sector Information			C	0, *
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Extension			BO	0, 1

6.11 Multiplicity of Features

Type: ComplexAttributeType

Definition: The number of features of identical character that exist as a colocated group.

CamelCase: multiplicityOfFeatures

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Multiplicity Known			BO	1, 1
Number of Features			IN	0, 1

6.12 Orientation

Type: ComplexAttributeType

Definition: The angular distance measured from true north to the major axis of the feature.

CamelCase: orientation

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Orientation Uncertainty			RE	0, 1
Orientation Value			RE	1, 1

6.13 Periodic Date Range

Type: ComplexAttributeType

Definition: The active period of a recurring event or occurrence.

CamelCase: periodicDateRange

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Date End			TD	1, 1
Date Start			TD	1, 1

6.14 Positioning Method

Type: ComplexAttributeType

Definition: A description of the method used to obtain a position.(proposed by CCG)

CamelCase: positioningMethod

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Positioning Equipment		1. DGPS Receiver 2. GLONASS Receiver 3. GPS Receiver 4. GPS/WAAS Receiver	EN	1, 1
NMEAString			TE	1, 1

6.15 Radar Wave Length

Type: ComplexAttributeType

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave in the radar band of the electromagnetic spectrum.

CamelCase: radarWaveLength

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Radar Band			TE	1, 1
Wave Length Value			RE	1, 1

6.16 Rhythm of Light

Type: ComplexAttributeType

Definition: The sequence of times occupied by intervals of light/sound and eclipse/silence for all light characteristics or sound signals.

CamelCase: rhythmOfLight

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash	EN	1, 1

		28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light 31. Composite Group-Occulting Light 32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light		
Signal Group			TE	0, 10
Signal Period			RE	0, 1
Signal Sequence			C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

6.17 Sector Characteristics

Type: ComplexAttributeType

Definition: Describes the characteristics of a light sector.

CamelCase: sectorCharacteristics

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting	EN	1, 1

		12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing 30. Group-Occulting Light 31. Composite Group-Occulting Light 32. Group flashing light 33. Composite Group-Flashing Light 34. Group Quick Light 35. Group Very Quick Light		
Light Sector			C	1, 10
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown	(S) EN	1, *

		9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink 14. Green A 15. Green B 16. White Temporary 17. Red Temporary 18. Yellow Temporary 19. Green Preferred 20. Green Temporary		
Directional Character			(S) C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	(S) EN	0, *
Sector Limit			(S) C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1

Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Value of Nominal Range			(S) RE	0, 1
Sector Information			(S) C	0, *
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Extension			(S) BO	0, 1
Signal Group			TE	0, 10
Signal Period			RE	0, 1
Signal Sequence			C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Candela			RE	0, 1

6.18 Sector Information

Type: ComplexAttributeType

Definition: Additional textual information about a light sector.

CamelCase: sectorInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Language			TE	0, 1
Text			TE	1, 1

6.19 Sector Limit

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. The sector limit specifies the limits of the sector In a clockwise direction around the central feature (for example a light).

CamelCase: sectorLimit

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Limit One			C	1, 1

Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two			C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) IN	0, 1

6.20 Sector Limit One

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit one specifies the first limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light).

CamelCase: sectorLimitOne

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Bearing			RE	1, 1
Sector Line Length			IN	0, 1

6.21 Sector Limit Two

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit two specifies the second limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light).

CamelCase: sectorLimitTwo

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Bearing			RE	1, 1
Sector Line Length			IN	0, 1

6.22 Shape Information

Type: ComplexAttributeType

Definition: Textual information about the shape of a non-standard topmark.

CamelCase: shapeInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Language			TE	0, 1
Text			TE	1, 1

6.23 Signal Sequence

Type: ComplexAttributeType

Definition: The sequence of times occupied by intervals of light and eclipse for all light characteristics.

CamelCase: signalSequence

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Signal Duration			RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	EN	1, 1

6.24 Sinker Dimensions

Type: ComplexAttributeType

Definition: The dimensions of a sinker/anchor to give its three dimensional shape measurements.

CamelCase: sinkerDimensions

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Height/Length Units		1. Metres 2. Feet 3. Kilometres 4. Hectometres 5. Statute Miles 6. Nautical Miles	EN	1, 1
Horizontal Length			RE	0, 1
Horizontal Width			RE	0, 1
Vertical Length			RE	0, 1

6.25 Spatial Accuracy

Type: ComplexAttributeType

Definition: Provides an indication of the vertical and horizontal positional uncertainty of bathymetric data, optionally within a specified date range.

CamelCase: spatialAccuracy

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
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Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Horizontal Position Uncertainty			C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Vertical Uncertainty			C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1

6.26 Textual Description

Type: ComplexAttributeType

Definition: Encodes the file name of a single external text file that contains the text in a defined language, which provides additional textual information that cannot be provided using other allowable attributes for the feature.

CamelCase: textualDescription

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
File Reference			TE	1, 1
Language			TE	0, 1

6.27 Vertical Uncertainty

Type: ComplexAttributeType

Definition: The best estimate of the vertical accuracy of depths, heights, vertical distances and vertical clearances.

CamelCase: verticalUncertainty

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed			RE	1, 1
Uncertainty Variable Factor			RE	0, 1

7 Roles

7.1 Aton Part

Type: Role

Definition: -

CamelCase: Atonpart

7.2 Status Part

Type: Role

Definition: -

CamelCase: Statuspart

7.3 Buoy Part

Type: Role

Definition: TBD

CamelCase: buoyPart

7.4 Topmark part

Type: Role

Definition: TBD

CamelCase: topmarkPart

7.5 Parent

Type: Role

Definition: -

CamelCase: parentFihoXyz

7.6 Child

Type: Role

Definition: -

CamelCase: child

7.7 Physical AIS Broadcast By

Type: Role

Definition: -

CamelCase: physicalAISbroadcastBy

7.8 Physical AIS Broadcasts

Type: Role

Definition: -

CamelCase: physicalAISbroadcasts

7.9 Synthetic AIS Broadcast By

Type: Role

Definition: -

CamelCase: syntheticAISbroadcastBy

7.10 Synthetic AIS Broadcasts

Type: Role

Definition: -

CamelCase: syntheticAISbroadcasts

7.11 Virtual AIS Broadcast By

Type: Role

Definition: -

CamelCase: virtualAISbroadcastBy

7.12 Virtual AIS Broadcasts

Type: Role

Definition: -

CamelCase: virtualAISbroadcasts

7.13 Buoy Attached

Type: Role

Definition: -

CamelCase: buoyattached

7.14 Counter Weighholds

Type: Role

Definition: -

CamelCase: counterWeighholds

7.15 Buoy Hangs

Type: Role

Definition: -

CamelCase: buoyhangs

7.16 Bridle Holds

Type: Role

Definition: -

CamelCase: bridleholds

7.17 Shackle To Cableconnected To